European Journal of Operational Research 336 (2027) 68–84

Contents lists available at ScienceDirect

European Journal of Operational Research

journal homepage: www.elsevier.com/locate/eor

Continuous Optimization

A causation-based framework for pricing and cost allocation of energy, reserves, and transmission in modern power systems

Luíza Ribeiro [a], Alexandre Street [a,d,e,*], José M. Arroyo [b], Rodrigo Moreno [c]

[a] Electrical Engineering Department, Pontifical Catholic University of Rio de Janeiro, Rio de Janeiro, Brazil
[b] E.T.S.I. Industrial, Universidad de Castilla-La Mancha, Ciudad Real, Spain
[c] Universidad Adolfo Ibáñez, Santiago, Chile
[d] Center for Latin American Studies (CLAS), Stanford University, Stanford, CA, USA
[e] Program on Energy and Sustainable Development (PESD), Stanford University, Stanford, CA, USA

ARTICLE INFO

Keywords:
OR in energy
Causation-based pricing
Co-optimized energy and reserve electricity markets
Security charges

ABSTRACT

The increasing vulnerability of power systems has heightened the need for operating reserves to manage contingencies such as generator outages, line failures, and sudden load variations. Unlike energy costs, driven by consumer demand, operating reserve costs arise from addressing the most critical contingencies—prompting the question: how should these costs be allocated through efficient pricing mechanisms? As an alternative to previously reported schemes, this paper presents a new causation-based pricing framework for electricity markets based on contingency-constrained energy and reserve scheduling models. Major salient features include a novel security charge mechanism along with the explicit definition of prices for up-spinning reserves, down-spinning reserves, and transmission services. These features ensure more comprehensive and efficient cost-reflective market operations. Moreover, the proposed nodal pricing scheme yields cost recovery and system revenue adequacy based on the cost-causation principle. Numerical results from three case studies illustrate the performance of the proposed pricing scheme.

1. Introduction

The optimization of power system operation and planning has been a central theme of research for more than half a century (Kaya et al., 2026). Over this time, a core and enduring topic in this field stands out, namely ensuring system security—the ability to withstand contingencies such as generator outages, line failures, or sudden load variations without involuntary load shedding (Arroyo & Galiana, 2005). Although security has been a consistent priority for the power industry, the challenge has increased significantly in recent years. In modern power systems, besides energy, ancillary services have become indispensable for meeting diverse security requirements. Among these, operating reserves (ORs) are crucial for maintaining system security by balancing supply and demand. The transition towards deregulated markets and the large-scale integration of renewable generation have introduced new dynamics and uncertainties, making the traditional security problem more complex and critical than ever (Kaya et al., 2026). It is worth emphasizing that the increasing penetration of non-synchronous, inverter-based generation (e.g., wind and solar) has led to a significant reduction in overall system inertia. This physical inertia, traditionally provided by the rotating mass of conventional generators, is critical for grid stability as it naturally dampens frequency fluctuations. In low-inertia systems, a sudden contingency causes a much faster and deeper frequency deviation, thereby requiring larger and faster-acting ORs to maintain system security. Compounded by a growing exposure to natural hazards, these new dynamics have heightened existing vulnerabilities and amplified the role of ancillary services. Consequently, the costs of ORs represent an increasing share of operational costs (Badesa et al., 2025; Matamala et al., 2024; Shi et al., 2023).

Despite the increasing reliance on ancillary services, existing electricity markets have failed to establish adequate frameworks to meet the evolving challenges of modern power systems (Billimoria et al., 2020). Markets generally follow either joint or sequential designs for energy and reserve clearing (Galiana et al., 2005; Ribeiro et al., 2023). Joint markets, like those run across the United States (Federal Energy Regulatory Commission (FERC), 2011), co-optimize energy and reserves simultaneously. Conversely, sequential markets, common in Europe, clear energy and reserves separately. While sequential markets aim to optimize reserves as market products, they often rely on *ad hoc* allocation of transmission capacity for reserves, leading to inefficiencies and suboptimal outcomes (European Union Agency for the Cooperation

* Corresponding author at: Electrical Engineering Department, Pontifical Catholic University of Rio de Janeiro, Rio de Janeiro, Brazil.
E-mail address: street@puc-rio.br (A. Street).

https://doi.org/10.1016/j.ejor.2026.08.005
Received 14 May 2025; Accepted 3 August 2026
Available online 11 August 2026
0377-2217/© 2026 The Authors. Published by Elsevier B.V. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

European Journal of Operational Research 336 (2027) 68–84

Contents lists available at ScienceDirect

European Journal of Operational Research

journal homepage: www.elsevier.com/locate/eor

连续优化

基于因果关系的现代电力系统能源、备用容量和输电定价与成本分配框架

Luíza Ribeiro [a], Alexandre Street [a,d,e]*, José M. Arroyo [b], Rodrigo Moreno [c]

[a] Electrical Engineering Department, Pontifical Catholic University of Rio de Janeiro, Rio de Janeiro, Brazil [b] E.T.S.I. Industrial, Universidad de Castilla-La Mancha, Ciudad Real, Spain [c] Universidad Adolfo Ibáñez, Santiago, Chile [d] Center for Latin American Studies (CLAS), Stanford University, Stanford, CA, USA [e] Program on Energy and Sustainable Development (PESD), Stanford University, Stanford, CA, USA

文章信息

Keywords:
能源因果定价中的或 联合优化的能源和备用
电力市场 安全费用

摘要

电力系统日益脆弱，这增加了对备用容量的需求，以应对如发电机故障、线路故障和突发负荷变化等突发事件。与受消费者需求驱动的能源成本不同，备用容量成本来源于应对最关键的突发事件——这引出了一个问题：应如何通过高效的定价机制分配这些成本？作为对先前报告方案的替代，本文提出了一种基于因果关系的电力市场定价框架，该框架基于考虑突发事件约束的能源和备用调度模型。主要显著特点包括一种新颖的安全收费机制，以及对上旋备用、下旋备用和输电服务价格的明确定义。这些特点确保了更全面和高效的成本反映市场运行。此外，所提出的节点定价方案实现了成本回收和系统收入充分性。

1. 引言

电力系统运行和规划的优化已经是半个多世纪以来研究的核心主题（Kaya 等，2026）。在这段时间里，该领域有一个核心且持久的话题尤为突出，即确保系统安全——在不进行非自愿负荷削减的情况下承受发电机故障、线路故障或突发负荷变化等偶发事件的能力（Arroyo & Galiana，2005）。虽然安全一直是电力行业的一贯优先事项，但近年来这一挑战显著增加。在现代电力系统中，除了能源之外，辅助服务已成为满足多样化安全需求不可或缺的组成部分。在这些服务中，运行备用（ORs）对于通过平衡供需来维持系统安全至关重要。向放松管制市场的过渡以及可再生能源的大规模接入引入了新的动态和不确定性，使传统的安全问题更加电网稳定性因为它能够自然地抑制频率波动。在低惯性系统中，突发事件会导致频率偏差更快且更深，因此需要更大且反应更快的操作储备（ORs）来维持系统安全。由于对自然灾害的暴露日益增加，这些新动态加剧了已有的脆弱性，并放大了辅助服务的作用。因此，操作储备的成本在运营成本中所占的比例越来越高（Badesa 等, 2025; Matamala 等, 2024; Shi 等, 2023）。

尽管对辅助服务的依赖日益增加，但现有电力市场未能建立足够的框架来应对现代电力系统不断变化的挑战（Billimoria 等，2020）。市场通常采用能源和备用电力清算的联合或顺序设计（Galiana 等，2005；Ribeiro 等，2023）。联合市场，如美国各地运行的市场（联邦能源监管委员会（FERC），2011），同时优化能源和备用电力。相比之下，欧洲常见的顺序市场则分别清算能源和备用电力。尽管顺序市场旨在将备用作为市场产品进行优化，但它们通常依赖于备用电力传输容量的分配，从而导致效率低下和次优结果（欧洲联盟合作机构）。

* Corresponding author at: Electrical Engineering Department, Pontifical Catholic University of Rio de Janeiro, Rio de Janeiro, Brazil.
E-mail address: street@puc-rio.br (A. Street).

https://doi.org/10.1016/j.ejor.2026.08.005
Received 14 May 2025; Accepted 3 August 2026
Available online 11 August 2026
0377-2217/© 2026 The Authors. Published by Elsevier B.V. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

of Energy Regulators (ACER), 2026). In contrast, joint markets typically achieve higher social welfare by better capturing the interdependencies between these products, as evidenced by several studies (Aganagic et al., 1998; Ahunbay et al., 2024; Arroyo & Galiana, 2005; Badesa et al., 2023; Bouffard et al., 2005; Byers & Hug, 2023; Chen et al., 2014; Galiana et al., 2005; Gan & Litvinov, 2003; Karangelos & Bouffard, 2012; Kirschen & Strbac, 2019; Morales et al., 2012; Wang & Chen, 2021; Wang et al., 2009; Wong & Fuller, 2007; Wu et al., 2004). Unfortunately, notwithstanding the significant improvements brought about by Independent System Operators (ISOs) in terms of coordination, transparency, and the centralized co-optimization of energy and ancillary services, many joint markets rely on static, exogenously defined reserve requirements, such as those in PJM (PJM Interconnection, L.L.C., 2025) and CAISO (California ISO, 2019). The primary motivation for these static methods lies in their simplicity. However, such simpler models fail to reflect the endogenous relationship between reserve needs and the energy dispatch under different system conditions. Moreover, reserve deliverability during contingencies is often overlooked, leading to *ad hoc* solutions like zonal divisions with separate reserve requirements and prices (Shi et al., 2023).

Another persistent challenge lies in the cost allocation of reserves. Most ISOs allocate reserve costs proportionally among load-serving entities based on energy consumption, which raises concerns about fairness and efficiency (Badesa et al., 2025; Matamala et al., 2024; Shi et al., 2023). Proportional allocation often results in cross-subsidies, as it fails to link costs directly to the entities responsible for reserve requirements. As far as the authors are aware, among existing markets, the cost-causation principle for reserve cost allocation is solely applied by the Australian Energy Market Operator (AEMO) across its two independent systems: the Wholesale Electricity Market (WEM) and the National Electricity Market (NEM). However, while the WEM employs a sequential approach that better aligns costs with responsible entities, the NEM still relies on proportional allocation (Australian Energy Market Operator (AEMO), 2023), perpetuating inefficiencies.

In light of these challenges, numerous studies have proposed deterministic frameworks, such as those relying on the $N-1$ security standard, as well as probabilistic approaches to improve reserve pricing and scheduling (Ahunbay et al., 2024; Arroyo & Galiana, 2005; Badesa et al., 2023; Bouffard et al., 2005; Byers & Hug, 2023; Galiana et al., 2005; Karangelos & Bouffard, 2012; Martin & Fanzeres, 2023; Mays et al., 2021; Morales et al., 2012; O'Neill et al., 2005; Roald et al., 2023; Street et al., 2025; Wang et al., 2009; Wong & Fuller, 2007). These approaches derive energy and reserve prices from the dual variables of power balance constraints, ensuring that prices reflect the marginal costs of maintaining security. However, most frameworks assume ancillary services costs are fully borne by consumers. Additionally, existing models fail to incorporate transmission revenues into reserve pricing, i.e., prices are derived without properly considering the transmission capacity required to deliver reserves, potentially leading to settlement imbalance. Furthermore, the nonconvex operational characteristics of generating units introduce additional distortions, often resulting in payments in addition to those resulting from the marginal prices, and several studies have sought to mitigate these inefficiencies (Ahunbay et al., 2024; Byers & Hug, 2023; O'Neill et al., 2017, 2005).

Early discussions on reserve cost allocation focused on generator outages. Proportional cost allocation based on generating units' capacity and unavailability was proposed in Strbac and Kirschen (2000). In Kirby and Hirst (2003), this idea was expanded with mechanisms incorporating historical outages and dispatch size. However, using historical outages as a proxy for predicting future events is suboptimal due to their low probability of occurrence, leading to erratic charges and misalignment between prospective reserve procurement and retrospective cost allocation. Specifically, if no outages occur, reserve costs lack clear entities to charge (Badesa et al., 2023). More recent studies, such as Liu et al. (2023) and Xiang et al. (2023), emphasize renewables and consumption variabilities as factors influencing reserve costs, relying on historical data. However, these variabilities are not the dominant driver of reserve needs in low-inertia systems (Matamala et al., 2024), and historical data often fail to predict future contingencies, underscoring the need for adaptive, causation-based allocation methods.

In Matamala et al. (2024), the costs of ORs in low-inertia systems are shown to be primarily driven by the generators with the largest capacity. The study evaluates proportional, Shapley value, and nucleolus methods, identifying the nucleolus as the one that prevents cross-subsidies, thereby promoting cooperative behavior. Building on this finding, the cost-causation principle is advocated in Badesa et al. (2025) to ensure reserve costs are borne by those creating the need for ancillary services. Furthermore, in Badesa et al. (2025), the authors highlight the need for cost-reflective frameworks that incentivize responsible behavior and investment in future grids. However, to date, no study combines a unified pricing model for energy, reserves, and transmission with a causation-based cost allocation methodology, leaving an important gap in the literature.

Within the context of an ISO-based market design, this paper addresses these issues by introducing a unified framework for the pricing and cost allocation of energy, reserves, and transmission services, grounded in causation-based principles. More specifically, the objective of this work is to extend the findings of Arroyo and Galiana (2005) to incorporate the cost-causation principle. To the best of the authors' knowledge, Arroyo and Galiana (2005) can be regarded as a foundational benchmark in the literature, as it is the first work to endogenously derive nodal marginal prices for both energy and reserves within a contingency-constrained co-optimization framework. This formulation has since underpinned a substantial body of subsequent research (Bouffard et al., 2005; Galiana et al., 2005; Karangelos & Bouffard, 2012; Wong & Fuller, 2007) and has also motivated practical developments in ISO-based market implementations, particularly in addressing reserve deliverability and post-contingency constraints (Chen et al., 2014; Wang & Chen, 2021).

Under a contingency-constrained setting, system operation is explicitly characterized for the pre-contingency state and for every contingency within the prescribed security criterion. Therefore, security is ensured as power balance is met without violating any technical limits for all system states under consideration. The adoption of such a setting is justified by the fact that the cost of secure operation is significantly lower than the cost of large-scale blackouts. The use of a contingency-constrained approach is consistent with established $N-k$ security standards, such as those of the North American Electric Reliability Corporation (NERC), which require system operators to restore a secure state immediately after a contingency, reflecting the critical nature of power grid infrastructure (North American Electric Reliability Corporation (NERC), 2026). These principles are also reflected in the operational procedures routinely implemented by system operators worldwide (Li & Hedman, 2019), including those of California (California ISO, 2025) and New York (New York ISO, 2026). It should be noted that the choice of the security standard is ultimately a regulatory decision, which directly shapes the cost-causation principle embedded in the market design.

The proposed adoption of the cost-causation principle prevents settlement imbalance, which is a major flaw of existing pricing methods even in the absence of nonconvexities in the market clearing. Such imbalances are here precluded by tying security charges to generating units that tighten security constraints. The security charge derived in this work can be understood as a causation-based remuneration adjustment to reflect the marginal contribution of each generator's outage to the total reserve cost. As a result, the proposed approach gives rise to a pricing system that aligns payments from consumers, which are the beneficiaries of security, with generation and transmission revenues while ensuring cost recovery for market participants. Moreover, this correction provides a sound basis for extensions to nonconvex market-clearing formulations and the analysis of market power abuse. We therefore contribute to knowledge with the following concepts:

能源监管机构（ACER），2026）。相比之下，联合市场通常通过更好地捕捉这些产品之间的相互依赖性来实现更高的社会福利，正如多项研究所证明的那样（Aganagic 等，1998；Ahunbay 等，2024；Arroyo & Galiana，2005；Badesa 等，2023；Bouffard 等，2005；Byers & Hug，2023；Chen 等，2014；Galiana 等，2005；Gan & Litvinov，2003；Karangelos & Bouffard，2012；Kirschen & Strbac，2019；Morales 等，2012；Wang & Chen，2021；Wang 等，2009；Wong & Fuller，2007；Wu 等，2004）。不幸的是，尽管独立系统运营商（ISO）在协调、透明度以及能源和辅助服务的集中优化方面带来了显著改进，许多联合市场仍依赖静态的、外生定义的备用要求，如PJM（PJM互联公司，2025）和CAISO（加州ISO，2019）。这些静态方法的主要动机

另一个持续存在的挑战在于备用容量成本的分配。大多数独立系统运营商（ISOs）根据能源消耗在负荷服务实体之间按比例分配备用容量成本，这引发了关于公平性和效率的担忧（Badesa 等, 2025；Matamala 等, 2024；Shi 等, 2023）。按比例分配通常会导致交叉补贴，因为它未能将成本直接与负责备用容量需求的实体联系起来。据作者所知，在现有市场中，备用容量成本分配的成本归因原则仅由澳大利亚能源市场运营商（AEMO）在其两个独立系统——批发电力市场（WEM）和国家电力市场（NEM）中应用。然而，尽管 WEM 采用了一种更好地将成本与负责实体对齐的顺序方法，NEM 仍然依赖按比例分配（澳大利亚能源市场运营商（AEMO）, 2023），从而延续了低效率问题。

鉴于这些挑战，许多研究提出了确定性框架，例如依赖于$N-1$安全标准的框架，以及为了改善备用定价和调度的概率方法（Ahunbay 等，2024；Arroyo & Galiana，2005；Badesa 等，2023；Bouffard 等，2005；Byers & Hug，2023；Galiana 等，2005；Karangelos & Bouffard，2012；Martin & Fanzeres，2023；Mays 等，2021；Morales 等，2012；O’Neill 等，2005；Roald 等，2023；Street 等，2025；Wang 等，2009；Wong & Fuller，2007）。这些方法从电力平衡约束的对偶变量中得出能源和备用价格，以确保价格反映维持安全的边际成本。然而，大多数框架假设额外服务成本完全由消费者承担。此外，现有模型未能将输电收入纳入备用定价，即价格的推导未能正确考虑输电能力。

关于备用容量成本分配的早期讨论集中在发电机故障上。Strbac 和 Kirschen (2000) 提出了基于发电机容量和不可用性按比例分配成本的方案。在 Kirby 和 Hirst (2003) 中，这一想法被扩展，引入了结合历史故障和调度规模的机制。然而，由于历史故障发生的概率较低，将其用作预测未来事件的替代方法并不理想，这会导致费用波动大，并且前瞻性的备用容量采购与事后的成本分配不一致。具体而言，如果没有发生故障，备用成本就缺乏明确的收费主体（Badesa 等人，2023）。近期的研究，如 Liu 等人 (2023) 和 Xiang 等人 (2023)，强调可再生能源和消费波动性作为影响备用成本的因素，并依赖于历史数据。然而，这些变动性并不是低惯性系统中备用电力需求的主要驱动因素（Matamala 等，2024），而且历史数据通常无法预测未来的突发事件，这凸显了基于因果关系的自适应分配方法的必要性。

在Matamala等人（2024年）的研究中，显示低惯性系统中调度储备（ORs）的成本主要由容量最大的发电机驱动。该研究评估了比例分配、Shapley值和核方法，确定核方法能够防止交叉补贴，从而促进合作行为。基于这一发现，Badesa等人（2025年）提倡成本归因原则，以确保储备成本由那些引起辅助服务需求的主体承担。此外，在Badesa等人（2025年）的研究中，作者强调了建立反映成本的框架的必要性，以激励负责任的行为和对未来电网的投资。然而，迄今为止，还没有研究将能源、储备和输电的统一定价模型与基于归因的成本分配方法结合起来，这在文献中留下了一个重要的空白。

在基于ISO的市场设计背景下，本文通过引入一个统一的框架来解决这些问题，该框架针对能源、备用容量和输电服务的定价与成本分配，并以因果性原则为基础。更具体地说，这项工作的目标是将Arroyo和Galiana（2005年的研究）成果扩展，以纳入成本因果原则。据作者所知，Arroyo和Galiana（2005年）可被视为文献中的基础性基准，因为它是首个在受应急约束的联合优化框架下，从内生角度推导出能源和备用容量节点边际价格的工作。这一理论框架随后支持了大量后续研究（Bouffard等，2005；Galiana等，2005；Karangelos & Bouffard, 2012；Wong & Fuller, 2007），并且也推动了基于ISO市场实施的实际应用发展，尤其是在解决

在应急约束的设定下，系统运行被明确描述为预应急状态以及在规定的安全标准内每一个应急情景下的状态。因此，在考虑的所有系统状态下，满足功率平衡而不违反任何技术限制，从而确保了安全性。采用这种设定的理由在于，安全运行的成本显著低于大规模停电的成本。采用应急约束方法与既定的安全标准一致，例如北美电力可靠性公司（NERC）的标准，这些标准要求系统运营商在应急事件发生后立即恢复到安全状态，反映了电网基础设施的重要性（北美电力可靠性公司（NERC），2026）。这些原则也体现在系统运营商日常执行的操作程序中。

所提出的采用成本归因原则可以防止结算不平衡，这是现有定价方法的一个主要缺陷，即使在市场清算中不存在非凸性的情况下也是如此。通过将安全费用与加紧安全约束的发电单元挂钩，可以预防这种不平衡。本文所推导的安全费用可以被理解为一种基于因果的报酬调整，用于反映每个发电机故障对总储备成本的边际贡献。因此，所提出的方法产生了一种定价系统，将受益于安全的消费者支付与发电及输电收入相一致，同时确保市场参与者的成本回收。此外，这种修正为扩展至非凸市场清算模型和分析市场滥用提供了可靠的基础。因此，我们通过以下概念对知识做出了贡献：

1. A security charge that complements the prices of energy and reserves provided by the benchmark scheme proposed in Arroyo and Galiana (2005) so that reserve costs are efficiently allocated to the entities responsible for causing them,
2. the differentiation between up- and down-spinning reserve prices based on system's opportunity costs, and
3. a pricing system that properly remunerates transmission lines for the capacity set aside for contingencies that is used to ensure reserve deliverability through the network.

These newly proposed concepts rely on Lagrange multipliers, ensuring a transparent, fair, and efficient cost allocation under marginal pricing (Schweppe et al., 1988). Based on those concepts, we achieve a consistent settlement process that adheres to key axiomatic principles for energy and reserve contingency-constrained models. These principles include: (1) cost recovery, ensuring non-negative profit (surplus) for producers and consumers, and (2) system revenue adequacy, whereby total consumers' payment equals total generation and transmission revenue.

It is important to note that, while this study focuses on generation-driven cost causation, the same principles apply to demand-driven cost causation. Although frequency drops are the primary concern in most power systems, over-frequency events can be modeled analogously by considering sudden load disconnections instead of generation outages. In practice, such events are usually less critical than generator contingencies, but, with the advent of large data centers, load outages could become comparably significant.

The remainder of this paper is structured as follows. Section 2 develops the pricing framework for a simplified model without network constraints, considering only generator contingencies to place the focus on the concept of security charges and highlight their role in efficient market operations. In Section 3, the model is extended to incorporate network constraints, demand-side bidding, and both generator and line outages. This extension reinforces the necessity of security charges in more complex systems, examines their impact on the revenues of different market participants, including transmission assets, and introduces a novel approach for separately pricing up- and down-spinning reserves. Each section presents the corresponding optimization model for market clearing, introduces an example that is addressed using the benchmark pricing system, develops the proposed pricing framework, and provides and discusses the new results using the same example previously considered in the section. Section 4 reports our numerical experience with a larger case study based on the IEEE 300-bus system. Section 5 draws relevant conclusions and suggests directions for future research. Finally, the nomenclature is given in the Appendix.

2. Introducing the notion of security charges

This section introduces the concept of security charges under marginal pricing through a simplified contingency-constrained market-clearing model for energy and reserves. Briefly, the security charge concept is a cost allocation mechanism that reflects the different contributions of generators to the endogenously defined reserve needs, thereby addressing the limitations of existing uniform pricing in contingency-constrained models. Thus, higher charges are incurred by generators whose potential outages create higher reserve needs and costs, leading to a pricing framework that more accurately aligns costs with system's opportunity costs. This concept will be further developed in the following sections.

To that end, we first formulate the stylized optimization model used to introduce the security charge concept. Then, we revisit the existing uniform pricing framework in Arroyo and Galiana (2005), hereinafter referred to as the benchmark pricing scheme, and illustrate its limitations. Finally, we develop the notion of security charges based on the study of the Lagrangian dual function and how each term in this function should be attributed to each agent to ensure the cost-causation principle. The benefits of the proposed pricing system are illustrated through the assessment with the benchmark scheme.

2.1. Optimization model

For expository purposes, generation outages are considered using a contingency-constrained model (Wood et al., 2014), which can accommodate any instance of the practical deterministic $N-k$ security criterion (North American Electric Reliability Corporation (NERC), 2026). Moreover, a single-bus, single-period model is employed, considering an inelastic demand and linear offer cost functions, with a focus on spinning or synchronized reserves. For the sake of clarity in illustrating the cost-causation principle, this framework solely characterizes generation contingencies as full generator outages. Note, however, that the model could be extended to account for partial generator outages (derating). Such an extension is left for future work.

In this simplified framework, only one type of reserve, namely upward spinning reserves, is required to ensure system security. Thus, the market-clearing procedure is given by the following optimization problem, which is an instance of linear programming:

$$\min_{\boldsymbol{x}\geq\mathbf{0}} \quad \boldsymbol{c}^\top \boldsymbol{g}_0 + (\boldsymbol{q}^{g,up})^\top \boldsymbol{r}^{g,up} \tag{1}$$

subject to:

$$\mathbb{1}^\top \boldsymbol{g}_0 = d \quad (\pi_0) \tag{2}$$

$$\mathbb{1}^\top \boldsymbol{g}_k = d \quad (\pi_k), \qquad \forall k \in \mathcal{K} \tag{3}$$

$$\boldsymbol{g}_k \leq \mathbb{D}(\boldsymbol{a}_k^g)(\boldsymbol{g}_0 + \boldsymbol{r}^{g,up}), \quad \forall k \in \mathcal{K} \tag{4}$$

$$\boldsymbol{g}_0 + \boldsymbol{r}^{g,up} \leq \overline{\boldsymbol{G}} \tag{5}$$

$$\boldsymbol{r}^{g,up} \leq \overline{\boldsymbol{R}}^{g,up} \tag{6}$$

where $\boldsymbol{x} = \{\boldsymbol{g}_0, \boldsymbol{r}^{g,up}, \{\boldsymbol{g}_k\}_{k\in\mathcal{K}}\}$, $\mathbb{D}(\cdot)$ denotes the diagonal matrix operator, and $\mathbb{1}$ represents the all-ones vector. The pre-contingency state is denoted by subscript 0 while contingency states are indexed by $k \in \mathcal{K}$, with $\mathcal{K} = \{1, 2, \ldots, |\mathcal{K}|\}$.

The objective function to be minimized (1) consists of the sum of the costs for generating power and providing up-spinning reserves offered by the generators.

Constraints (2) and (3) ensure the power balance between generation and consumption under both pre-contingency and contingency states, respectively. Accordingly, the total output of all generators under every state must equal the load demand, which is assumed to be constant. Note that π_0 and π_k represent the associated Lagrange multipliers.

Constraint (4) relates the up-spinning reserve contributions to the power levels produced under the pre-contingency and contingency states through availability factors a_{ik}^g. Thus, for contingency states k where a given unit i is in service ($a_{ik}^g = 1$), the post-contingency output g_{ik} is bounded by the pre-contingency production plus the up-spinning reserve allocation. On the other hand, for contingency states k where the unit is out of service ($a_{ik}^g = 0$), the post-contingency production g_{ik} is forced to be zero. Lastly, constraints (5) and (6) set the operational limits of the generator outputs and reserves.

2.2. Challenges of the benchmark pricing framework

For contingency-constrained models such as problem (1)–(6), prices under a marginal pricing framework (Schweppe et al., 1988) can be computed using the benchmark scheme described in Arroyo and Galiana (2005), which is based on the use of Karush-Kuhn-Tucker (KKT) optimality conditions. Accordingly, the energy price is defined as the sum of the Lagrange multipliers corresponding to the power balance in both pre-contingency and contingency states. Moreover, reserves are priced through the so-called security price, which is defined as the sum of the Lagrange multipliers related to the power balance equations under contingency. As a result, for problem (1)–(6), generators collect revenues for their pre-contingency power outputs and scheduled up-spinning reserves using the aforementioned energy and security prices, respectively. Analogously, consumers are charged the energy price for

1. 一种安全费用，用以补充由Arroyo和Galiana（2005）提出的基准方案提供的能源和备用容量价格，从而将备用成本有效地分配给导致这些成本的实体，2. 基于系统机会成本的上旋备用和下旋备用价格的区分，3. 一种定价系统，适当补偿用于应急预留容量的输电线路，以确保通过网络提供备用能力。

这些新提出的概念依赖于拉格朗日乘子，从而确保在边际定价下实现透明、公平且高效的成本分配（Schweppe 等，1988）。基于这些概念，我们实现了一种一致的结算流程，遵循能源和备用应急约束模型的关键公理原则。这些原则包括：(1) 成本回收，确保生产者和消费者的利润（盈余）非负；(2) 系统收入充足，即消费者总支付等于总发电和传输收入。

需要注意的是，虽然本研究关注的是由发电引起的成本原因，但相同的原理也适用于需求驱动的成本原因。尽管在大多数电力系统中，频率下降是主要问题，但过频事件可以通过考虑负荷突然断开而不是发电机故障进行类似建模。在实践中，这类事件通常不如发电机突发事件关键，但随着大型数据中心的出现，负荷中断可能变得同样重要。

本文的其余部分结构如下。第2节为一个简化模型建立定价框架，该模型不考虑网络约束，仅考虑发电机的应急情况，以便将重点放在安全费用的概念上并突出其在高效市场运作中的作用。在第3节中，该模型扩展以纳入网络约束、需求方竞价以及发电机和线路的故障。这一扩展强调了在更复杂系统中安全费用的必要性，考察其对不同市场参与者（包括输电资产）收入的影响，并引入了一种分别定价上旋和下旋备用容量的新方法。每一节都展示了用于市场清算的相应优化模型，介绍了使用基准定价系统处理的示例，发展了所提出的定价框架，并使用之前考虑的相同示例提供并讨论了新的结果。

2. 介绍安全费用的概念

本节介绍了在边际定价下通过简化的受应急约束的能源和备用市场清算模型引入安全费用的概念。简言之，安全费用的概念是一种成本分配机制，反映了发电机对内生定义的备用需求的不同贡献，从而解决了现有受应急约束模型中统一定价的局限性。因此，那些潜在停机会产生更高备用需求和成本的发电机会承担更高的费用，从而形成一种定价框架，更准确地将成本与系统的机会成本对齐。这个概念将在后续章节中进一步展开。

为此，我们首先制定了用于引入安全费用概念的程式化优化模型。然后，我们重新审视了Arroyo和Galiana（2005）中现有的统一定价框架，在本文中将其称为基准定价方案，并说明其局限性。最后，我们基于对拉格朗日对偶函数的研究，发展了安全费用的概念，并说明如何将该函数中的每一项归因于每个代理，以确保成本归因原则。通过与基准方案的比较评估，说明了所提出定价系统的优势。

2.1. Optimization model

出于说明目的，发电故障被视为使用受应急约束的模型（Wood 等，2014），该模型可以适应实际确定性 $N-k$ 安全标准的任何实例（北美电力可靠性公司（NERC），2026）。此外，采用单母线、单时间段模型，考虑刚性需求和线性报价成本函数，重点关注旋转备用或同步备用。为了清楚地说明成本归因原则，该框架仅将发电应急情况描述为发电机完全停机。然而，需要注意的是，该模型可以扩展以考虑发电机部分停机（降额）。这种扩展留作未来工作。

在这个简化的框架中，系统安全只需要一种类型的备用，即向上旋转备用。因此，市场清算过程由以下优化问题给出，这是一种线性规划实例：

$$\min_{\boldsymbol{x}\geq \mathbf{0}} \quad \boldsymbol{c}^{\top}\boldsymbol{g}_0 + (\boldsymbol{q}^{g,up})^{\top}\boldsymbol{r}^{g,up} \tag{1}$$

subject to:

$$\mathbb{1}^{\top}\boldsymbol{g}_0 = d \quad (\pi_0) \tag{2}$$

$$\mathbb{1}^{\top}\boldsymbol{g}_k = d \quad (\pi_k), \qquad \forall k \in \mathcal{K} \tag{3}$$

$$\boldsymbol{g}_k \leq \mathbb{D}(\boldsymbol{a}_k^g)(\boldsymbol{g}_0 + \boldsymbol{r}^{g,up}), \quad \forall k \in \mathcal{K} \tag{4}$$

$$\boldsymbol{g}_0 + \boldsymbol{r}^{g,up} \leq \overline{\boldsymbol{G}} \tag{5}$$

$$\boldsymbol{r}^{g,up} \leq \overline{\boldsymbol{R}}^{g,up} \tag{6}$$

其中 $\boldsymbol{x} = \{\boldsymbol{g}_0, \boldsymbol{r}^{g,up}, \{\boldsymbol{g}_k\}_{k\in\mathcal{K}}\}$，$\mathbb{D}(\cdot)$ 表示对角矩阵算子，1 表示全 1 向量。预应急状态用下标 0 表示，而应急状态由 $k \in \mathcal{K}$ 索引，$\mathcal{K} = \{1, 2, \ldots, |\mathcal{K}|\}$。

要最小化的目标函数 (1) 包括发电成本和发电机提供的上旋备用容量成本的总和。

约束条件 (2) 和 (3) 分别确保在预应急状态和应急状态下发电与消费之间的功率平衡。因此，在每种状态下，所有发电机的总输出必须等于负荷需求，假设其为常数。注意，π_0 和 π_k 表示相应的拉格朗日乘子。

约束条件(4)通过可用性因子 a_{ik}^g 将上旋备用贡献与突发前和突发状态下的发电水平联系起来。因此，对于某些单元 i 在运行状态 ($a_{ik}^g = = 1$) 的突发状态 k，突发后的发电输出 g_{ik} 受限于突发前发电量加上上旋备用分配。另一方面，对于单元停运 ($a_{ik}^g = = 0$) 的突发状态 k，突发后的发电输出 g_{ik} 被强制为零。最后，约束条件(5)和(6)设定了发电机输出和备用的运行限制。

2.2. Challenges of the benchmark pricing framework

对于像问题 (1)–(6) 这样的应急约束模型，在边际定价框架下的价格（Schweppe 等，1988）可以使用 Arroyo 和 Galiana（2005）描述的基准方案来计算，该方案基于使用 Karush-Kuhn-Tucker (KKT) 最优性条件。因此，能源价格定义为对应于应急前和应急状态下功率平衡的拉格朗日乘子的总和。此外，储备通过所谓的安全价格定价，安全价格被定义为应急情况下与功率平衡方程相关的拉格朗日乘子总和。因此，对于问题 (1)–(6)，发电机分别使用上述能源价格和安全价格收取其应急前功率输出和计划的增量调峰储备的收入。类比地，消费者按照能源价格支付

Table 1
Prices and settlement for problem (1)–(6) according to the benchmark pricing system.

Energy Price	$p^e = \pi_0^* + \sum_{k\in\mathcal{K}} \pi_k^*$
Security Price	$p^s = \sum_{k\in\mathcal{K}} \pi_k^*$
Generation Energy Revenue	$R_i^{g,e} = p^e g_{i0}$
Generation Up-Spinning Reserve Revenue	$R_i^{g,up} = p^s r_i^{g,up}$
Generation Total Revenue	$R_i^{g,t} = R_i^{g,e} + R_i^{g,up}$
Consumer Payment	$CP = p^e d$
Generation Energy Cost	$C_i^{g,e} = c_i g_{i0}$
Generation Up-Spinning Reserve Cost	$C_i^{g,up} = q_i^{g,up} r_i^{g,up}$
Generation Total Cost	$C_i^{g,t} = C_i^{g,e} + C_i^{g,up}$
Generation Profit	$Profit_i^g = R_i^{g,t} - C_i^{g,t}$

Table 2
Single-bus example – Generation data.

i	$\overline{G}_i$ (MW)	$\overline{R}_i^{g,up}$ (MW)	c_i ($/MWh)	$q_i^{g,up}$ ($/MW/h)
1	100	50	20	2
2	60	30	50	5
3	70	35	100	10

Table 3
Single-bus example – Optimal results (MW).

Generator	Pre-Contingency Generation	Up-Spinning Reserve	Generation under Contingency		
			Outage of Generator 1	Outage of Generator 2	Outage of Generator 3
1	65	0	0	65	65
2	30	30	60	0	55
3	25	35	60	55	0
Total	120	65	120	120	120

Table 4
Single-bus example – Lagrange multipliers ($/MWh).

Pre-Contingency State	Outage of Generator 1	Outage of Generator 2	Outage of Generator 3
20	80	0	0

their consumption. Table 1 summarizes the prices and settlement for the market-clearing model (1)–(6) according to the benchmark pricing scheme (Arroyo & Galiana, 2005).

We now apply this benchmark pricing framework to an illustrative example involving three generators. Generation data are provided in Table 2. The load is 120 MW. Three contingencies are considered here, each defined by the outage of each generator. Using the simplex algorithm of CPLEX, problem (1)–(6) has been solved to optimality. Table 3 presents the optimal solution, which features a total cost equal to $5,800. As can be seen, the optimization prioritizes energy and reserve offers in reverse order, scheduling the least-cost generator (generator 1) exclusively to supply power in the pre-contingency state. In contrast, generators 2 and 3 are used for both power provision in the pre-contingency state and up-spinning reserve. Note that the up-spinning reserve contributions of generators 2 and 3, which are respectively equal to their corresponding upper bounds, amount to the pre-contingency power output of generator 1. It is also worth mentioning that the up-spinning reserve contributions of generators 2 and 3 exceed the reserve needed to guard against the loss of generators 3 and 2, respectively. In other words, the outage of generator 1 is the critical contingency state.

The Lagrange multipliers associated with the power balance equations in the pre-contingency and contingency states are listed in Table 4. For the outages of generators 2 and 3, the associated multipliers are zero, as the loss of these generators does not constrain the system. In contrast, the multiplier for the outage of generator 1 is different from zero as an infinitesimal perturbation of the corresponding power balance equation would yield a different solution, thereby resulting in a change in the value of the objective function.

Table 5
Single-bus example – Prices according to the benchmark pricing system ($/MWh).

Energy Price, p^e	Security Price, p^s
100	80

Table 6
Single-bus example – Generation settlement according to the benchmark pricing system ($).

i	$R_i^{g,e}$	$R_i^{g,up}$	$R_i^{g,t}$	$C_i^{g,e}$	$C_i^{g,up}$	$C_i^{g,t}$	$Profit_i^g$
1	6,500	0	6,500	1,300	0	1,300	5,200
2	3,000	2,400	5,400	1,500	150	1,650	3,750
3	2,500	2,800	5,300	2,500	350	2,850	2,450
Total	**12,000**	**5,200**	**17,200**	**5,300**	**500**	**5,800**	**11,400**

Table 7
Single-bus example – System settlement according to the benchmark pricing system ($).

Generation Revenue	17,200
Consumer Payment	12,000
Balance	–5,200

Tables 5 and 6 respectively report the prices and settlement for the generators as per the definitions of Table 1. Also, according to Table 1, the consumer payment is given by $p^e d = 100 \times 120 = \$12,000$. Interestingly, the benchmark pricing scheme yields a total generation revenue exceeding the total consumer payment by $5,200, a distortion generated by the pricing system, as highlighted in Table 7. This mismatch is addressed by the novel approach described in the next section.

2.3. Proposed pricing system

As an alternative to the benchmark scheme, the proposed pricing system acknowledges the contribution of each generator to keep the power balance under each contingency state. To that end, we explore the structure of the Lagrangian dual (LD) problem to ensure a market with system revenue adequacy where the total generation revenue is equal to the consumer payment. When strong duality holds, which is guaranteed for feasible linear programs, the LD problem yields the optimal solution to the original problem. Moreover, the LD problem can be interpreted as a price-based coordination mechanism among market agents, where Lagrange multipliers play the role of prices (Conejo et al., 1999). For problem (1)–(6), the LD function is built as follows (Conejo et al., 1999):

$$\phi(\boldsymbol{\Pi}) = \min_{\boldsymbol{x}\in\mathcal{X}} \; \boldsymbol{c}^\top \boldsymbol{g}_0 + \left(\boldsymbol{q}^{g,up}\right)^\top \boldsymbol{r}^{g,up} + \pi_0 \left(d - \mathbb{1}^\top \boldsymbol{g}_0\right) + \sum_{k\in\mathcal{K}} \pi_k \left(d - \mathbb{1}^\top \boldsymbol{g}_k\right) \tag{7}$$

where $\boldsymbol{\Pi} = \{\pi_0, \{\pi_k\}_{k\in\mathcal{K}}\}$ and $\mathcal{X}$ includes constraints (4)–(6) and $\boldsymbol{x} \geq \boldsymbol{0}$.

Taking the terms that do not depend on $\boldsymbol{x}$ out of the minimization problem and factoring out yields:

$$\phi(\boldsymbol{\Pi}) = \left(\pi_0 + \sum_{k\in\mathcal{K}} \pi_k\right) d + \min_{\boldsymbol{x}\in\mathcal{X}} \left\{ \left(\boldsymbol{c}^\top - \pi_0 \mathbb{1}^\top\right) \boldsymbol{g}_0 + \left(\boldsymbol{q}^{g,up}\right)^\top \boldsymbol{r}^{g,up} - \sum_{k\in\mathcal{K}} \pi_k \mathbb{1}^\top \boldsymbol{g}_k \right\} \tag{8}$$

表 1 根据基准定价系统的问题 (1)–(6) 的价格和结算。

Energy Price	$p^e = \pi_0^* + \sum_{k\in\mathcal{K}} \pi_k^*$
Security Price	$p^s = \sum_{k\in\mathcal{K}} \pi_k^*$
Generation Energy Revenue	$R_i^{g,e} = p^e g_{i0}$
Generation Up-Spinning Reserve Revenue	$R_i^{g,up} = p^s r_i^{g,up}$
Generation Total Revenue	$R_i^{g,t} = R_i^{g,e} + R_i^{g,up}$
Consumer Payment	$CP = p^e d$
Generation Energy Cost	$C_i^{g,e} = c_i g_{i0}$
Generation Up-Spinning Reserve Cost	$C_i^{g,up} = q_i^{g,up} r_i^{g,up}$
Generation Total Cost	$C_i^{g,t} = C_i^{g,e} + C_i^{g,up}$
Generation Profit	$Profit_i^g = R_i^{g,t} - C_i^{g,t}$

表2
单母线示例——发电数据。

i	$\overline{G}_i$ (MW)	$\overline{R}_i^{g,up}$ (MW)	c_i (\$/MWh)	$q_i^{g,up}$ (\$/MW/h)
1	100	50	20	2
2	60	30	50	5
3	70	35	100	10

表3 单母线示例 – 最优结果（兆瓦）

Generator	Pre-Contingency Generation	Up-Spinning Reserve	Generation under Contingency		
			Outage of Generator 1	Outage of Generator 2	Outage of Generator 3
1	65	0	0	65	65
2	30	30	60	0	55
3	25	35	60	55	0
Total	120	65	120	120	120

表 4 单母线示例 – 拉格朗日乘子（\$/MWh）。

Pre-Contingency State	Outage of Generator 1	Outage of Generator 2	Outage of Generator 3
20	80	0	0

他们的消费。表 1 汇总了根据基准定价方案（Arroyo & Galiana, 2005）对市场清算模型 (1)–(6) 的价格和结算情况。

我们现在将这个基准定价框架应用于一个涉及三台发电机的示例。发电数据见表2。负载为120兆瓦。这里考虑三种应急情况，每种情况由每台发电机的停运定义。使用CPLEX的单纯形算法，问题(1)–(6)已经求得最优解。表3展示了最优解，其总成本为5800美元。可以看出，优化优先安排能源和备用出价的顺序相反，仅安排成本最低的发电机（发电机1）在应急前状态供电。相比之下，发电机2和3不仅在应急前状态提供电力，还提供上旋备用。需要注意的是，发电机2和3的上旋备用贡献，分别等于其对应的上限，总量相当于发电机1的应急前电力输出。同样值得一提的是

预事件状态和应急状态下与功率平衡方程相关的拉格朗日乘子列在表4中。对于发电机2和3的停机，相关的乘子为零，因为这些发电机的损失不构成约束。

表5 单母线示例 – 根据基准定价系统的价格（\$/MWh）。

Energy Price, p^e	Security Price, p^s
100	80

表6 单母线示例 – 根据基准定价系统的发电结算（美元）。

i	$R_i^{g,e}$	$R_i^{g,up}$	$R_i^{g,t}$	$C_i^{g,e}$	$C_i^{g,up}$	$C_i^{g,t}$	$Profit_i^g$
1	6,500	0	6,500	1,300	0	1,300	5,200
2	3,000	2,400	5,400	1,500	150	1,650	3,750
3	2,500	2,800	5,300	2,500	350	2,850	2,450
Total	**12,000**	**5,200**	**17,200**	**5,300**	**500**	**5,800**	**11,400**

表7 单母线示例 – 根据基准定价系统的系统结算（美元）。

Generation Revenue	17,200
Consumer Payment	12,000
Balance	–5,200

系统。相比之下，发电机1停运的乘数不同于零，因为相应功率平衡方程的无穷小扰动会产生不同的解，从而导致目标函数值的变化。

表 5 和表 6 分别根据表 1 的定义报告了发电机的价格和结算。此外，根据表 1，消费者支付由 $p^e d = 100 \times 120 = \$12,000$ 给出。有趣的是，基准定价方案产生的总发电收入比消费者总支付多 5,200 美元，这一扭曲由定价系统产生，如表 7 所示。下节所述的新方法解决了这一不匹配问题。

2.3. Proposed pricing system

作为基准方案的替代方案，所提出的定价系统承认每个发电机在每种应急状态下保持电力平衡所作的贡献。为此，我们探讨拉格朗日对偶（LD）问题的结构，以确保一个系统收入充足的市场，其中总发电收入等于消费者支付。当强对偶成立时，这对于可行的线性规划是有保障的，LD问题将产生原始问题的最优解。此外，LD问题可以被解释为市场主体之间基于价格的协调机制，其中拉格朗日乘子起到价格的作用（Conejo 等，1999）。对于问题 (1)–(6)，LD函数构建如下（Conejo 等，1999）：

$$\phi(\boldsymbol{\Pi}) = \min_{\boldsymbol{x}\in\mathcal{X}} \; \boldsymbol{c}^\top \boldsymbol{g}_0 + \left(\boldsymbol{q}^{\boldsymbol{g},\boldsymbol{up}}\right)^\top \boldsymbol{r}^{\boldsymbol{g},\boldsymbol{up}} + \pi_0 \left(d - \mathbb{1}^\top \boldsymbol{g}_0\right) + \sum_{k\in\mathcal{K}} \pi_k \left(d - \mathbb{1}^\top \boldsymbol{g}_{\boldsymbol{k}}\right) \tag{7}$$

其中 $\boldsymbol{\Pi} = \{\pi_0, \{\pi_k\}_{k\in\mathcal{K}}\}$ 和 $\mathcal{X}$ 包括约束条件 (4)–(6) 和 $\boldsymbol{x} \geq \boldsymbol{0}$。将不依赖于 $\boldsymbol{x}$ 的项从最小化问题中提取出来并提取因子得到：

$$\phi(\boldsymbol{\Pi}) = \left(\pi_0 + \sum_{k\in\mathcal{K}} \pi_k\right) d + \min_{\boldsymbol{x}\in\mathcal{X}} \left\{ \left(\boldsymbol{c}^\top - \pi_0 \mathbb{1}^\top\right) \boldsymbol{g}_0 + \left(\boldsymbol{q}^{\boldsymbol{g},\boldsymbol{up}}\right)^\top \boldsymbol{r}^{\boldsymbol{g},\boldsymbol{up}} - \sum_{k\in\mathcal{K}} \pi_k \mathbb{1}^\top \boldsymbol{g}_{\boldsymbol{k}} \right\} \tag{8}$$

The minimization problem in (8) is generator-wise separable, as no coupling constraints are present, i.e., there are no system-wide constraints that link actions across all generators. This separability reveals the incentive compatibility of the proposed pricing framework, as each generator maximizes its individual profit at the optimal solution.

Besides, by factoring out the negative sign in (8), we can transform the *min* operator into a *max* operator. Thus, rewriting Eq. (8) accordingly, we have Eq. (9) as follows:

$$\phi(\boldsymbol{\Pi}) = \left(\pi_0 + \sum_{k\in\mathcal{K}} \pi_k\right) d - \sum_{i\in\mathcal{I}} \max_{\boldsymbol{x}_i\in\mathcal{X}_i} \left[\left(\pi_0 - c_i\right) g_{i0} - q_i^{g,up} r_i^{g,up} + \sum_{k\in\mathcal{K}} \psi_{ik}^{g}(\pi_k, \boldsymbol{x}_i) \right] \tag{9}$$

where $\boldsymbol{x}_i = \{g_{i0}, r_i^{g,up}\}$, $\mathcal{X}_i$ is the subset of constraints related to $\boldsymbol{x}_i$, i.e., constraints (5), (6), and $\boldsymbol{x}_i \geq \mathbf{0}$, whereas $\psi_{ik}^{g}(\pi_k, \boldsymbol{x}_i)$ stands for the revenue fraction of generator i under contingency k due to its best response against post-contingency Lagrange multiplier π_k, given its pre-contingency dispatch and reserve schedule, $\boldsymbol{x}_i$, and availability status at state k. Note that $\psi_{ik}^{g}(\pi_k, \boldsymbol{x}_i)$ can be cast as:

$$\psi_{ik}^{g}(\pi_k, \boldsymbol{x}_i) = \max_{g_{ik}\geq 0} \quad \pi_k g_{ik} \tag{10}$$

subject to:

$$g_{ik} \leq a_{ik}^{g} \left(g_{i0} + r_i^{g,up}\right) \tag{11}$$

Based on KKT conditions, π_k can only take non-negative values at the optimal solution to (1)–(6). As a consequence, we have two possible outputs for $\psi_{ik}^{g}(\pi_k, \boldsymbol{x}_i)$: (1) If π_k is positive, g_{ik}^{*} will take its maximum possible value, given by $g_{ik}^{*} = a_{ik}^{g}\left(g_{i0} + r_i^{g,up}\right)$, and $\psi_{ik}^{g}(\pi_k, \boldsymbol{x}_i)$ will be equal to $\pi_k a_{ik}^{g}\left(g_{i0} + r_i^{g,up}\right)$; (2) If π_k is zero, $\psi_{ik}^{g}(\pi_k, \boldsymbol{x}_i)$ will be 0. Thus, $\psi_{ik}^{g}(\pi_k, \boldsymbol{x}_i)$ will only take values different from zero when π_k is positive. Hence, $\psi_{ik}^{g}(\pi_k, \boldsymbol{x}_i)$ can be equivalently replaced with $\pi_k a_{ik}^{g}\left(g_{i0} + r_i^{g,up}\right)$. Rewriting (9) accordingly, we have Eq. (12) as follows:

$$\phi(\boldsymbol{\Pi}) = \left(\pi_0 + \sum_{k\in\mathcal{K}} \pi_k\right) d - \sum_{i\in\mathcal{I}} \max_{\boldsymbol{x}_i\in\mathcal{X}_i} \left[\left(\pi_0 - c_i\right) g_{i0} - q_i^{g,up} r_i^{g,up} + \sum_{k\in\mathcal{K}} \pi_k a_{ik}^{g}\left(g_{i0} + r_i^{g,up}\right) \right] \tag{12}$$

By rearranging Eq. (12), we obtain Eq. (13):

$$\phi(\boldsymbol{\Pi}) = \left(\pi_0 + \sum_{k\in\mathcal{K}} \pi_k\right) d - \sum_{i\in\mathcal{I}} \max_{\boldsymbol{x}_i\in\mathcal{X}_i} \left[\left(\pi_0 + \sum_{k\in\mathcal{K}} \pi_k a_{ik}^{g} - c_i\right) g_{i0} + \left(\sum_{k\in\mathcal{K}} \pi_k a_{ik}^{g} - q_i^{g,up}\right) r_i^{g,up} \right] \tag{13}$$

The first term in the right-hand side of (13) represents the consumer payment. The terms in square brackets represent the profit of generator i, which is made up of the net revenue, i.e., revenue minus cost, from both selling energy and providing up-spinning reserve. In conventional models, all generators are paid uniform prices for energy and reserves. However, when contingencies are explicitly modeled, cost causation also becomes explicit in the final optimal reserve allocation cost. In this case, generators receive no remuneration in the contingency states where they fail to be available to contribute to energy and reserves. This is reflected in the second and third terms of expression (13), where the binary availability parameter a_{ik}^{g} nullifies the contribution of the Lagrange multipliers in the contingency states where the generator is not available. Nevertheless, uniform pricing does not account for availability parameters. Therefore, rewriting (13) accordingly, we have expression (14):

$$\phi(\boldsymbol{\Pi}) = \left(\pi_0 + \sum_{k\in\mathcal{K}} \pi_k\right) d - \sum_{i\in\mathcal{I}} \max_{\boldsymbol{x}_i\in\mathcal{X}_i} \left[\left(\pi_0 + \sum_{k\in\mathcal{K}} \pi_k - c_i\right) g_{i0} + \left(\sum_{k\in\mathcal{K}} \pi_k - q_i^{g,up}\right) r_i^{g,up} - \sum_{k\in\mathcal{K}_i^{OFF}} \pi_k \left(g_{i0} + r_i^{g,up}\right) \right] \tag{14}$$

Table 8
Proposed pricing system and settlement for problem (1)–(6).

Energy Price	$p^{e} = \pi_0^{*} + \sum_{k\in\mathcal{K}} \pi_k^{*}$
Up-Spinning Reserve Price	$p^{up} = \sum_{k\in\mathcal{K}} \pi_k^{*}$
Generation Energy Revenue	$R_i^{g,e} = p^{e} g_{i0}$
Generation Up-Spinning Reserve Revenue	$R_i^{g,up} = p^{up} r_i^{g,up}$
Generation Security Charge	$C_i^{g,s} = \sum_{k\in\mathcal{K}_i^{OFF}} \pi_k^{*}\left(g_{i0} + r_i^{g,up}\right)$
Generation Total Revenue	$R_i^{g,t} = R_i^{g,e} + R_i^{g,up} - C_i^{g,s}$
Consumer Payment	$CP = p^{e} d$
Generation Energy Cost	$C_i^{g,e} = c_i g_{i0}$
Generation Up-Spinning Reserve Cost	$C_i^{g,up} = q_i^{g,up} r_i^{g,up}$
Generation Total Cost	$C_i^{g,t} = C_i^{g,e} + C_i^{g,up}$
Generation Profit	$Profit_i^{g} = R_i^{g,t} - C_i^{g,t}$

where the set $\mathcal{K}_i^{OFF}$ explicitly identifies the contingency states in which generator i is out of service. The first term within the square brackets represents the net revenue of generator i from selling energy. Note that, at the optimal solution, energy is priced at $\pi_0^{*} + \sum_{k\in\mathcal{K}} \pi_k^{*}$, which is identical to the price charged to consumers (first term in the right-hand side of (14)). Moreover, this price is identical to that derived in the benchmark pricing system (Table 1). The second term within the square brackets represents the net revenue from up-spinning reserve, where such a commodity is priced at $\sum_{k\in\mathcal{K}} \pi_k^{*}$, which is consistent with the security price defined in the benchmark pricing system (Table 1). The last term in (14) is here coined as "security charge" and can be viewed as what generator i should pay back for being responsible for system required reserves, which is aligned with the cost-causation principle. In other words, this term can be interpreted as the payback of the portion of the security revenue not secured by this generator, i.e., the part involving Lagrange multipliers associated with an availability parameter equal to 0. Importantly, while energy and reserve prices are uniform system-wide, the security charge is generator-specific.

Therefore, different from the settlement associated with the benchmark pricing system, the total revenue for each generator consists of three components: the energy revenue (energy price times pre-contingency power output), the up-spinning reserve revenue (up-spinning reserve price times up-spinning reserve), and the security charge (sum of the corresponding post-contingency Lagrange multipliers times the sum of pre-contingency power output and up-spinning reserve). Table 8 shows the proposed pricing scheme and the corresponding settlement.

The proposed pricing system has been applied to the illustrative example described in Section 2.2. Using the results provided in Tables 3 and 4 in the expressions listed in Table 8 gives rise to the results reported in Table 9, which lists the breakdown of revenues, costs, charges, and profits for the three generators. Compared to the results reported in Table 6, the revenues from energy and reserves and the associated costs are identical. However, as a major salient result, generator 1 is subject to a security charge under the proposed pricing scheme. Note that this generator is the primary driver of the reserve requirement, as indicated by the non-zero Lagrange multiplier in Table 4. This charge accounts for the increased system costs associated with the need for up-spinning reserves. Specifically, generator 1's failure requires 65 MW of up-spinning reserve. As a result, under the proposed method, the total generation revenue is $12,000, which exactly matches the consumer payment (Table 10), thereby overcoming the issue featured by the benchmark approach. As can be seen in Table 9, the consideration of the security charge reduces the profit for generator 1 compared to that reported in Table 6. Note, however, that the new resulting profit is non-negative.

The following theorems are stated for the proposed pricing framework, with their respective proofs provided subsequently.

公式（8）中的最小化问题是按发电机分开的，因为不存在耦合约束，即没有将所有发电机的动作联系起来的系统范围约束。这种可分性揭示了所提出定价框架的激励兼容性，因为每个发电机在最优解时都最大化其个人利润。

此外，通过在 (8) 中提取负号，我们可以将 *min* 算子变换为 *max* 算子。因此，相应地重写方程 (8)，我们得到方程 (9) 如下：

$$\phi(\boldsymbol{\Pi}) = \left(\pi_0 + \sum_{k\in\mathcal{K}} \pi_k\right) d - \sum_{i\in\mathcal{I}} \max_{\boldsymbol{x}_i\in\mathcal{X}_i} \left[\left(\pi_0 - c_i\right) g_{i0} - q_i^{g,up} r_i^{g,up} + \sum_{k\in\mathcal{K}} \psi_{ik}^{g}(\pi_k, \boldsymbol{x}_i) \right] \tag{9}$$

其中 $\boldsymbol{x}_i = \{g_{i0}, r_i^{g,up}\}$，$\mathcal{X}_i$ 是与 $\boldsymbol{x}_i$ 相关的约束子集，即约束 (5)、(6) 和 $\boldsymbol{x}_i \geq \mathbf{0}$，而 $\psi_{ik}^{g}(\pi_k, \boldsymbol{x}_i)$ 表示发电机 i 在应急 k 下的收入比例，这是由于其在后应急拉格朗日乘子 π_k 下的最佳响应，给定其前应急调度和备用计划 $\boldsymbol{x}_i$ 以及在状态 k 下的可用性。注意 $\psi_{ik}^{g}(\pi_k, \boldsymbol{x}_i)$ 可以表示为：

$$\psi_{ik}^{g}(\pi_k, \boldsymbol{x}_i) = \max_{g_{ik}\geq 0} \quad \pi_k g_{ik} \tag{10}$$

受以下条件约束：

$$g_{ik} \leq a_{ik}^{g} \left(g_{i0} + r_i^{g,up}\right) \tag{11}$$

根据KKT条件，π_k在(1)–(6)的最优解中只能取非负值。因此，我们有两个可能的$\psi_{ik}^{g}(\pi_k, \boldsymbol{x}_i)$输出：(1) 如果$\pi_k$为正，$g_{ik}^{*}$将取其可能的最大值，即$g_{ik}^{*} = a_{ik}^{g}\left(g_{i0} + r_i^{g,up}\right)$，并且$\psi_{ik}^{g}(\pi_k, \boldsymbol{x}_i)$将等于$\pi_k a_{ik}^{g}\left(g_{i0} + r_i^{g,up}\right)$；(2) 如果$\pi_k$为零，$\psi_{ik}^{g}(\pi_k, \boldsymbol{x}_i)$将为0。因此，$\psi_{ik}^{g}(\pi_k, \boldsymbol{x}_i)$只有在$\pi_k$为正时才会取非零值。因此，$\psi_{ik}^{g}(\pi_k, \boldsymbol{x}_i)$可以等效地替换为$\pi_k a_{ik}^{g}\left(g_{i0} + r_i^{g,up}\right)$。相应地重写 (9)，我们得到如下的方程(12)：

$$\phi(\boldsymbol{\Pi}) = \left(\pi_0 + \sum_{k\in\mathcal{K}} \pi_k\right) d - \sum_{i\in\mathcal{I}} \max_{\boldsymbol{x}_i\in\mathcal{X}_i} \left[\left(\pi_0 - c_i\right) g_{i0} - q_i^{g,up} r_i^{g,up} + \sum_{k\in\mathcal{K}} \pi_k a_{ik}^{g}\left(g_{i0} + r_i^{g,up}\right) \right] \tag{12}$$

通过重新排列方程 (12)，我们得到方程 (13)：

$$\phi(\boldsymbol{\Pi}) = \left(\pi_0 + \sum_{k\in\mathcal{K}} \pi_k\right) d - \sum_{i\in\mathcal{I}} \max_{\boldsymbol{x}_i\in\mathcal{X}_i} \left[\left(\pi_0 + \sum_{k\in\mathcal{K}} \pi_k a_{ik}^{g} - c_i\right) g_{i0} + \left(\sum_{k\in\mathcal{K}} \pi_k a_{ik}^{g} - q_i^{g,up}\right) r_i^{g,up} \right] \tag{13}$$

(13) 右侧的第一项代表消费者支付。方括号内的各项代表发电机 i 的利润，该利润由净收入（即销售能源和提供上旋储备的收入减去成本）组成。在传统模型中，所有发电机的能源和储备支付价格都是统一的。然而，当明确建模意外情况时，成本因果关系在最终的最优储备分配成本中也会变得明确。在这种情况下，当发电机无法参与能源和储备贡献时，它们在意外状态下不会获得报酬。这在公式 (13) 的第二项和第三项中有所体现，其中二进制可用性参数 a_{ik}^{g} 在发电机不可用的意外状态中将拉格朗日乘子的贡献置零。然而，统一定价并未考虑可用性参数。因此，相应地重写 (13)

$$\phi(\boldsymbol{\Pi}) = \left(\pi_0 + \sum_{k\in\mathcal{K}} \pi_k\right) d - \sum_{i\in\mathcal{I}} \max_{\boldsymbol{x}_i\in\mathcal{X}_i} \left[\left(\pi_0 + \sum_{k\in\mathcal{K}} \pi_k - c_i\right) g_{i0} + \left(\sum_{k\in\mathcal{K}} \pi_k - q_i^{g,up}\right) r_i^{g,up} - \sum_{k\in\mathcal{K}_i^{OFF}} \pi_k \left(g_{i0} + r_i^{g,up}\right) \right] \tag{14}$$

表 8 问题 (1)–(6) 的拟议定价系统及结算。

Energy Price	$p^e = \pi_0^* + \sum_{k\in\mathcal{K}} \pi_k^*$
Up-Spinning Reserve Price	$p^{up} = \sum_{k\in\mathcal{K}} \pi_k^*$
Generation Energy Revenue	$R_i^{g,e} = p^e g_{i0}$
Generation Up-Spinning Reserve Revenue	$R_i^{g,up} = p^{up} r_i^{g,up}$
Generation Security Charge	$C_i^{g,s} = \sum_{k\in\mathcal{K}_i^{OFF}} \pi_k^* \left(g_{i0} + r_i^{g,up}\right)$
Generation Total Revenue	$R_i^{g,t} = R_i^{g,e} + R_i^{g,up} - C_i^{g,s}$
Consumer Payment	$CP = p^e d$
Generation Energy Cost	$C_i^{g,e} = c_i g_{i0}$
Generation Up-Spinning Reserve Cost	$C_i^{g,up} = q_i^{g,up} r_i^{g,up}$
Generation Total Cost	$C_i^{g,t} = C_i^{g,e} + C_i^{g,up}$
Generation Profit	$Profit_i^{g} = R_i^{g,t} - C_i^{g,t}$

集合 $\mathcal{K}_i^{OFF}$ 明确标识了发电机 i 停机的应急状态。方括号内的第一项表示发电机 i 通过售电获得的净收入。请注意，在最优解下，电价为 $\pi_0^* + \sum_{k\in\mathcal{K}} \pi_k^*$，这与消费者支付的价格（公式 (14) 右侧的第一项）相同。此外，这一价格与基准定价系统中得出的价格相同（表 1）。方括号内的第二项表示上旋备用容量的净收入，这类商品的价格为 $\sum_{k\in\mathcal{K}} \pi_k^*$，与基准定价系统中定义的安全价格一致（表 1）。(14) 式中的最后一项称为“安全费”，可以视为发电机 i 应为系统所需备用容量负责而支付的费用，这与成本归因原则一致。换句话说，这一项可以理解为应付的费用。

因此，不同于与基准定价系统相关的结算，每个发电机的总收入由三个部分组成：能源收入（能源价格乘以事故前功率输出）、上旋备用收入（上旋备用价格乘以上旋备用）以及安全费（相应事故后拉格朗日乘子与事故前功率输出和上旋备用之和的乘积）。表8显示了拟议的定价方案及相应的结算。

拟议的定价系统已应用于第2.2节中描述的示例。使用表3和表4中提供的结果，并将其代入表8列出的表达式，得出了表9中报告的结果，该表列出了三个发电机的收入、成本、收费和利润的明细。与表6报告的结果相比，能源和备用容量的收入及其相关成本是相同的。然而，作为一个重要的显著结果，根据拟议的定价方案，发电机1需支付安全费用。请注意，该发电机是备用容量需求的主要驱动力，如表4中非零的拉格朗日乘子所示。该费用反映了因需要上调备用容量而增加的系统成本。具体而言，发电机1的故障需要65兆瓦的上调备用容量。因此，根据拟议的方法，总发电收入为12,000美元，这与

以下定理是针对所提出的定价框架陈述的，其各自的证明随后提供。

Table 9
Single-bus example – Proposed generation settlement ($).

i	$R_i^{g,e}$	$R_i^{g,up}$	$C_i^{g,s}$	$R_i^{g,t}$	$C_i^{g,e}$	$C_i^{g,up}$	$C_i^{g,t}$	$Profit_i^g$
1	6,500	0	5,200	1,300	1,300	0	1,300	0
2	3,000	2,400	0	5,400	1,500	150	1,650	3,750
3	2,500	2,800	0	5,300	2,500	350	2,850	2,450
Total	**12,000**	**5,200**	**5,200**	**12,000**	**5,300**	**500**	**5,800**	**6,200**

Table 10
Single-bus example – Proposed system settlement ($).

Generation Revenue	12,000
Consumer Payment	12,000
Balance	0

Theorem 1 (*Cost Recovery*)**.** *The proposed pricing framework ensures that generators' profits are non-negative.*

Proof. Let us rewrite the LD function in Eq. (13) as follows:

$$\phi(\boldsymbol{\Pi}) = CP(\boldsymbol{\Pi}) - \sum_{i\in\mathcal{I}} \max_{\boldsymbol{x}_i\in\mathcal{X}_i} \left\{ Profit_i^g(\boldsymbol{\Pi}, \boldsymbol{x}_i) \right\} \tag{15}$$

where $CP(\boldsymbol{\Pi})$ and $Profit_i^g(\boldsymbol{\Pi}, \boldsymbol{x}_i)$ represent the consumer payment and the profit of generator i, respectively. Since $Profit_i^g(\boldsymbol{\Pi}, \boldsymbol{x}_i)$ is a linear objective function maximized over $\boldsymbol{x}_i$, with $\boldsymbol{x}_i \geq \mathbf{0}$, the optimal value of $Profit_i^g(\boldsymbol{\Pi}, \boldsymbol{x}_i)$ is guaranteed to be non-negative. In other words, if π_0 and π_k fail to cover generator i's costs, the generator will not produce any output, resulting in $g_{i0}^* = r_i^{g,up*} = 0$. Likewise, at the optimal solution $\boldsymbol{x}_i^*$ and $\boldsymbol{\Pi}^*$:

$$Profit_i^g(\boldsymbol{\Pi}^*, \boldsymbol{x}_i^*) \geq 0, \ \forall i \in \mathcal{I} \tag{16}$$

Therefore, the theorem holds. □

Theorem 2 (*System Revenue Adequacy*)**.** *The total consumer payment is equal to the total generation revenue.*

Proof. By the strong duality theorem, the optimal value of the primal objective function in (1) is equal to the optimal value of the LD function in Eq. (14). Thus, at the optimal solution $\boldsymbol{x}^*$ and $\boldsymbol{\Pi}^*$, we have:

$$\begin{aligned}\sum_{i\in\mathcal{I}} c_i g_{i0}^* + \sum_{i\in\mathcal{I}} q_i^{g,up} r_i^{g,up*} = {} & \left(\pi_0^* + \sum_{k\in\mathcal{K}} \pi_k^*\right) d \\ & - \sum_{i\in\mathcal{I}} \left[\left(\pi_0^* + \sum_{k\in\mathcal{K}} \pi_k^* - c_i\right) g_{i0}^* \right. \\ & + \left(\sum_{k\in\mathcal{K}} \pi_k^* - q_i^{g,up}\right) r_i^{g,up*} \\ & \left. - \sum_{k\in\mathcal{K}_i^{OFF}} \pi_k^* \left(g_{i0}^* + r_i^{g,up*}\right) \right]\end{aligned} \tag{17}$$

By canceling out identical terms in both sides of Eq. (17) and rearranging terms, we simplify the expression as follows:

$$\begin{aligned}\left(\pi_0^* + \sum_{k\in\mathcal{K}} \pi_k^*\right) d = \sum_{i\in\mathcal{I}} & \left[\left(\pi_0^* + \sum_{k\in\mathcal{K}} \pi_k^*\right) g_{i0}^* + \sum_{k\in\mathcal{K}} \pi_k^* r_i^{g,up*} \right. \\ & \left. - \sum_{k\in\mathcal{K}_i^{OFF}} \pi_k^* \left(g_{i0}^* + r_i^{g,up*}\right) \right]\end{aligned} \tag{18}$$

Therefore, at the optimal solution, the consumer payment equals the sum of all generator revenues. □

3. Separately pricing up and down reserves and transmission pricing

In this section, the scope is broadened to consider a more realistic setting. To that end, network constraints and demand-side bidding are now incorporated into the market-clearing model, whereas both generation and line outages are accounted for. As a consequence, the set of ancillary services is extended to also include downward reserves, which are co-optimized with energy and upward reserves. This model is useful to comprehensively present the salient features of the proposed pricing scheme compared to the benchmark pricing system, which, aside from the concept of security charges already discussed in Section 2, comprise (1) separately pricing upward and downward reserves, and (2) transmission pricing.

3.1. Optimization model

The market-clearing model is formulated as the following linear program:

$$\begin{aligned}\min_{\substack{x\geq 0, y\geq 0,\\ \theta_0, \theta_k}} \quad & \boldsymbol{c}^\top \boldsymbol{g}_0 + (\boldsymbol{q}^{g,up})^\top \boldsymbol{r}^{g,up} + (\boldsymbol{q}^{g,dn})^\top \boldsymbol{r}^{g,dn} - \boldsymbol{w}^\top \boldsymbol{d}_0 \\ & + (\boldsymbol{q}^{d,up})^\top \boldsymbol{r}^{d,up} + (\boldsymbol{q}^{d,dn})^\top \boldsymbol{r}^{d,dn}\end{aligned} \tag{19}$$

subject to:

$$\boldsymbol{M}^g \boldsymbol{g}_0 + \boldsymbol{A}\boldsymbol{H}\boldsymbol{\theta}_0 = \boldsymbol{M}^d \boldsymbol{d}_0 \quad (\boldsymbol{\pi_0}) \tag{20}$$

$$\boldsymbol{M}^g \boldsymbol{g}_k + \boldsymbol{A}_k \boldsymbol{H}_k \boldsymbol{\theta}_k = \boldsymbol{M}^d \boldsymbol{d}_k \quad (\boldsymbol{\pi}_k), \qquad \forall k \in \mathcal{K} \tag{21}$$

$$-\boldsymbol{F}^{ss} \leq \boldsymbol{H}\boldsymbol{\theta}_0 \leq \boldsymbol{F}^{ss} \quad \left(\boldsymbol{\pi}_0^{f+}, \boldsymbol{\pi}_0^{f-}\right) \tag{22}$$

$$-\boldsymbol{F}^{st} \leq \boldsymbol{H}_k \boldsymbol{\theta}_k \leq \boldsymbol{F}^{st} \quad \left(\boldsymbol{\pi}_k^{f+}, \boldsymbol{\pi}_k^{f-}\right), \qquad \forall k \in \mathcal{K} \tag{23}$$

$$\boldsymbol{g}_k \leq \mathbb{D}(\boldsymbol{a}_k^g)(\boldsymbol{g}_0 + \boldsymbol{r}^{g,up}), \qquad \forall k \in \mathcal{K} \tag{24}$$

$$\boldsymbol{g}_k \geq \mathbb{D}(\boldsymbol{a}_k^g)(\boldsymbol{g}_0 - \boldsymbol{r}^{g,dn}), \qquad \forall k \in \mathcal{K} \tag{25}$$

$$\boldsymbol{g}_0 + \boldsymbol{r}^{g,up} \leq \overline{\boldsymbol{G}} \tag{26}$$

$$\boldsymbol{g}_0 - \boldsymbol{r}^{g,dn} \geq \mathbf{0} \tag{27}$$

$$\boldsymbol{r}^{g,up} \leq \overline{\boldsymbol{R}}^{g,up} \tag{28}$$

$$\boldsymbol{r}^{g,dn} \leq \overline{\boldsymbol{R}}^{g,dn} \tag{29}$$

$$\boldsymbol{d}_k \geq \boldsymbol{d}_0 - \boldsymbol{r}^{d,up}, \qquad \forall k \in \mathcal{K} \tag{30}$$

$$\boldsymbol{d}_k \leq \boldsymbol{d}_0 + \boldsymbol{r}^{d,dn}, \qquad \forall k \in \mathcal{K} \tag{31}$$

$$\boldsymbol{d}_0 - \boldsymbol{r}^{d,up} \geq \mathbf{0} \tag{32}$$

$$\boldsymbol{d}_0 + \boldsymbol{r}^{d,dn} \leq \overline{\boldsymbol{D}} \tag{33}$$

$$\boldsymbol{r}^{d,up} \leq \overline{\boldsymbol{R}}^{d,up} \tag{34}$$

$$\boldsymbol{r}^{d,dn} \leq \overline{\boldsymbol{R}}^{d,dn} \tag{35}$$

where $\boldsymbol{x} = \{\boldsymbol{g}_0, \boldsymbol{r}^{g,up}, \boldsymbol{r}^{g,dn}, \{\boldsymbol{g}_k\}_{k\in\mathcal{K}}\}$, $\boldsymbol{y} = \{\boldsymbol{d}_0, \boldsymbol{r}^{d,up}, \boldsymbol{r}^{d,dn}, \{\boldsymbol{d}_k\}_{k\in\mathcal{K}}\}$, $\boldsymbol{\theta_0}$ and $\boldsymbol{\theta}_k$ are the unbounded nodal phase angles, $\boldsymbol{H} = \boldsymbol{X}^{-1}\boldsymbol{A}^\top$, and $\boldsymbol{H}_k = \boldsymbol{X}^{-1}\boldsymbol{A}_k^\top$. Note also that relevant Lagrange multipliers are shown in parentheses.

As per (19), the optimization goal is the minimization of the sum of the costs for generating power and providing up- and down-spinning reserves offered by the generators minus the sum of the bid utility functions for consuming power plus the costs for providing up- and down-spinning reserves by the loads. This optimization goal is equivalent to maximizing the social welfare.

As done in Arroyo and Galiana (2005), and for presentation convenience, a dc power flow model relying on nodal phase angles (Wood et al., 2014) is used. Note, however, that the alternative dc power flow formulation based on sensitivity factors, which is typically adopted in industry practice, can be equivalently used. Constraints (20) and (21) represent the nodal power balance equations under the pre-contingency and contingency states, respectively. Constraints (22) and (23) enforce the line flow capacity limits, thereby explicitly bounding the differences of phase angles of connected buses. Additionally, to represent a possible flow-relief adjustment under stressed operating conditions, a flow-relief vector $\boldsymbol{\gamma}$ is introduced to relate short-term and steady-state line ratings through the entry-wise product $\boldsymbol{\gamma}\boldsymbol{F}^{ss}$, i.e., $\boldsymbol{F}^{st} = \boldsymbol{\gamma}\boldsymbol{F}^{ss}$. Line outages

表9 单母线示例 – 提议的发电结算（$）。

i	$R_i^{g,e}$	$R_i^{g,up}$	$C_i^{g,s}$	$R_i^{g,t}$	$C_i^{g,e}$	$C_i^{g,up}$	$C_i^{g,t}$	$Profit_i^g$
1	6,500	0	5,200	1,300	1,300	0	1,300	0
2	3,000	2,400	0	5,400	1,500	150	1,650	3,750
3	2,500	2,800	0	5,300	2,500	350	2,850	2,450
Total	**12,000**	**5,200**	**5,200**	**12,000**	**5,300**	**500**	**5,800**	**6,200**

表10 单母线示例 – 拟议系统结算（$）

Generation Revenue	12,000
Consumer Payment	12,000
Balance	0

定理 1 (*Cost Recovery*)。*The proposed pricing framework ensures that generators' profits are non-negative.*

证明。我们将第 (13) 式中的 LD 函数重写如下：

$$\phi(\boldsymbol{\Pi}) = CP(\boldsymbol{\Pi}) - \sum_{i\in\mathcal{I}} \max_{\boldsymbol{x}_i\in\mathcal{X}_i} \left\{ Profit_i^g(\boldsymbol{\Pi}, \boldsymbol{x}_i) \right\} \tag{15}$$

其中 $CP(\boldsymbol{\Pi})$ 和 $Profit_i^g(\boldsymbol{\Pi}, \boldsymbol{x}_i)$ 分别表示消费者支付金额和发电机 i 的利润。由于 $Profit_i^g(\boldsymbol{\Pi}, \boldsymbol{x}_i)$ 是一个在线性目标函数 $\boldsymbol{x}_i$ 上进行最大化的问题，并且有 $\boldsymbol{x}_i \geq \boldsymbol{0}$ 约束，$Profit_i^g(\boldsymbol{\Pi}, \boldsymbol{x}_i)$ 的最优值保证为非负。换句话说，如果 π_0 和 π_k 无法覆盖发电机 i 的成本，发电机将不会产生任何输出，从而导致 $g_{i0}^* = r_i^{g,up*} =$ 为 0。同样地，在最优解下 $\boldsymbol{x}_i^*$ 和 $\boldsymbol{\Pi}^*$：

$$Profit_i^g(\boldsymbol{\Pi}^*, \boldsymbol{x}_i^*) \geq 0, \ \forall i \in \mathcal{I} \tag{16}$$

因此，该定理成立。□

定理 2（*System Revenue Adequacy*）。*The total consumer payment is equal to the total generation revenue.*

证明。根据强对偶定理，(1) 中原始目标函数的最优值等于式 (14) 中 LD 函数的最优值。因此，在最优解 $\boldsymbol{x}^*$ 和 $\boldsymbol{\Pi}^*$ 处，我们有：

$$\begin{aligned}\sum_{i\in\mathcal{I}} c_i g_{i0}^* + \sum_{i\in\mathcal{I}} q_i^{g,up} r_i^{g,up*} = &\left(\pi_0^* + \sum_{k\in\mathcal{K}} \pi_k^*\right) d \\ &- \sum_{i\in\mathcal{I}} \left[\left(\pi_0^* + \sum_{k\in\mathcal{K}} \pi_k^* - c_i\right) g_{i0}^* \right. \\ &+ \left(\sum_{k\in\mathcal{K}} \pi_k^* - q_i^{g,up}\right) r_i^{g,up*} \\ &\left. - \sum_{k\in\mathcal{K}_i^{OFF}} \pi_k^* \left(g_{i0}^* + r_i^{g,up*}\right) \right]\end{aligned} \tag{17}$$

通过消去方程（17）两边的相同项并重新排列各项，我们将表达式简化如下：

$$\begin{aligned}\left(\pi_0^* + \sum_{k\in\mathcal{K}} \pi_k^*\right) d = \sum_{i\in\mathcal{I}} &\left[\left(\pi_0^* + \sum_{k\in\mathcal{K}} \pi_k^*\right) g_{i0}^* + \sum_{k\in\mathcal{K}} \pi_k^* r_i^{g,up*} \right. \\ &\left. - \sum_{k\in\mathcal{K}_i^{OFF}} \pi_k^* \left(g_{i0}^* + r_i^{g,up*}\right) \right]\end{aligned} \tag{18}$$

因此，在最优解处，消费者支付等于所有发电机收入的总和。□

3. 分别定价上下调备用容量和输电定价

在本节中，范围被扩大以考虑一个更现实的环境。为此，网络约束和需求方投标现在已被纳入市场清算模型，同时考虑了发电机和线路的故障。因此，辅助服务的范围也扩大到包括下行备用，其与电能和上行备用共同优化。该模型有助于全面展示所提出定价方案的突出特征，与基准定价系统相比，基准系统除了在第2节已讨论的安全费用概念外，还包括（1）分别对上行和下行备用进行定价，以及（2）输电定价。

3.1. Optimization model

市场清算模型被表述为如下线性规划：

$$\begin{aligned}\min_{\substack{x\geq 0, y\geq 0,\\ \theta_0, \theta_k}} \quad & \boldsymbol{c}^\top \boldsymbol{g}_0 + (\boldsymbol{q}^{g,up})^\top \boldsymbol{r}^{g,up} + (\boldsymbol{q}^{g,dn})^\top \boldsymbol{r}^{g,dn} - \boldsymbol{w}^\top \boldsymbol{d}_0 \\ & + (\boldsymbol{q}^{d,up})^\top \boldsymbol{r}^{d,up} + (\boldsymbol{q}^{d,dn})^\top \boldsymbol{r}^{d,dn}\end{aligned} \tag{19}$$

受以下条件约束：

$$\boldsymbol{M}^g \boldsymbol{g}_0 + \boldsymbol{A}\boldsymbol{H}\boldsymbol{\theta}_0 = \boldsymbol{M}^d \boldsymbol{d}_0 \quad (\boldsymbol{\pi_0}) \tag{20}$$

$$\boldsymbol{M}^g \boldsymbol{g}_k + \boldsymbol{A}_k \boldsymbol{H}_k \boldsymbol{\theta}_k = \boldsymbol{M}^d \boldsymbol{d}_k \quad (\boldsymbol{\pi}_k), \qquad \forall k \in \mathcal{K} \tag{21}$$

$$-\boldsymbol{F}^{ss} \leq \boldsymbol{H}\boldsymbol{\theta}_0 \leq \boldsymbol{F}^{ss} \quad \left(\boldsymbol{\pi}_0^{f+}, \boldsymbol{\pi}_0^{f-}\right) \tag{22}$$

$$-\boldsymbol{F}^{st} \leq \boldsymbol{H}_k \boldsymbol{\theta}_k \leq \boldsymbol{F}^{st} \quad \left(\boldsymbol{\pi}_k^{f+}, \boldsymbol{\pi}_k^{f-}\right), \qquad \forall k \in \mathcal{K} \tag{23}$$

$$\boldsymbol{g}_k \leq \mathbb{D}(\boldsymbol{a}_k^g)(\boldsymbol{g}_0 + \boldsymbol{r}^{g,up}), \qquad \forall k \in \mathcal{K} \tag{24}$$

$$\boldsymbol{g}_k \geq \mathbb{D}(\boldsymbol{a}_k^g)(\boldsymbol{g}_0 - \boldsymbol{r}^{g,dn}), \qquad \forall k \in \mathcal{K} \tag{25}$$

$$\boldsymbol{g}_0 + \boldsymbol{r}^{g,up} \leq \overline{\boldsymbol{G}} \tag{26}$$

$$\boldsymbol{g}_0 - \boldsymbol{r}^{g,dn} \geq \boldsymbol{0} \tag{27}$$

$$\boldsymbol{r}^{g,up} \leq \overline{\boldsymbol{R}}^{g,up} \tag{28}$$

$$\boldsymbol{r}^{g,dn} \leq \overline{\boldsymbol{R}}^{g,dn} \tag{29}$$

$$\boldsymbol{d}_k \geq \boldsymbol{d}_0 - \boldsymbol{r}^{d,up}, \qquad \forall k \in \mathcal{K} \tag{30}$$

$$\boldsymbol{d}_k \leq \boldsymbol{d}_0 + \boldsymbol{r}^{d,dn}, \qquad \forall k \in \mathcal{K} \tag{31}$$

$$\boldsymbol{d}_0 - \boldsymbol{r}^{d,up} \geq \boldsymbol{0} \tag{32}$$

$$\boldsymbol{d}_0 + \boldsymbol{r}^{d,dn} \leq \overline{\boldsymbol{D}} \tag{33}$$

$$\boldsymbol{r}^{d,up} \leq \overline{\boldsymbol{R}}^{d,up} \tag{34}$$

$$\boldsymbol{r}^{d,dn} \leq \overline{\boldsymbol{R}}^{d,dn} \tag{35}$$

其中 $\boldsymbol{x} = \{\boldsymbol{g}_0, \boldsymbol{r}^{g,up}, \boldsymbol{r}^{g,dn}, \{\boldsymbol{g}_k\}_{k\in\mathcal{K}}\}$、$\boldsymbol{y} = \{\boldsymbol{d}_0, \boldsymbol{r}^{d,up}, \boldsymbol{r}^{d,dn}, \{\boldsymbol{d}_k\}_{k\in\mathcal{K}}\}$、$\boldsymbol{\theta_0}$ 和 $\boldsymbol{\theta}_k$ 是无界节点相角，$\boldsymbol{H} = \boldsymbol{X}^{-1}\boldsymbol{A}^\top$ 和 $\boldsymbol{H}_k = \boldsymbol{X}^{-1}\boldsymbol{A}_k^\top$。另请注意，相关的拉格朗日乘子在括号中显示。

根据(19)，优化目标是最小化发电机提供的发电成本以及上旋和下旋备用容量成本的总和，减去负荷消耗电力的出价效用函数总和，再加上负荷提供上旋和下旋备用容量的成本。这个优化目标等同于最大化社会福利。

如同在 Arroyo 和 Galiana (2005) 中所做的，为了展示的便利性，使用了依赖节点相角的直流潮流模型（Wood 等人, 2014）。然而需要注意的是，基于灵敏度因子的替代直流潮流公式，通常在工业实践中采用，也可以同样使用。约束条件 (20) 和 (21) 分别表示预故障和故障状态下的节点功率平衡方程。约束条件 (22) 和 (23) 强制线路潮流容量限制，从而明确地限制了相连节点的相角差。此外，为了表示在压力操作条件下可能的流量缓解调整，引入了流量缓解向量 $\boldsymbol{\gamma}$，通过逐元素相乘 $\boldsymbol{\gamma}\boldsymbol{F}^{ss}$ 将短期和稳态线路额定值联系起来，即 $\boldsymbol{F}^{st} = \boldsymbol{\gamma}\boldsymbol{F}^{ss}$。线路停运

Table 11
Nodal prices for problem (19)–(35) according to the benchmark pricing system.

Nodal Energy Price	$p^e_b = \pi^*_{b0} + \sum_{k\in\mathcal{K}} \pi^*_{bk}$
Nodal Security Price	$p^s_b = \sum_{k\in\mathcal{K}} \pi^*_{bk}$

Table 12
Generation settlement for problem (19)–(35) according to the benchmark pricing system.

Generation Energy Revenue	$R^{g,e}_i = p^e_{b(i)} g_{i0}$
Generation Up-Spinning Reserve Revenue	$R^{g,up}_i = p^s_{b(i)} r^{g,up}_i$
Generation Down-Spinning Reserve Revenue	$R^{g,dn}_i = p^s_{b(i)} r^{g,dn}_i$
Generation Total Revenue	$R^{g,t}_i = R^{g,e}_i + R^{g,up}_i + R^{g,dn}_i$
Generation Energy Cost	$C^{g,e}_i = c_i g_{i0}$
Generation Up-Spinning Reserve Cost	$C^{g,up}_i = q^{g,up}_i r^{g,up}_i$
Generation Down-Spinning Reserve Cost	$C^{g,dn}_i = q^{g,dn}_i r^{g,dn}_i$
Generation Total Cost	$C^{g,t}_i = C^{g,e}_i + C^{g,up}_i + C^{g,dn}_i$
Generation Profit	$Profit^g_i = R^{g,t}_i - C^{g,t}_i$

are incorporated by explicitly modifying the network topology for each contingency k. This is achieved by constructing a contingency-specific incidence matrix, $\boldsymbol{A_k}$, obtained from the base-case incidence matrix $\boldsymbol{A}$ by setting to zero the column associated with the out-of-service line. The modified matrix $\boldsymbol{A_k}$ ensures that the disconnected line carries no flow and does not contribute to the nodal power balance under that contingency. In such a case, the flow-mapping matrix $\boldsymbol{H_k}$, which relates bus voltage angles to line flows, is updated accordingly. The superscripts "+" and "−" on the Lagrange multipliers for (22) and (23) indicate their association with the left- and right-hand inequalities, respectively. According to KKT conditions, the Lagrange multipliers for the left inequalities ($\boldsymbol{\pi}^{f+}_0, \boldsymbol{\pi}^{f+}_k$) are non-negative, while those for the right inequalities ($\boldsymbol{\pi}^{f-}_0, \boldsymbol{\pi}^{f-}_k$) are non-positive.

Admittedly, the precise characterization of the effect of the transmission network should rely on an ac power flow model, especially for systems under stress (Frank & Rebennack, 2016). Notwithstanding, the dc model is the standard network characterization in market design and pricing practice worldwide, including state-of-the-art contingency-constrained models such as that presented in Arroyo and Galiana (2005), upon which this work is built.

Constraints (24) and (25) relate the up- and down-spinning reserve contributions from generators to the production levels under the pre-contingency and contingency states.

Furthermore, constraints (26)–(29) ensure that such generation and reserve levels remain within their respective operational boundaries.

Constraints (30) and (31) establish the relationship between the up- and down-spinning reserve contributions from consumers and the power levels consumed under the pre-contingency and contingency states. Finally, demand-related bounds are set in constraints (32)–(35).

3.2. Challenges of the benchmark pricing framework

According to Arroyo and Galiana (2005), nodal prices for energy and security can be defined for problem (19)–(35), as summarized in Table 11. The corresponding generation and consumer settlements are provided in Tables 12 and 13, respectively. As can be seen, for each bus, a single price, namely the nodal security price, is used for all spinning reserves provided by the generators and consumers at that bus.

For illustration purposes, the benchmark pricing system presented in Table 11 is applied to the two-bus, one-line system depicted in Fig. 1, where the generation fleet is based on that considered in Section 2.2. The line reactance is 1 p.u. on a base of 100 MVA and 138 kV, whereas the steady-state and short-term line capacities are both 70 MW, i.e., the

Table 13
Consumer settlement for problem (19)–(35) according to the benchmark pricing system.

Consumer Payment for Energy	$CP^e_j = p^e_{b(j)} d_{j0}$
Consumer Up-Spinning Reserve Revenue	$R^{d,up}_j = p^s_{b(j)} r^{d,up}_j$
Consumer Down-Spinning Reserve Revenue	$R^{d,dn}_j = p^s_{b(j)} r^{d,dn}_j$
Consumer Utility	$U^d_j = w_j d_{j0}$
Consumer Up-Spinning Reserve Cost	$C^{d,up}_j = q^{d,up}_j r^{d,up}_j$
Consumer Down-Spinning Reserve Cost	$C^{d,dn}_j = q^{d,dn}_j r^{d,dn}_j$
Consumer Total Cost	$C^{d,t}_j = C^{d,up}_j + C^{d,dn}_j$
Consumer Payment	$CP_j = CP^e_j - R^{d,up}_j - R^{d,dn}_j$
Consumer Profit	$Profit^d_j = U^d_j - C^{d,t}_j - CP_j$

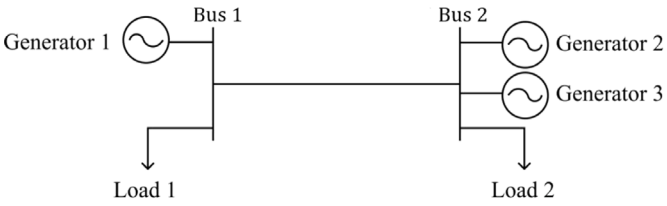


Fig. 1. Two-bus example – One-line diagram.

Table 14
Two-bus example – Generation data.

i	$\overline{G}_i$ (MW)	$\overline{R}^{g,up}_i$ (MW)	$\overline{R}^{g,dn}_i$ (MW)	c_i ($/MWh)	$q^{g,up}_i$ ($/MW/h)	$q^{g,dn}_i$ ($/MW/h)
1	100	50	50	20	2	2
2	60	30	30	50	5	5
3	70	35	35	100	10	10

Table 15
Two-bus example – Load data.

j	$\overline{D}_j$ (MW)	$\overline{R}^{d,up}_j$ (MW)	$\overline{R}^{d,dn}_j$ (MW)	w_j ($/MWh)	$q^{d,up}_j$ ($/MW/h)	$q^{d,dn}_j$ ($/MW/h)
1	90	10	10	200	150	300
2	40	10	10	150	100	250

corresponding flow-relief parameter is equal to 1. In Table 14, the generation data presented in Table 2 are extended with the information for down-spinning reserve offers, which is the same as for up-spinning reserves. Load data are provided in Table 15. The analysis considers four contingencies: the failure of each generator and the outage of the line.

Problem (19)–(35) has been solved to optimality using the simplex algorithm of CPLEX. Table 16 displays the optimal results, which feature a total value of the objective function equal to −$15,475. In other words, the optimal social welfare amounts to $15,475. It should be noted that the optimal pre-contingency system demand amounts to 120 MW, which is identical to the value of the inelastic demand in the single-bus example. Consequently, the optimal generation schedule is similar to that attained for the single-bus example, the main difference being the 5 MW of down-spinning reserve scheduled to generator 2. It is also worth highlighting that load 1 is scheduled to provide upward spinning reserve, which is deployed in response to contingencies involving the loss of generator 1 or line 1–2.

The outage of generator 1 constitutes a critical contingency for the system, requiring the full deployment of the up-spinning reserves scheduled to generators 2 and 3 and load 1. Analogously, the line outage is another critical contingency that compels generator 2 to reduce its output by fully utilizing its down-spinning reserve. Thus, for

表11 根据基准定价系统的问题(19)–(35)的节点价格。

Nodal Energy Price	$p_b^e = \pi_{b0}^* + \sum_{k\in\mathcal{K}} \pi_{bk}^*$
Nodal Security Price	$p_b^s = \sum_{k\in\mathcal{K}} \pi_{bk}^*$

表12 根据基准定价系统对问题(19)–(35)的发电结算。

Generation Energy Revenue	$R_i^{g,e} = p_{b(i)}^e g_{i0}$
Generation Up-Spinning Reserve Revenue	$R_i^{g,up} = p_{b(i)}^s r_i^{g,up}$
Generation Down-Spinning Reserve Revenue	$R_i^{g,dn} = p_{b(i)}^s r_i^{g,dn}$
Generation Total Revenue	$R_i^{g,t} = R_i^{g,e} + R_i^{g,up} + R_i^{g,dn}$
Generation Energy Cost	$C_i^{g,e} = c_i g_{i0}$
Generation Up-Spinning Reserve Cost	$C_i^{g,up} = q_i^{g,up} r_i^{g,up}$
Generation Down-Spinning Reserve Cost	$C_i^{g,dn} = q_i^{g,dn} r_i^{g,dn}$
Generation Total Cost	$C_i^{g,t} = C_i^{g,e} + C_i^{g,up} + C_i^{g,dn}$
Generation Profit	$Profit_i^g = R_i^{g,t} - C_i^{g,t}$

通过明确修改每个应急情况下的网络拓扑来纳入。这是通过构建特定于应急的关联矩阵 $\boldsymbol{A_k}$ 实现的，该矩阵从基准情况下的关联矩阵 $\boldsymbol{A}$ 得到，通过将与停用线路相关的列设置为零。修改后的矩阵 $\boldsymbol{A_k}$ 确保断开的线路不承载任何功率流，并且在该应急情况下不影响节点功率平衡。在这种情况下，将母线电压角与线路功率流关联的功率流映射矩阵 $\boldsymbol{H_k}$ 会相应更新。拉格朗日乘子在公式 (22) 和 (23) 上的上标 ‘‘+’’ 和 ‘‘−’’ 表示它们分别与左侧和右侧不等式相关。根据 KKT 条件，左侧不等式 ($\boldsymbol{\pi}_0^{f+}, \boldsymbol{\pi}_{\boldsymbol{k}}^{f+}$) 的拉格朗日乘子为非负，而右侧不等式 ($\boldsymbol{\pi}^{f-}$) 的拉格朗日乘子为非负。

零 , $\boldsymbol{\pi}_{\boldsymbol{k}}^{f-}$) 是非正的。

诚然，对输电网络影响的精确描述应依赖交流功率流模型，尤其是在系统处于应力状态下（Frank & Rebennack, 2016）。尽管如此，直流模型仍然是全球市场设计和定价实践中标准的网络表征方法，包括如Arroyo和Galiana（2005）所提出的最先进的受应急限制模型，本研究即基于该模型。

约束条件(24)和(25)将发电机的上调和下调备用贡献与故障前状态和故障状态下的生产水平联系起来。

此外，约束条件 (26)–(29) 确保此类发电和备用水平保持在各自的操作边界内。

约束条件（30）和（31）建立了消费者的上旋和下旋备用贡献与在事故前和事故状态下消耗的功率水平之间的关系。最后，需求相关的界限在约束条件（32）–（35）中设定。

3.2. Challenges of the benchmark pricing framework

根据Arroyo和Galiana（2005）的研究，对于问题(19)–(35)，能源和安全的节点价格可以如表11所总结进行定义。相应的发电和消费者结算分别见表12和表13。可以看出，对于每个母线，单一价格，即节点安全价格，被用于该母线上所有由发电机和消费者提供的旋转备用。

为了说明目的，表11中给出的基准定价系统应用于图1所示的两母线一线路系统，其中发电机组基于第2.2节所考虑的情况。线路电抗为1 p.u.，基准为100 MVA和138 kV，而稳态和短期线路容量均为70 MW，即

表13 根据基准定价系统对问题(19)–(35)的消费者结算。

Consumer Payment for Energy	$CP_j^e = p_{b(j)}^e d_{j0}$
Consumer Up-Spinning Reserve Revenue	$R_j^{d,up} = p_{b(j)}^s r_j^{d,up}$
Consumer Down-Spinning Reserve Revenue	$R_j^{d,dn} = p_{b(j)}^s r_j^{d,dn}$
Consumer Utility	$U_j^d = w_j d_{j0}$
Consumer Up-Spinning Reserve Cost	$C_j^{d,up} = q_j^{d,up} r_j^{d,up}$
Consumer Down-Spinning Reserve Cost	$C_j^{d,dn} = q_j^{d,dn} r_j^{d,dn}$
Consumer Total Cost	$C_j^{d,t} = C_j^{d,up} + C_j^{d,dn}$
Consumer Payment	$CP_j = CP_j^e - R_j^{d,up} - R_j^{d,dn}$
Consumer Profit	$Profit_j^d = U_j^d - C_j^{d,t} - CP_j$

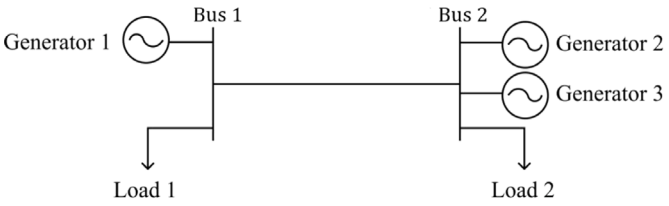


图1. 双母线示例——单线图。

表 14 双母线示例 – 发电数据。

i	$\overline{G}_i$ (MW)	$\overline{R}_i^{g,up}$ (MW)	$\overline{R}_i^{g,dn}$ (MW)	c_i ($/MWh)	$q_i^{g,up}$ ($/MW/h)	$q_i^{g,dn}$ ($/MW/h)
1	100	50	50	20	2	2
2	60	30	30	50	5	5
3	70	35	35	100	10	10

表 15 双母线示例 – 负载数据。

j	$\overline{D}_j$ (MW)	$\overline{R}_j^{d,up}$ (MW)	$\overline{R}_j^{d,dn}$ (MW)	w_j ($/MWh)	$q_j^{d,up}$ ($/MW/h)	$q_j^{d,dn}$ ($/MW/h)
1	90	10	10	200	150	300
2	40	10	10	150	100	250

相应的流量缓解参数等于 1。在表 14 中，表 2 中呈现的发电数据被扩展以包含下旋备用出价信息，这与上旋备用相同。负荷数据见表 15。分析考虑了四种应急情况：每个发电机的故障以及线路的停运。

问题 (19)–(35) 已使用 CPLEX 的单纯形算法求解到最优。表 16 显示了最优结果，其目标函数的总值为 –$15,475。换句话说，最优社会福利为 15,475 美元。值得注意的是，最优事故前系统的需求为 120 兆瓦，这与单母线示例中非弹性需求的数值相同。因此，最优发电计划与单母线示例中获得的计划相似，主要区别是为发电机 2 安排了 5 兆瓦的下旋储备。还值得强调的是，负荷 1 被安排提供上旋储备，该储备在发生涉及发电机 1 或 线路 1–2 的事故时会被启用。

发电机1的停机构成了系统的关键应急情况，需要全面调动计划用于发电机2和3以及负载1的上旋储备。类似地，线路停运是另一种关键应急情况，迫使发电机2通过充分利用其下旋储备来减少输出。因此，对于

Table 16
Two-bus example – Optimal results (MW).

Generator	Pre-Contingency Generation	Up-Spinning Reserve	Down-Spinning Reserve	Generation under Contingency			
				Outage of Generator 1	Outage of Generator 2	Outage of Generator 3	Outage of Line 1–2
1	75	0	0	0	75	75	75
2	30	30	5	60	0	45	25
3	15	35	0	50	45	0	15
Total	**120**	**65**	**5**	**110**	**120**	**120**	**115**
Load	Pre-Contingency Demand	Up-Spinning Reserve	Down-Spinning Reserve	Demand under Contingency			
				Outage of Generator 1	Outage of Generator 2	Outage of Generator 3	Outage of Line 1–2
1	80	10	0	70	80	80	75
2	40	0	0	40	40	40	40
Total	**120**	**10**	**0**	**110**	**120**	**120**	**115**
Line	Pre-Contingency Power Flow 1–2			Power Flow 1–2 under Contingency			
				Outage of Generator 1	Outage of Generator 2	Outage of Generator 3	Outage of Line 1–2
1–2	−5			−70	−5	−5	0

Table 17
Two-bus example – Lagrange multipliers for the nodal power balance equations ($/MWh).

Bus	Pre-Contingency State	Outage of Generator 1	Outage of Generator 2	Outage of Generator 3	Outage of Line 1–2
1	20	180	0	0	0
2	20	85	0	0	−5

this particular example, the line outage is the sole reason for downward reserves. Note, however, that under this contingency state a marginal increase in load 2 would actually reduce the need for down-spinning reserves at this bus. This explains why the corresponding Lagrange multiplier is negative (Table 17). Finally, it is worth pointing out that line congestion is solely experienced during the outage of generator 1. In fact, the pre-contingency power flow is well below the line capacity to allow generators 2 and 3 to ramp up during this critical outage.

The nodal prices and the settlement for generators and consumers calculated according to Tables 11, 12, and 13 are presented in Tables 18, 19, and 20, respectively. As can be seen, the profit distribution among market agents is economically inconsistent as the total profit of generators and consumers is 44.6% greater than the optimal social welfare. Furthermore, similar to the previous example, the benchmark pricing scheme results in total generation revenue exceeding total consumer payment, as summarized in Table 21. This issue becomes more important in a network-constrained setting because the absence of a transmission surplus to compensate for lines creates an additional financial burden. For this particular example, the consumer payment is insufficient to cover the generation revenue and the transmission revenue associated with the line congestion under the outage of generator 1. The resulting settlement imbalance may be addressed by a compensation in addition to the remuneration from the marginal prices that may give rise to discrimination among market participants and inadequate economic signals, among other issues. Alternatively, in Section 3.3 we propose a new pricing scheme that does not feature settlement imbalance while relying on a sound mathematical framework.

3.3. Proposed pricing system

Following the procedure described in Section 2.3, we start by building the LD function through the dualization of constraints (20)–(23),

Table 18
Two-bus example – Nodal prices according to the benchmark pricing system ($/MWh).

b	Nodal Energy Price, p^e_b	Nodal Security Price, p^s_b
1	200	180
2	100	80

Table 19
Two-bus example – Generation settlement according to the benchmark pricing system ($).

i	$R^{g,e}_i$	$R^{g,up}_i$	$R^{g,dn}_i$	$R^{g,t}_i$	$C^{g,e}_i$	$C^{g,up}_i$	$C^{g,dn}_i$	$C^{g,t}_i$	$Profit^g_i$
1	15,000	0	0	15,000	1,500	0	0	1,500	13,500
2	3,000	2,400	400	5,800	1,500	150	25	1,675	4,125
3	1,500	2,800	0	4,300	1,500	350	0	1,850	2,450
Total	**19,500**	**5,200**	**400**	**25,100**	**4,500**	**500**	**25**	**5,025**	**20,075**

Table 20
Two-bus example – Consumer settlement according to the benchmark pricing system ($).

j	CP^e_j	$R^{d,up}_j$	$R^{d,dn}_j$	CP_j	U^d_j	$C^{d,up}_j$	$C^{d,dn}_j$	$C^{d,t}_j$	$Profit^d_j$
1	16,000	1,800	0	14,200	16,000	1,500	0	1,500	300
2	4,000	0	0	4,000	6,000	0	0	0	2,000
Total	**20,000**	**1,800**	**0**	**18,200**	**22,000**	**1,500**	**0**	**1,500**	**2,300**

Table 21
Two-bus example – System settlement according to the benchmark pricing system ($).

Generation Revenue	25,100
Consumer Payment	18,200
Balance	−6,900

表16 两母线示例 – 最优结果（兆瓦）

Generator	Pre-Contingency Generation	Up-Spinning Reserve	Down-Spinning Reserve	Generation under Contingency			
				Outage of Generator 1	Outage of Generator 2	Outage of Generator 3	Outage of Line 1–2
1	75	0	0	0	75	75	75
2	30	30	5	60	0	45	25
3	15	35	0	50	45	0	15
Total	**120**	**65**	**5**	**110**	**120**	**120**	**115**
Load	Pre-Contingency Demand	Up-Spinning Reserve	Down-Spinning Reserve	Demand under Contingency			
				Outage of Generator 1	Outage of Generator 2	Outage of Generator 3	Outage of Line 1–2
1	80	10	0	70	80	80	75
2	40	0	0	40	40	40	40
Total	**120**	**10**	**0**	**110**	**120**	**120**	**115**
Line	Pre-Contingency Power Flow 1–2			Power Flow 1–2 under Contingency			
				Outage of Generator 1	Outage of Generator 2	Outage of Generator 3	Outage of Line 1–2
1–2	−5			−70	−5	−5	0

表17 两母线示例 – 节点功率平衡方程的拉格朗日乘子（$/MWh）。

Bus	Pre-Contingency State	Outage of Generator 1	Outage of Generator 2	Outage of Generator 3	Outage of Line 1–2
1	20	180	0	0	0
2	20	85	0	0	−5

在这个具体例子中，线路停运是下行备用容量不足的唯一原因。不过，需要注意的是，在这种应急状态下，对负荷2的边际增加实际上会减少该母线下旋备用容量的需求。这解释了为什么相应的拉格朗日乘子为负值（表17）。最后，值得指出的是，线路拥堵仅在发电机1停运期间出现。实际上，抢修前的潮流远低于线路容量，以允许发电机2和3在这一关键停运期间加大出力。

根据表11、12和13计算的节点价格以及发电机和消费者的结算分别列在表18、19和20中。可以看出，市场参与者之间的利润分配在经济上是不一致的，因为发电机和消费者的总利润比最优社会福利高出44.6%。此外，与前一个例子类似，基准定价方案导致总发电收入超过总消费者支付，如表21所总结。在网络受限的情况下，这一问题变得更加重要，因为缺乏用于补偿线路的传输盈余会产生额外的财务负担。对于这个特定的例子，消费者支付不足以覆盖发电收入以及在发电机1停运情况下与线路拥堵相关的传输收入。由此产生的结算不平衡可以通过

3.3.Proposed pricing system

按照第2.3节中描述的程序，我们首先通过对约束条件(20)–(23)进行对偶化来构建LD函数，

表18 双母线示例——根据基准定价系统的节点价格（$/MWh）。

b	Nodal Energy Price, p^e_b	Nodal Security Price, p^s_b
1	200	180
2	100	80

表19 两母线示例 – 按基准定价系统的发电结算（美元）。

i	$R^{g,e}_i$	$R^{g,up}_i$	$R^{g,dn}_i$	$R^{g,t}_i$	$C^{g,e}_i$	$C^{g,up}_i$	$C^{g,dn}_i$	$C^{g,t}_i$	$Profit^g_i$
1	15,000	0	0	15,000	1,500	0	0	1,500	13,500
2	3,000	2,400	400	5,800	1,500	150	25	1,675	4,125
3	1,500	2,800	0	4,300	1,500	350	0	1,850	2,450
Total	**19,500**	**5,200**	**400**	**25,100**	**4,500**	**500**	**25**	**5,025**	**20,075**

表20 双母线示例 – 根据基准定价系统的消费者结算（美元）。

j	CP^e_j	$R^{d,up}_j$	$R^{d,dn}_j$	CP_j	U^d_j	$C^{d,up}_j$	$C^{d,dn}_j$	$C^{d,t}_j$	$Profit^d_j$
1	16,000	1,800	0	14,200	16,000	1,500	0	1,500	300
2	4,000	0	0	4,000	6,000	0	0	0	2,000
Total	**20,000**	**1,800**	**0**	**18,200**	**22,000**	**1,500**	**0**	**1,500**	**2,300**

表21 双母线示例 – 根据基准定价系统的系统结算（美元）。

Generation Revenue	25,100
Consumer Payment	18,200
Balance	−6,900

giving rise to Eq. (36):

$$\begin{aligned}\phi(\boldsymbol{\Pi}) = \min_{\substack{\boldsymbol{x}\in\mathcal{X},\\ \boldsymbol{y}\in\mathcal{Y},\\ \boldsymbol{\theta_0},\boldsymbol{\theta_k}}} \Bigg\{ & \boldsymbol{c}^\top \boldsymbol{g}_0 + (\boldsymbol{q}^{g,up})^\top \boldsymbol{r}^{g,up} + (\boldsymbol{q}^{g,dn})^\top \boldsymbol{r}^{g,dn} \\ & - \boldsymbol{w}^\top \boldsymbol{d}_0 + (\boldsymbol{q}^{d,up})^\top \boldsymbol{r}^{d,up} + (\boldsymbol{q}^{d,dn})^\top \boldsymbol{r}^{d,dn} \\ & + \boldsymbol{\pi}_0^\top \left(\boldsymbol{M}^d \boldsymbol{d}_0 - \boldsymbol{M}^g \boldsymbol{g}_0 - \boldsymbol{A}\boldsymbol{H}\boldsymbol{\theta}_0\right) \\ & + \sum_{k\in\mathcal{K}} \left[\boldsymbol{\pi}_k^\top \left(\boldsymbol{M}^d \boldsymbol{d}_k - \boldsymbol{M}^g \boldsymbol{g}_k - \boldsymbol{A}_k \boldsymbol{H}_k \boldsymbol{\theta}_k\right)\right] \\ & + \left(\boldsymbol{\pi}_0^{f+}\right)^\top \left(-\boldsymbol{F}^{ss} - \boldsymbol{H}\boldsymbol{\theta}_0\right) + \left(\boldsymbol{\pi}_0^{f-}\right)^\top \left(\boldsymbol{F}^{ss} - \boldsymbol{H}\boldsymbol{\theta}_0\right) \\ & + \sum_{k\in\mathcal{K}} \Bigg[\left(\boldsymbol{\pi}_k^{f+}\right)^\top \left(-\boldsymbol{F}^{st} - \boldsymbol{H}_k\, \boldsymbol{\theta}_k\right) \\ & + \left(\boldsymbol{\pi}_k^{f-}\right)^\top \left(\boldsymbol{F}^{st} - \boldsymbol{H}_k \boldsymbol{\theta}_k\right) \Bigg] \Bigg\} \end{aligned} \tag{36}$$

where $\boldsymbol{\Pi} = \left\{\boldsymbol{\pi}_0, \boldsymbol{\pi}_0^{f+}, \boldsymbol{\pi}_0^{f-}, \left\{\boldsymbol{\pi}_k, \boldsymbol{\pi}_k^{f+}, \boldsymbol{\pi}_k^{f-}\right\}_{k\in\mathcal{K}}\right\}$, $\mathcal{X}$ represents the feasibility set associated with generation-related constraints $\boldsymbol{x} \geq \boldsymbol{0}$ and (24)–(29), whereas $\mathcal{Y}$ is the feasibility set corresponding to consumer-related constraints $\boldsymbol{y} \geq \boldsymbol{0}$ and (30)–(35). Rearranging (36), we have:

$$\begin{aligned}\phi(\boldsymbol{\Pi}) = \min_{\substack{\boldsymbol{x}\in\mathcal{X},\\ \boldsymbol{y}\in\mathcal{Y},\\ \boldsymbol{\theta_0},\boldsymbol{\theta_k}}} \Bigg\{ & \left(\boldsymbol{c}^\top - \boldsymbol{\pi}_0^\top \boldsymbol{M}^g\right) \boldsymbol{g}_0 - \sum_{k\in\mathcal{K}} \boldsymbol{\pi}_k^\top \boldsymbol{M}^g \boldsymbol{g}_k + (\boldsymbol{q}^{g,up})^\top \boldsymbol{r}^{g,up} \\ & + (\boldsymbol{q}^{g,dn})^\top \boldsymbol{r}^{g,dn} + \left(\boldsymbol{\pi}_0^\top \boldsymbol{M}^d - \boldsymbol{w}^\top\right) \boldsymbol{d}_0 \\ & + \sum_{k\in\mathcal{K}} \boldsymbol{\pi}_k^\top \boldsymbol{M}^d \boldsymbol{d}_k + (\boldsymbol{q}^{d,up})^\top \boldsymbol{r}^{d,up} + (\boldsymbol{q}^{d,dn})^\top \boldsymbol{r}^{d,dn} \\ & - \left[\boldsymbol{\pi}_0^\top \boldsymbol{A} + (\boldsymbol{\pi}_0^{f+} + \boldsymbol{\pi}_0^{f-})^\top\right] \boldsymbol{H}\boldsymbol{\theta}_0 \\ & - \sum_{k\in\mathcal{K}} \left[\boldsymbol{\pi}_k^\top \boldsymbol{A}_k + (\boldsymbol{\pi}_k^{f+} + \boldsymbol{\pi}_k^{f-})^\top\right] \boldsymbol{H}_k \boldsymbol{\theta}_k \\ & - \left[(\boldsymbol{\pi}_0^{f+} - \boldsymbol{\pi}_0^{f-})^\top \boldsymbol{F}^{ss} + \sum_{k\in\mathcal{K}} (\boldsymbol{\pi}_k^{f+} - \boldsymbol{\pi}_k^{f-})^\top \boldsymbol{F}^{st}\right] \Bigg\} \end{aligned} \tag{37}$$

The last square-bracketed term in Eq. (37) is independent of $\boldsymbol{x}$ and $\boldsymbol{y}$, allowing it to be factored out of the minimization problem. Moreover, leveraging the separability of the resulting minimization problem, Eq. (37) can be equivalently rewritten as Eq. (38):

$$\begin{aligned}\phi(\boldsymbol{\Pi}) = & -\left[\left(\boldsymbol{\pi}_0^{f+} - \boldsymbol{\pi}_0^{f-}\right)^\top \boldsymbol{F}^{ss} + \sum_{k\in\mathcal{K}} \left(\boldsymbol{\pi}_k^{f+} - \boldsymbol{\pi}_k^{f-}\right)^\top \boldsymbol{F}^{st}\right] \\ & + \min_{\boldsymbol{x}\in\mathcal{X}} \Big[\left(\boldsymbol{c}^\top - \boldsymbol{\pi}_0^\top \boldsymbol{M}^g\right) \boldsymbol{g}_0 - \sum_{k\in\mathcal{K}} \boldsymbol{\pi}_k^\top \boldsymbol{M}^g \boldsymbol{g}_k + (\boldsymbol{q}^{g,up})^\top \boldsymbol{r}^{g,up} \\ & + (\boldsymbol{q}^{g,dn})^\top \boldsymbol{r}^{g,dn}\Big] + \min_{\boldsymbol{y}\in\mathcal{Y}} \Big[\left(\boldsymbol{\pi}_0^\top \boldsymbol{M}^d - \boldsymbol{w}^\top\right) \boldsymbol{d}_0 + \sum_{k\in\mathcal{K}} \boldsymbol{\pi}_k^\top \boldsymbol{M}^d \boldsymbol{d}_k \\ & + (\boldsymbol{q}^{d,up})^\top \boldsymbol{r}^{d,up} + (\boldsymbol{q}^{d,dn})^\top \boldsymbol{r}^{d,dn}\Big] \\ & - \min_{\boldsymbol{\theta_0},\boldsymbol{\theta_k}} \Bigg\{ \left[\boldsymbol{\pi}_0^\top \boldsymbol{A} + \left(\boldsymbol{\pi}_0^{f+} + \boldsymbol{\pi}_0^{f-}\right)^\top\right] \boldsymbol{H}\boldsymbol{\theta}_0 \\ & + \sum_{k\in\mathcal{K}} \left[\boldsymbol{\pi}_k^\top \boldsymbol{A}_k + \left(\boldsymbol{\pi}_k^{f+} + \boldsymbol{\pi}_k^{f-}\right)^\top\right] \boldsymbol{H}_k \boldsymbol{\theta}_k \Bigg\} \end{aligned} \tag{38}$$

For quick reference, the last minimization term in (38), related to phase angles, is expressed in a compact way as $\phi_\theta(\boldsymbol{\Pi})$. In addition, using a component-wise formulation for the other terms making up $\phi(\boldsymbol{\Pi})$, Eq. (38) is equivalently cast as:

$$\begin{aligned}\phi(\boldsymbol{\Pi}) = & -\sum_{l\in\mathcal{L}} \left[\left(\pi_{l0}^{f+} - \pi_{l0}^{f-}\right) F_l^{ss} + \sum_{k\in\mathcal{K}} \left(\pi_{lk}^{f+} - \pi_{lk}^{f-}\right) F_l^{st}\right] \\ & - \sum_{i\in\mathcal{I}} \max_{\boldsymbol{x}_i\in\mathcal{X}_i} \Bigg[\left(\pi_{b(i)0} - c_i\right) g_{i0} - q_i^{g,up} r_i^{g,up} - q_i^{g,dn} r_i^{g,dn} \\ & + \sum_{k\in\mathcal{K}} \psi_{ik}^g(\pi_{b(i)k}, \boldsymbol{x}_i) \Bigg] - \sum_{j\in\mathcal{J}} \max_{\boldsymbol{y}_j\in\mathcal{Y}_j} \Bigg[\left(w_j - \pi_{b(j)0}\right) d_{j0} \\ & - q_j^{d,up} r_j^{d,up} - q_j^{d,dn} r_j^{d,dn} - \sum_{k\in\mathcal{K}} \psi_{jk}^d(\pi_{b(j)k}, \boldsymbol{y}_j) \Bigg] - \phi_\theta(\boldsymbol{\Pi}) \end{aligned} \tag{39}$$

where $\boldsymbol{x}_i = \{g_{i0}, r_i^{g,up}, r_i^{g,dn}\}$ and $\mathcal{X}_i$ is the subset of constraints related to $\boldsymbol{x}_i$, i.e., constraints (26)–(29) and $\boldsymbol{x}_i \geq \boldsymbol{0}$. Analogously, $\boldsymbol{y}_j = \{d_{j0}, r_j^{d,up}, r_j^{d,dn}\}$ and $\mathcal{Y}_j$ represents the subset of constraints related to $\boldsymbol{y}_j$, i.e., constraints (32)–(35) and $\boldsymbol{y}_j \geq \boldsymbol{0}$. Additionally, $\psi_{ik}^g(\pi_{b(i)k}, \boldsymbol{x}_i)$ and $\psi_{jk}^d(\pi_{b(j)k}, \boldsymbol{y}_j)$ respectively represent the revenue fraction earned by generator i and the payment fraction of consumer j under contingency k due to their best response against post-contingency Lagrange multiplier π_{bk}.

$\psi_{ik}^g(\pi_{b(i)k}, \boldsymbol{x}_i)$ depends on the generator availability status a_{ik}^g and on the pre-contingency dispatch and reserve schedule $\boldsymbol{x}_i$:

$$\psi_{ik}^g(\pi_{b(i)k}, \boldsymbol{x}_i) = \max_{g_{ik}\geq 0} \ \pi_{b(i)k} g_{ik} \tag{40}$$

subject to:

$$g_{ik} \leq a_{ik}^g (g_{i0} + r_i^{g,up}) \tag{41}$$

$$g_{ik} \geq a_{ik}^g (g_{i0} - r_i^{g,dn}) \tag{42}$$

Analogously, $\psi_{jk}^d(\pi_{b(j)k}, \boldsymbol{y}_j)$ depends on the pre-contingency dispatch and reserve schedule $\boldsymbol{y}_j$:

$$\psi_{jk}^d(\pi_{b(j)k}, \boldsymbol{y}_j) = \min_{d_{jk}\geq 0} \ \pi_{b(j)k} d_{jk} \tag{43}$$

subject to:

$$d_{jk} \geq d_{j0} - r_j^{d,up} \tag{44}$$

$$d_{jk} \leq d_{j0} + r_j^{d,dn} \tag{45}$$

The optimal solutions to problems (40)–(42) and (43)–(45) give rise to three possible outcomes for $\psi_{ik}^g(\pi_{b(i)k}, \boldsymbol{x}_i)$ and $\psi_{jk}^d(\pi_{b(j)k}, \boldsymbol{y}_j)$:

1. If π_{bk} is positive, g_{ik}^* and d_{jk}^* will be equal to the upper and lower bounds respectively set in (41) and (44), i.e., all generators and consumers at bus b will be willing to deploy as much up-spinning reserve as possible under contingency k. As a result:
   - $g_{ik}^* = a_{ik}^g (g_{i0} + r_i^{g,up})$ and $\psi_{ik}^g(\pi_{b(i)k}, \boldsymbol{x}_i) = \pi_{b(i)k} a_{ik}^g (g_{i0} + r_i^{g,up})$
   - $d_{jk}^* = d_{j0} - r_j^{d,up}$ and $\psi_{jk}^d(\pi_{b(j)k}, \boldsymbol{y}_j) = \pi_{b(j)k} (d_{j0} - r_j^{d,up})$
2. If π_{bk} is negative, g_{ik}^* and d_{jk}^* will be equal to the lower and upper bounds respectively set in (42) and (45), i.e., all generators and consumers at bus b will be willing to deploy as much down-spinning reserve as possible under contingency k. As a result:
   - $g_{ik}^* = a_{ik}^g (g_{i0} - r_i^{g,dn})$ and $\psi_{ik}^g(\pi_{b(i)k}, \boldsymbol{x}_i) = \pi_{b(i)k} a_{ik}^g (g_{i0} - r_i^{g,dn})$
   - $d_{jk}^* = d_{j0} + r_j^{d,dn}$ and $\psi_{jk}^d(\pi_{b(j)k}, \boldsymbol{y}_j) = \pi_{b(j)k} (d_{j0} + r_j^{d,dn})$
3. If π_{bk} equals zero, $\psi_{ik}^g(\pi_{b(i)k}, \boldsymbol{x}_i)$ and $\psi_{jk}^d(\pi_{b(j)k}, \boldsymbol{y}_j)$ will be both equal to zero for the market agents at bus b.

导致方程 (36):

$$\begin{aligned}\phi(\boldsymbol{\Pi}) = \min_{\substack{\boldsymbol{x}\in\mathcal{X},\\ \boldsymbol{y}\in\mathcal{Y},\\ \theta_0,\boldsymbol{\theta_k}}} \Bigg\{ & \boldsymbol{c}^\top \boldsymbol{g}_0 + (\boldsymbol{q}^{g,up})^\top \boldsymbol{r}^{g,up} + (\boldsymbol{q}^{g,dn})^\top \boldsymbol{r}^{g,dn} \\ & - \boldsymbol{w}^\top \boldsymbol{d}_0 + (\boldsymbol{q}^{d,up})^\top \boldsymbol{r}^{d,up} + (\boldsymbol{q}^{d,dn})^\top \boldsymbol{r}^{d,dn} \\ & + \boldsymbol{\pi}_0^\top \left(\boldsymbol{M}^d \boldsymbol{d}_0 - \boldsymbol{M}^g \boldsymbol{g}_0 - \boldsymbol{A}\boldsymbol{H}\theta_0\right) \\ & + \sum_{k\in\mathcal{K}} \left[\boldsymbol{\pi}_k^\top \left(\boldsymbol{M}^d \boldsymbol{d}_k - \boldsymbol{M}^g \boldsymbol{g}_k - \boldsymbol{A}_k \boldsymbol{H}_k \theta_k\right)\right] \\ & + \left(\boldsymbol{\pi}_0^{f+}\right)^\top \left(-\boldsymbol{F}^{ss} - \boldsymbol{H}\theta_0\right) + \left(\boldsymbol{\pi}_0^{f-}\right)^\top \left(\boldsymbol{F}^{ss} - \boldsymbol{H}\theta_0\right) \\ & + \sum_{k\in\mathcal{K}} \Bigg[\left(\boldsymbol{\pi}_k^{f+}\right)^\top \left(-\boldsymbol{F}^{st} - \boldsymbol{H}_k\, \theta_k\right) \\ & + \left(\boldsymbol{\pi}_k^{f-}\right)^\top \left(\boldsymbol{F}^{st} - \boldsymbol{H}_k \theta_k\right) \Bigg] \Bigg\} \end{aligned} \tag{36}$$

其中 $\boldsymbol{\Pi} = \left\{\boldsymbol{\pi}_0, \boldsymbol{\pi}_0^{f+}, \boldsymbol{\pi}_0^{f-}, \left\{\boldsymbol{\pi}_k, \boldsymbol{\pi}_k^{f+}, \boldsymbol{\pi}_k^{f-}\right\}_{k\in\mathcal{K}}\right\}$、$\mathcal{X}$ 表示与发电相关约束 $\boldsymbol{x} \geq \boldsymbol{0}$ 以及 (24)–(29) 相关的可行集合，而 $\mathcal{Y}$ 是与消费者相关约束 $\boldsymbol{y} \geq \boldsymbol{0}$ 以及 (30)–(35) 相关的可行集合。重排 (36)，我们得到:

$$\begin{aligned}\phi(\boldsymbol{\Pi}) = \min_{\substack{\boldsymbol{x}\in\mathcal{X},\\ \boldsymbol{y}\in\mathcal{Y},\\ \theta_0,\boldsymbol{\theta_k}}} \Bigg\{ & \left(\boldsymbol{c}^\top - \boldsymbol{\pi}_0^\top \boldsymbol{M}^g\right) \boldsymbol{g}_0 - \sum_{k\in\mathcal{K}} \boldsymbol{\pi}_k^\top \boldsymbol{M}^g \boldsymbol{g}_k + (\boldsymbol{q}^{g,up})^\top \boldsymbol{r}^{g,up} \\ & + (\boldsymbol{q}^{g,dn})^\top \boldsymbol{r}^{g,dn} + \left(\boldsymbol{\pi}_0^\top \boldsymbol{M}^d - \boldsymbol{w}^\top\right) \boldsymbol{d}_0 \\ & + \sum_{k\in\mathcal{K}} \boldsymbol{\pi}_k^\top \boldsymbol{M}^d \boldsymbol{d}_k + (\boldsymbol{q}^{d,up})^\top \boldsymbol{r}^{d,up} + (\boldsymbol{q}^{d,dn})^\top \boldsymbol{r}^{d,dn} \\ & - \left[\boldsymbol{\pi}_0^\top \boldsymbol{A} + (\boldsymbol{\pi}_0^{f+} + \boldsymbol{\pi}_0^{f-})^\top\right] \boldsymbol{H}\theta_0 \\ & - \sum_{k\in\mathcal{K}} \left[\boldsymbol{\pi}_k^\top \boldsymbol{A}_k + (\boldsymbol{\pi}_k^{f+} + \boldsymbol{\pi}_k^{f-})^\top\right] \boldsymbol{H}_k \boldsymbol{\theta}_k \\ & - \left[(\boldsymbol{\pi}_0^{f+} - \boldsymbol{\pi}_0^{f-})^\top \boldsymbol{F}^{ss} + \sum_{k\in\mathcal{K}} (\boldsymbol{\pi}_k^{f+} - \boldsymbol{\pi}_k^{f-})^\top \boldsymbol{F}^{st}\right] \Bigg\} \end{aligned} \tag{37}$$

式 (37) 中最后一个方括号内的项与 $\boldsymbol{x}$ 和 $\boldsymbol{y}$ 无关，因此可以将其从最小化问题中提取出来。此外，利用由此产生的最小化问题的可分性，式 (37) 可以等价地改写为式 (38):

$$\begin{aligned}\phi(\boldsymbol{\Pi}) = & -\left[\left(\boldsymbol{\pi}_0^{f+} - \boldsymbol{\pi}_0^{f-}\right)^\top \boldsymbol{F}^{ss} + \sum_{k\in\mathcal{K}} \left(\boldsymbol{\pi}_k^{f+} - \boldsymbol{\pi}_k^{f-}\right)^\top \boldsymbol{F}^{st}\right] \\ & + \min_{\boldsymbol{x}\in\mathcal{X}}\Big[\left(\boldsymbol{c}^\top - \boldsymbol{\pi}_0^\top \boldsymbol{M}^g\right) \boldsymbol{g}_0 - \sum_{k\in\mathcal{K}} \boldsymbol{\pi}_k^\top \boldsymbol{M}^g \boldsymbol{g}_k + (\boldsymbol{q}^{g,up})^\top \boldsymbol{r}^{g,up} \\ & + (\boldsymbol{q}^{g,dn})^\top \boldsymbol{r}^{g,dn}\Big] + \min_{\boldsymbol{y}\in\mathcal{Y}}\Big[\left(\boldsymbol{\pi}_0^\top \boldsymbol{M}^d - \boldsymbol{w}^\top\right) \boldsymbol{d}_0 + \sum_{k\in\mathcal{K}} \boldsymbol{\pi}_k^\top \boldsymbol{M}^d \boldsymbol{d}_k \\ & + (\boldsymbol{q}^{d,up})^\top \boldsymbol{r}^{d,up} + (\boldsymbol{q}^{d,dn})^\top \boldsymbol{r}^{d,dn}\Big] \\ & - \min_{\boldsymbol{\theta_0},\boldsymbol{\theta_k}} \Bigg\{ \left[\boldsymbol{\pi}_0^\top \boldsymbol{A} + \left(\boldsymbol{\pi}_0^{f+} + \boldsymbol{\pi}_0^{f-}\right)^\top\right] \boldsymbol{H}\theta_0 \\ & + \sum_{k\in\mathcal{K}} \left[\boldsymbol{\pi}_k^\top \boldsymbol{A}_k + \left(\boldsymbol{\pi}_k^{f+} + \boldsymbol{\pi}_k^{f-}\right)^\top\right] \boldsymbol{H}_k \boldsymbol{\theta}_k \Bigg\} \end{aligned} \tag{38}$$

为了便于快速参考，(38) 中与相位角相关的最后一个最小化项以紧凑的方式表示为 $\phi_\theta(\boldsymbol{\Pi})$。此外，使用其他构成 $\phi(\boldsymbol{\Pi})$ 项的逐分量形式，方程 (38) 等价地表示为:

$$\begin{aligned}\phi(\boldsymbol{\Pi}) = & -\sum_{l\in\mathcal{L}} \left[\left(\pi_{l0}^{f+} - \pi_{l0}^{f-}\right) F_l^{ss} + \sum_{k\in\mathcal{K}} \left(\pi_{lk}^{f+} - \pi_{lk}^{f-}\right) F_l^{st}\right] \\ & - \sum_{i\in\mathcal{I}} \max_{\boldsymbol{x}_i\in\mathcal{X}_i} \Bigg[\left(\pi_{b(i)0} - c_i\right) g_{i0} - q_i^{g,up} r_i^{g,up} - q_i^{g,dn} r_i^{g,dn} \\ & + \sum_{k\in\mathcal{K}} \psi_{ik}^g(\pi_{b(i)k}, \boldsymbol{x}_i) \Bigg] - \sum_{j\in\mathcal{J}} \max_{\boldsymbol{y}_j\in\mathcal{Y}_j} \Bigg[\left(w_j - \pi_{b(j)0}\right) d_{j0} \\ & - q_j^{d,up} r_j^{d,up} - q_j^{d,dn} r_j^{d,dn} - \sum_{k\in\mathcal{K}} \psi_{jk}^d(\pi_{b(j)k}, \boldsymbol{y}_j) \Bigg] - \phi_\theta(\boldsymbol{\Pi}) \end{aligned} \tag{39}$$

其中 $\boldsymbol{x}_i = \{g_{i0}, r_i^{g,up}, r_i^{g,dn}\}$ 和 $\mathcal{X}_i$ 是与 $\boldsymbol{x}_i$ 相关的约束子集，即约束 (26)–(29) 和 $\boldsymbol{x}_i \geq \boldsymbol{0}$。类似地，$\boldsymbol{y}_j = \{d_{j0}, r_j^{d,up}, r_j^{d,dn}\}$ 和 $\mathcal{Y}_j$ 表示与 $\boldsymbol{y}_j$ 相关的约束子集，即约束 (32)–(35) 和 $\boldsymbol{y}_j \geq \boldsymbol{0}$。此外，$\psi_{ik}^g(\pi_{b(i)k}, \boldsymbol{x}_i)$ 和 $\psi_{jk}^d(\pi_{b(j)k}, \boldsymbol{y}_j)$ 分别表示在应急 k 下，发电机 i 获得的收入比例以及消费者 j 的支付比例，这是由于他们针对事后应急拉格朗日乘子 π_{bk} 的最佳响应所致。

$\psi_{ik}^g(\pi_{b(i)k}, \boldsymbol{x}_i)$取决于发电机可用性状态 a_{ik}^g 以及预应急调度和备用计划 $\boldsymbol{x}_i$:

$$\psi_{ik}^g(\pi_{b(i)k}, \boldsymbol{x}_i) = \max_{g_{ik}\geq 0} \ \pi_{b(i)k} g_{ik} \tag{40}$$

subject to:

$$g_{ik} \leq a_{ik}^g (g_{i0} + r_i^{g,up}) \tag{41}$$

$$g_{ik} \geq a_{ik}^g (g_{i0} - r_i^{g,dn}) \tag{42}$$

类似地，$\psi_{jk}^d(\pi_{b(j)k}, \boldsymbol{y}_j)$ 依赖于前应急调度和备用计划 $\boldsymbol{y}_j$:

$$\psi_{jk}^d(\pi_{b(j)k}, \boldsymbol{y}_j) = \min_{d_{jk}\geq 0} \ \pi_{b(j)k} d_{jk} \tag{43}$$

subject to:

$$d_{jk} \geq d_{j0} - r_j^{d,up} \tag{44}$$

$$d_{jk} \leq d_{j0} + r_j^{d,dn} \tag{45}$$

问题 (40)–(42) 和 (43)–(45) 的最优解会导致 $\psi_{ik}^g(\pi_{b(i)k}, \boldsymbol{x}_i)$ 和 $\psi_{jk}^d(\pi_{b(j)k}, \boldsymbol{y}_j)$ 的三种可能结果:

1. 如果 π_{bk} 是正数，g_{ik}^* 和 d_{jk}^* 将分别等于在 (41) 和 (44) 中设定的上限和下限，即 b 母线的所有发电机和负荷都将愿意在 k 事故下尽可能多地部署上旋备用容量。因此:

- $g_{ik}^* = a_{ik}^g(g_{i0} + r_i^{g,up})$ 和 $\psi_{ik}^g(\pi_{b(i)k}, \boldsymbol{x}_i) = \pi_{b(i)k} a_{ik}^g(g_{i0} + r_i^{g,up})$
- $d_{jk}^* = d_{j0} - r_j^{d,up}$ 和 $\psi_{jk}^d(\pi_{b(j)k}, \boldsymbol{y}_j) = \pi_{b(j)k}(d_{j0} - r_j^{d,up})$

2. 如果 π_{bk} 为负，g_{ik}^* 和 d_{jk}^* 将分别等于 (42) 和 (45) 中设定的下限和上限，即总线 b 的所有发电机和负荷都将愿意在应急情况 k 下部署尽可能多的下旋备用。因此:

- $g_{ik}^* = a_{ik}^g(g_{i0} - r_i^{g,dn})$ and $\psi_{ik}^g(\pi_{b(i)k}, \boldsymbol{x}_i) = \pi_{b(i)k} a_{ik}^g(g_{i0} - r_i^{g,dn})$
- $d_{jk}^* = d_{j0} + r_j^{d,dn}$ and $\psi_{jk}^d(\pi_{b(j)k}, \boldsymbol{y}_j) = \pi_{b(j)k}(d_{j0} + r_j^{d,dn})$

3. 如果 π_{bk} 等于零，位于节点 b 的市场参与者的 $\psi_{ik}^g(\pi_{b(i)k}, \boldsymbol{x}_i)$ 和 $\psi_{jk}^d(\pi_{b(j)k}, \boldsymbol{y}_j)$ 都将等于零。

Moreover, defining $\pi^{+}_{bk} = \max\{\pi_{bk}, 0\}$ and $\pi^{-}_{bk} = -\min\{\pi_{bk}, 0\}$, and using the results presented above, $\psi^{g}_{ik}(\pi_{b(i)k}, \boldsymbol{x_i})$ and $\psi^{d}_{jk}(\pi_{b(j)k}, \boldsymbol{y_j})$ can be expressed as:

$$\psi^{g}_{ik}(\pi_{b(i)k}, \boldsymbol{x_i}) = \pi_{b(i)k} a^{g}_{ik} g_{i0} + \pi^{+}_{b(i)k} a^{g}_{ik} r^{g,up}_{i} + \pi^{-}_{b(i)k} a^{g}_{ik} r^{g,dn}_{i} \tag{46}$$

$$\psi^{d}_{jk}(\pi_{b(j)k}, \boldsymbol{y_j}) = \pi_{b(j)k} d_{j0} - \pi^{+}_{b(j)k} r^{d,up}_{j} - \pi^{-}_{b(j)k} r^{d,dn}_{j} \tag{47}$$

According to the strong duality theorem (Conejo et al., 1999), the minimization problem (19)–(35) and the maximization of the Lagrangian dual function (37) over $\boldsymbol{\Pi}$ yield identical values for their respective objective functions. Additionally, at the optimal dual solution $\boldsymbol{\Pi}^{*}$, the terms associated with sign-unconstrained variables $\boldsymbol{\theta_0}$ and $\boldsymbol{\theta_k}$, represented by $\phi_{\theta}(\boldsymbol{\Pi}^{*})$, are equal to zero (Bertsimas & Tsitsiklis, 1997).

Moreover, note that $\boldsymbol{\pi}^{f+}_{0}, \boldsymbol{\pi}^{f+}_{\boldsymbol{k}} \geq \mathbf{0}$ and $\boldsymbol{\pi}^{f-}_{0}, \boldsymbol{\pi}^{f-}_{\boldsymbol{k}} \leq \mathbf{0}$. The complementary slackness condition further ensures that, at the optimal dual solution, entry-wise products $\boldsymbol{\pi}^{f+*}_{0}\boldsymbol{\pi}^{f-*}_{0}$ and $\boldsymbol{\pi}^{f+*}_{\boldsymbol{k}}\boldsymbol{\pi}^{f-*}_{\boldsymbol{k}}, \forall k \in \mathcal{K}$, are all equal to $\mathbf{0}$. Thus, defining $\boldsymbol{\pi}^{f*}_{0} = \boldsymbol{\pi}^{f+*}_{0} + \boldsymbol{\pi}^{f-*}_{0}$ and $\boldsymbol{\pi}^{f*}_{\boldsymbol{k}} = \boldsymbol{\pi}^{f+*}_{\boldsymbol{k}} + \boldsymbol{\pi}^{f-*}_{\boldsymbol{k}}$, the following equalities hold:

$$\boldsymbol{\pi}^{f+*}_{0} - \boldsymbol{\pi}^{f-*}_{0} = |\boldsymbol{\pi}^{f*}_{0}| \tag{48}$$

$$\boldsymbol{\pi}^{f+*}_{\boldsymbol{k}} - \boldsymbol{\pi}^{f-*}_{\boldsymbol{k}} = |\boldsymbol{\pi}^{f*}_{\boldsymbol{k}}|, \forall k \in \mathcal{K} \tag{49}$$

Consequently, Eq. (39) at the optimal solution is recast as Eq. (50) by using Eqs. (46)–(49), dropping $\phi_{\theta}(\boldsymbol{\Pi}^{*})$, and replacing F^{st}_{l} by $\gamma_l F^{ss}_{l}$:

$$\begin{aligned}
\phi\left(\boldsymbol{\Pi}^{*}\right) = &- \sum_{l\in\mathcal{L}} \left(\left|\pi^{f*}_{l0}\right| + \sum_{k\in\mathcal{K}} \left|\pi^{f*}_{lk}\right| \gamma_l \right) F^{ss}_{l} \\
&- \sum_{i\in\mathcal{I}} \max_{\boldsymbol{x_i}\in\mathcal{X}_i} \left[\left(\pi^{*}_{b(i)0} + \sum_{k\in\mathcal{K}} \pi^{*}_{b(i)k} a^{g}_{ik} - c_i \right) g_{i0} \right. \\
&\left. + \left(\sum_{k\in\mathcal{K}} \pi^{+*}_{b(i)k} a^{g}_{ik} - q^{g,up}_{i} \right) r^{g,up}_{i} + \left(\sum_{k\in\mathcal{K}} \pi^{-*}_{b(i)k} a^{g}_{ik} - q^{g,dn}_{i} \right) r^{g,dn}_{i} \right] \\
&- \sum_{j\in\mathcal{J}} \max_{\boldsymbol{y_j}\in\mathcal{Y}_j} \left[\left(w_j - \pi^{*}_{b(j)0} - \sum_{k\in\mathcal{K}} \pi^{*}_{b(j)k} \right) d_{j0} \right. \\
&+ \left(\sum_{k\in\mathcal{K}} \pi^{+*}_{b(j)k} - q^{d,up}_{j} \right) r^{d,up}_{j} \\
&\left. + \left(\sum_{k\in\mathcal{K}} \pi^{-*}_{b(j)k} - q^{d,dn}_{j} \right) r^{d,dn}_{j} \right]
\end{aligned} \tag{50}$$

Similar to the approach described in Section 2, Eq. (50) can be recast by implicitly considering generation availability statuses a^{g}_{ik} in set $\mathcal{K}^{OFF}_{i}$ as follows:

$$\begin{aligned}
\phi\left(\boldsymbol{\Pi}^{*}\right) = &- \sum_{l\in\mathcal{L}} \left(\left|\pi^{f*}_{l0}\right| + \sum_{k\in\mathcal{K}} \left|\pi^{f*}_{lk}\right| \gamma_l \right) F^{ss}_{l} \\
&- \sum_{i\in\mathcal{I}} \max_{\boldsymbol{x_i}\in\mathcal{X}_i} \left[\left(\pi^{*}_{b(i)0} + \sum_{k\in\mathcal{K}} \pi^{*}_{b(i)k} - c_i \right) g_{i0} \right. \\
&+ \left(\sum_{k\in\mathcal{K}} \pi^{+*}_{b(i)k} - q^{g,up}_{i} \right) r^{g,up}_{i} + \left(\sum_{k\in\mathcal{K}} \pi^{-*}_{b(i)k} - q^{g,dn}_{i} \right) r^{g,dn}_{i} \\
&\left. - \sum_{k\in\mathcal{K}^{OFF}_{i}} \left(\pi^{*}_{b(i)k} g_{i0} + \pi^{+*}_{b(i)k} r^{g,up}_{i} + \pi^{-*}_{b(i)k} r^{g,dn}_{i} \right) \right] \\
&- \sum_{j\in\mathcal{J}} \max_{\boldsymbol{y_j}\in\mathcal{Y}_j} \left[\left(w_j - \pi^{*}_{b(j)0} - \sum_{k\in\mathcal{K}} \pi^{*}_{b(j)k} \right) d_{j0} \right. \\
&\left. + \left(\sum_{k\in\mathcal{K}} \pi^{+*}_{b(j)k} - q^{d,up}_{j} \right) r^{d,up}_{j} + \left(\sum_{k\in\mathcal{K}} \pi^{-*}_{b(j)k} - q^{d,dn}_{j} \right) r^{d,dn}_{j} \right]
\end{aligned} \tag{51}$$

The summation in the first term in the right-hand side of (51) represents the transmission congestion rent, which is made up of the revenues collected by transmission lines. Note that transmission lines are compensated through the Lagrange multipliers associated with the constraints bounding line power flows. More specifically, transmission prices result from the summation of $\left|\pi^{f*}_{l0}\right|$ and $\sum_{k\in\mathcal{K}} \left|\pi^{f*}_{lk}\right| \gamma_l$.

The total profit of generator i is characterized by the four terms within the first pair of square brackets in (51). The first term is the energy profit, from which the energy price at bus b is defined as $\pi^{*}_{b0} + \sum_{k\in\mathcal{K}} \pi^{*}_{bk}$, as done in the benchmark pricing system (Table 11). Interestingly, the three other generator profit terms involve salient features compared to the benchmark pricing system. First, from the second and third terms respectively related to up- and down-spinning reserves, separate prices are defined. Thus, up- and down-spinning reserve prices at bus b are equal to $\sum_{k\in\mathcal{K}} \pi^{+*}_{bk}$ and $\sum_{k\in\mathcal{K}} \pi^{-*}_{bk}$, respectively. Additionally, the fourth generator profit term models the security charge.

Finally, the terms within the second pair of square brackets in (51) represent the profit of consumer j associated with energy consumption and up- and down-spinning reserve procurement. As can be observed, the above price definitions also hold for the corresponding consumer profit terms.

It is worth noting that the maximization of the individual profit terms in (51) backs the incentive compatibility of the proposed framework.

Tables 22, 23, and 24 present the proposed pricing scheme and the corresponding generation and consumer settlements, respectively.

Tables 25–29 summarize the results from the application of the proposed pricing scheme to the two-bus example examined in Section 3.2.

As can be seen in Table 25, the proposed nodal energy prices are the same as those attained by the benchmark methodology presented

Table 22
Proposed pricing system for problem (19)–(35).

Nodal Energy Price	$p^{e}_{b} = \pi^{*}_{b0} + \sum_{k\in\mathcal{K}} \pi^{*}_{bk}$
Nodal Up-Spinning Reserve Price	$p^{up}_{b} = \sum_{k\in\mathcal{K}} \pi^{+*}_{bk}$
Nodal Down-Spinning Reserve Price	$p^{dn}_{b} = \sum_{k\in\mathcal{K}} \pi^{-*}_{bk}$
Generation Security Charge	$C^{g,s}_{i} = \sum_{k\in\mathcal{K}^{OFF}_{i}} \left(\pi^{*}_{b(i)k} g_{i0} + \pi^{+*}_{b(i)k} r^{g,up}_{i} + \pi^{-*}_{b(i)k} r^{g,dn}_{i} \right)$
Transmission Price	$p^{f}_{l} = \left|\pi^{f*}_{l0}\right| + \sum_{k\in\mathcal{K}} \left|\pi^{f*}_{lk}\right| \gamma_l$

Table 23
Generation settlement for problem (19)–(35) according to the proposed pricing system.

Generation Energy Revenue	$R^{g,e}_{i} = p^{e}_{b(i)} g_{i0}$
Generation Up-Spinning Reserve Revenue	$R^{g,up}_{i} = p^{up}_{b(i)} r^{g,up}_{i}$
Generation Down-Spinning Reserve Revenue	$R^{g,dn}_{i} = p^{dn}_{b(i)} r^{g,dn}_{i}$
Generation Total Revenue	$R^{g,t}_{i} = R^{g,e}_{i} + R^{g,up}_{i} + R^{g,dn}_{i} - C^{g,s}_{i}$
Generation Energy Cost	$C^{g,e}_{i} = c_i g_{i0}$
Generation Up-Spinning Reserve Cost	$C^{g,up}_{i} = q^{g,up}_{i} r^{g,up}_{i}$
Generation Down-Spinning Reserve Cost	$C^{g,dn}_{i} = q^{g,dn}_{i} r^{g,dn}_{i}$
Generation Total Cost	$C^{g,t}_{i} = C^{g,e}_{i} + C^{g,up}_{i} + C^{g,dn}_{i}$
Generation Profit	$Profit^{g}_{i} = R^{g,t}_{i} - C^{g,t}_{i}$

Table 24
Consumer settlement for problem (19)–(35) according to the proposed pricing system.

Consumer Payment for Energy	$CP^{e}_{j} = p^{e}_{b(j)} d_{j0}$
Consumer Up-Spinning Reserve Revenue	$R^{d,up}_{j} = p^{up}_{b(j)} r^{d,up}_{j}$
Consumer Down-Spinning Reserve Revenue	$R^{d,dn}_{j} = p^{dn}_{b(j)} r^{d,dn}_{j}$
Consumer Utility	$U^{d}_{j} = w_j d_{j0}$
Consumer Up-Spinning Reserve Cost	$C^{d,up}_{j} = q^{d,up}_{j} r^{d,up}_{j}$
Consumer Down-Spinning Reserve Cost	$C^{d,dn}_{j} = q^{d,dn}_{j} r^{d,dn}_{j}$
Consumer Total Cost	$C^{d,t}_{j} = C^{d,up}_{j} + C^{d,dn}_{j}$
Consumer Payment	$CP_j = CP^{e}_{j} - R^{d,up}_{j} - R^{d,dn}_{j}$
Consumer Profit	$Profit^{d}_{j} = U^{d}_{j} - C^{d,t}_{j} - CP_j$

此外，定义 $\pi_{bk}^{+} = \max\{\pi_{bk}, 0\}$ 和 $\pi_{bk}^{-} = -\min\{\pi_{bk}, 0\}$，并使用上面给出的结果，$\psi_{ik}^{g}(\pi_{b(i)k}, \boldsymbol{x_i})$ 和 $\psi_{jk}^{d}(\pi_{b(j)k}, \boldsymbol{y_j})$ 可以表示为：

$$\psi_{ik}^{g}(\pi_{b(i)k}, \boldsymbol{x_i}) = \pi_{b(i)k} a_{ik}^{g} g_{i0} + \pi_{b(i)k}^{+} a_{ik}^{g} r_i^{g,up} + \pi_{b(i)k}^{-} a_{ik}^{g} r_i^{g,dn} \tag{46}$$

$$\psi_{jk}^{d}(\pi_{b(j)k}, \boldsymbol{y_j}) = \pi_{b(j)k} d_{j0} - \pi_{b(j)k}^{+} r_j^{d,up} - \pi_{b(j)k}^{-} r_j^{d,dn} \tag{47}$$

根据强对偶定理（Conejo 等，1999），最小化问题 (19)–(35) 与在 $\boldsymbol{\Pi}$ 上最大化拉格朗日对偶函数 (37) 对应的目标函数值相同。此外，在最优对偶解 $\boldsymbol{\Pi}^{*}$ 时，与符号无约束变量 $\boldsymbol{\theta_0}$ 和 $\boldsymbol{\theta_k}$ 相关的项 $\phi_{\theta}(\boldsymbol{\Pi}^{*})$ 等于零（Bertsimas & Tsitsiklis，1997）。

此外，请注意 $\boldsymbol{\pi_0^{f+}}, \boldsymbol{\pi_k^{f+}} \geq \mathbf{0}$ 和 $\boldsymbol{\pi_0^{f-}}, \boldsymbol{\pi_k^{f-}} \leq \mathbf{0}$。互补松弛条件进一步确保，在最优对偶解处，每个元素的乘积 $\boldsymbol{\pi_0^{f+*}}\boldsymbol{\pi_0^{f-*}}$ 和 $\boldsymbol{\pi_k^{f+*}}\boldsymbol{\pi_k^{f-*}}, \forall k \in \mathcal{K}$ 全部等于 $\mathbf{0}$。因此，定义 $\boldsymbol{\pi_0^{f*}} = \boldsymbol{\pi_0^{f+*}} + \boldsymbol{\pi_0^{f-*}}$ 和 $\boldsymbol{\pi_k^{f*}} = \boldsymbol{\pi_k^{f+*}} + \boldsymbol{\pi_k^{f-*}}$，则以下等式成立：

$$\boldsymbol{\pi_0^{f+*}} - \boldsymbol{\pi_0^{f-*}} = |\boldsymbol{\pi_0^{f*}}| \tag{48}$$

$$\boldsymbol{\pi_k^{f+*}} - \boldsymbol{\pi_k^{f-*}} = |\boldsymbol{\pi_k^{f*}}|, \forall k \in \mathcal{K} \tag{49}$$

因此，通过使用方程 (46)–(49)、去掉 $\phi_{\theta}(\boldsymbol{\Pi}^{*})$ 并用 $\gamma_l F_l^{ss}$ 替换 F_l^{st}，方程 (39) 在最优解处被重新写为方程 (50)：

$$\begin{aligned}
\phi\left(\boldsymbol{\Pi}^{*}\right) = &- \sum_{l\in\mathcal{L}} \left(\left|\pi_{l0}^{f*}\right| + \sum_{k\in\mathcal{K}} \left|\pi_{lk}^{f*}\right| \gamma_l \right) F_l^{ss} \\
&- \sum_{i\in\mathcal{I}} \max_{\boldsymbol{x_i}\in\mathcal{X}_i} \left[\left(\pi_{b(i)0}^{*} + \sum_{k\in\mathcal{K}} \pi_{b(i)k}^{*} a_{ik}^{g} - c_i \right) g_{i0} \right. \\
&\left. + \left(\sum_{k\in\mathcal{K}} \pi_{b(i)k}^{+*} a_{ik}^{g} - q_i^{g,up} \right) r_i^{g,up} + \left(\sum_{k\in\mathcal{K}} \pi_{b(i)k}^{-*} a_{ik}^{g} - q_i^{g,dn} \right) r_i^{g,dn} \right] \\
&- \sum_{j\in\mathcal{J}} \max_{\boldsymbol{y_j}\in\mathcal{Y}_j} \left[\left(w_j - \pi_{b(j)0}^{*} - \sum_{k\in\mathcal{K}} \pi_{b(j)k}^{*} \right) d_{j0} \right. \\
&+ \left(\sum_{k\in\mathcal{K}} \pi_{b(j)k}^{+*} - q_j^{d,up} \right) r_j^{d,up} \\
&\left. + \left(\sum_{k\in\mathcal{K}} \pi_{b(j)k}^{-*} - q_j^{d,dn} \right) r_j^{d,dn} \right]
\end{aligned} \tag{50}$$

类似于第2节中描述的方法，方程(50)可以通过隐式考虑集合$\mathcal{K}_i^{OFF}$中发电可用状态a_{ik}^{g}来重新表达，如下所示：

$$\begin{aligned}
\phi\left(\boldsymbol{\Pi}^{*}\right) = &- \sum_{l\in\mathcal{L}} \left(\left|\pi_{l0}^{f*}\right| + \sum_{k\in\mathcal{K}} \left|\pi_{lk}^{f*}\right| \gamma_l \right) F_l^{ss} \\
&- \sum_{i\in\mathcal{I}} \max_{\boldsymbol{x_i}\in\mathcal{X}_i} \left[\left(\pi_{b(i)0}^{*} + \sum_{k\in\mathcal{K}} \pi_{b(i)k}^{*} - c_i \right) g_{i0} \right. \\
&+ \left(\sum_{k\in\mathcal{K}} \pi_{b(i)k}^{+*} - q_i^{g,up} \right) r_i^{g,up} + \left(\sum_{k\in\mathcal{K}} \pi_{b(i)k}^{-*} - q_i^{g,dn} \right) r_i^{g,dn} \\
&\left. - \sum_{k\in\mathcal{K}_i^{OFF}} \left(\pi_{b(i)k}^{*} g_{i0} + \pi_{b(i)k}^{+*} r_i^{g,up} + \pi_{b(i)k}^{-*} r_i^{g,dn} \right) \right] \\
&- \sum_{j\in\mathcal{J}} \max_{\boldsymbol{y_j}\in\mathcal{Y}_j} \left[\left(w_j - \pi_{b(j)0}^{*} - \sum_{k\in\mathcal{K}} \pi_{b(j)k}^{*} \right) d_{j0} \right. \\
&\left. + \left(\sum_{k\in\mathcal{K}} \pi_{b(j)k}^{+*} - q_j^{d,up} \right) r_j^{d,up} + \left(\sum_{k\in\mathcal{K}} \pi_{b(j)k}^{-*} - q_j^{d,dn} \right) r_j^{d,dn} \right]
\end{aligned} \tag{51}$$

(51)右边第一项中的求和表示输电拥塞租金，该租金由输电线路收取的收入组成。请注意，输电线路是通过与之相关的拉格朗日乘子进行补偿的

表22 问题(19)–(35)的拟议定价系统。

Nodal Energy Price	$p_b^{e} = \pi_{b0}^{*} + \sum_{k\in\mathcal{K}} \pi_{bk}^{*}$
Nodal Up-Spinning Reserve Price	$p_b^{up} = \sum_{k\in\mathcal{K}} \pi_{bk}^{+*}$
Nodal Down-Spinning Reserve Price	$p_b^{dn} = \sum_{k\in\mathcal{K}} \pi_{bk}^{-*}$
Generation Security Charge	$C_i^{g,s} = \sum_{k\in\mathcal{K}_i^{OFF}} \left(\pi_{b(i)k}^{*} g_{i0} + \pi_{b(i)k}^{+*} r_i^{g,up} + \pi_{b(i)k}^{-*} r_i^{g,dn} \right)$
Transmission Price	$p_l^{f} = \left\|\pi_{l0}^{f*}\right\| + \sum_{k\in\mathcal{K}} \left\|\pi_{lk}^{f*}\right\| \gamma_l$

表23 根据所提出的定价系统对问题(19)–(35)的代际结算。

Generation Energy Revenue	$R_i^{g,e} = p_{b(i)}^{e} g_{i0}$
Generation Up-Spinning Reserve Revenue	$R_i^{g,up} = p_{b(i)}^{up} r_i^{g,up}$
Generation Down-Spinning Reserve Revenue	$R_i^{g,dn} = p_{b(i)}^{dn} r_i^{g,dn}$
Generation Total Revenue	$R_i^{g,t} = R_i^{g,e} + R_i^{g,up} + R_i^{g,dn} - C_i^{g,s}$
Generation Energy Cost	$C_i^{g,e} = c_i g_{i0}$
Generation Up-Spinning Reserve Cost	$C_i^{g,up} = q_i^{g,up} r_i^{g,up}$
Generation Down-Spinning Reserve Cost	$C_i^{g,dn} = q_i^{g,dn} r_i^{g,dn}$
Generation Total Cost	$C_i^{g,t} = C_i^{g,e} + C_i^{g,up} + C_i^{g,dn}$
Generation Profit	$Profit_i^{g} = R_i^{g,t} - C_i^{g,t}$

表24 根据所提出的定价系统，问题(19)–(35)的消费者结算。

Consumer Payment for Energy	$CP_j^{e} = p_{b(j)}^{e} d_{j0}$
Consumer Up-Spinning Reserve Revenue	$R_j^{d,up} = p_{b(j)}^{up} r_j^{d,up}$
Consumer Down-Spinning Reserve Revenue	$R_j^{d,dn} = p_{b(j)}^{dn} r_j^{d,dn}$
Consumer Utility	$U_j^{d} = w_j d_{j0}$
Consumer Up-Spinning Reserve Cost	$C_j^{d,up} = q_j^{d,up} r_j^{d,up}$
Consumer Down-Spinning Reserve Cost	$C_j^{d,dn} = q_j^{d,dn} r_j^{d,dn}$
Consumer Total Cost	$C_j^{d,t} = C_j^{d,up} + C_j^{d,dn}$
Consumer Payment	$CP_j = CP_j^{e} - R_j^{d,up} - R_j^{d,dn}$
Consumer Profit	$Profit_j^{d} = U_j^{d} - C_j^{d,t} - CP_j$

限制约束线路功率流。更具体地说，输电价格是由 $\left|\pi_{l0}^{f*}\right|$ 和 $\sum_{k\in\mathcal{K}} \left|\pi_{lk}^{f*}\right| \gamma_l$ 的总和得出的。

发电机 i 的总利润由公式 (51) 中第一对方括号内的四个项表示。第一项是能源利润，其中 b 总线的能源价格定义为 $\pi_{b0}^{*} + \sum_{k\in\mathcal{K}} \pi_{bk}^{*}$，如基准定价系统（表 11）中所示。有趣的是，另外三个发电机利润项相比基准定价系统包含显著特征。首先，分别与上旋和下旋备用相关的第二项和第三项定义了单独的价格。因此，b 总线的上旋和下旋备用价格分别等于 $\sum_{k\in\mathcal{K}} \pi_{bk}^{+*}$ 和 $\sum_{k\in\mathcal{K}} \pi_{bk}^{-*}$。此外，第四个发电机利润项模拟了安全费用。

最后，(51) 中第二对方括号内的项表示与能源消耗及上下调储备采购相关的消费者 j 的利润。如可以观察到的，上述价格定义同样适用于相应的消费者利润项。

值得注意的是，公式 (51) 中个人利润项的最大化支持了所提出框架的激励相容性。

表 22、23 和 24 分别展示了拟议的定价方案以及相应的发电和消费者结算。

表 25–29 总结了所提出的定价方案在第 3.2 节中研究的两节点示例中的应用结果。

如表25所示，所提出的节点能源价格与所呈现的基准方法所获得的价格相同

Table 25
Two-bus example – Proposed nodal energy and reserve prices ($/MWh).

b	Nodal Energy Price, p_b^e	Nodal Up-Spinning Reserve Price, p_b^{up}	Nodal Down-Spinning Reserve Price, p_b^{dn}
1	200	180	0
2	100	85	5

Table 26
Two-bus example – Values of $\boldsymbol{\pi}_0^{f*}$, $\boldsymbol{\pi}_{\boldsymbol{k}}^{f*}$, and transmission price ($/MWh).

l	$\boldsymbol{\pi}_0^{f*}$	$\boldsymbol{\pi}_{\boldsymbol{k}}^{f*}$				Transmission Price, p_l^f
		Outage of Generator 1	Outage of Generator 2	Outage of Generator 3	Outage of Line 1–2	
1–2	0	95	0	0	–	95

Table 27
Two-bus example – Proposed generation settlement ($).

i	$R_i^{g,e}$	$R_i^{g,up}$	$R_i^{g,dn}$	$C_i^{g,s}$	$R_i^{g,t}$	$C_i^{g,e}$	$C_i^{g,up}$	$C_i^{g,dn}$	$C_i^{g,t}$	$Profit_i^g$
1	15,000	0	0	13,500	1,500	1,500	0	0	1,500	0
2	3,000	2,550	25	0	5,575	1,500	150	25	1,675	3,900
3	1,500	2,975	0	0	4,475	1,500	350	0	1,850	2,625
Total	**19,500**	**5,525**	**25**	**13,500**	**11,550**	**4,500**	**500**	**25**	**5,025**	**6,525**

Table 28
Two-bus example – Proposed consumer settlement ($).

j	CP_j^e	$R_j^{d,up}$	$R_j^{d,dn}$	CP_j	U_j^d	$C_j^{d,up}$	$C_j^{d,dn}$	$C_j^{d,t}$	$Profit_j^d$
1	16,000	1,800	0	14,200	16,000	1,500	0	1,500	300
2	4,000	0	0	4,000	6,000	0	0	0	2,000
Total	**20,000**	**1,800**	**0**	**18,200**	**22,000**	**1,500**	**0**	**1,500**	**2,300**

Table 29
Two-bus example – Proposed system settlement ($).

Generation Revenue	11,550
Transmission Revenue	6,650
Consumer Payment	18,200
Balance	0

in Arroyo and Galiana (2005), thereby leading to identical energy revenues for generators and energy payments by consumers. By contrast, the use of different nodal prices for up- and down-spinning reserves (Table 25) constitutes a significant departure from the single nodal price defined in the benchmark pricing system for both services, thus yielding substantial differences for the corresponding revenues. For this particular example, the comparison of Tables 27 and 19 shows that generators 2 and 3 increase their up-spinning reserve revenues by 6.25% whereas the down-spinning reserve revenue of generator 2 decreases by a factor of 16. Additionally, as generator 1 is responsible for the need for up-spinning reserve, a security charge is levied on this generator.

As for consumers, the settlement remains unaltered for this particular example (Tables 28 and 20). Note that the only load contributing to security is load 1 at bus 1, in the form of up-spinning reserve. For this bus, the proposed up-spinning reserve price happens to be the same as the security price resulting from the benchmark method, thereby giving rise to identical revenues.

Moreover, unlike the benchmark pricing system, line congestion under contingency is acknowledged by the proposed transmission price p_l^f (Table 26) and the corresponding transmission revenue, $p_l^f F_l^{ss}$, which, for this case, amounts to $95 \times 70 = \$6,650$.

As expected, the total generation profit, the total consumer profit, and the transmission revenue sum up the aforementioned optimal social welfare, i.e., $15,475. Remarkably, different from the benchmark pricing scheme, the proposed method renders the total consumer payment equal to the sum of the generation revenue and the transmission revenue (Table 29), as is desirable. These results provide empirical support for the cost recovery and system revenue adequacy featured by the proposed pricing scheme for the market-clearing problem analyzed in this section. The corresponding extensions of Theorems 1 and 2 and their proofs are provided next.

Theorem 3 (*Cost Recovery*). *The proposed pricing framework ensures that generators' and consumers' profits are non-negative.*

Proof. Let us rewrite the LD function in Eq. (51) as follows:

$$\begin{aligned}\phi\left(\boldsymbol{\Pi}^*\right) = &-\sum_{l\in\mathcal{L}}\left(\left|\pi_{l0}^{f*}\right| + \sum_{k\in\mathcal{K}}\left|\pi_{lk}^{f*}\right|\gamma_l\right)F_l^{ss} \\ &- \sum_{i\in\mathcal{I}}\max_{\boldsymbol{x_i}\in\mathcal{X}_i}\left[Profit_i^g\left(\boldsymbol{\Pi}_{\boldsymbol{b(i)}}^*, \boldsymbol{x_i}\right)\right] \\ &- \sum_{j\in\mathcal{J}}\max_{\boldsymbol{y_j}\in\mathcal{Y}_j}\left[Profit_j^d\left(\boldsymbol{\Pi}_{\boldsymbol{b(j)}}^*, \boldsymbol{y_j}\right)\right]\end{aligned} \tag{52}$$

The terms $Profit_i^g\left(\boldsymbol{\Pi}_{\boldsymbol{b(i)}}^*, \boldsymbol{x_i}\right)$ and $Profit_j^d\left(\boldsymbol{\Pi}_{\boldsymbol{b(j)}}^*, \boldsymbol{y_j}\right)$ respectively represent the profits of generator i and consumer j. It should be noted that $Profit_i^g\left(\boldsymbol{\Pi}_{\boldsymbol{b(i)}}^*, \boldsymbol{x_i}\right)$ and $Profit_j^d\left(\boldsymbol{\Pi}_{\boldsymbol{b(j)}}^*, \boldsymbol{y_j}\right)$ are linear objective functions respectively maximized over $\boldsymbol{x_i}$ and $\boldsymbol{y_j}$ with $\boldsymbol{x_i} \geq \boldsymbol{0}$ and $\boldsymbol{y_j} \geq \boldsymbol{0}$. Therefore, at the optimal solution, $Profit_i^g\left(\boldsymbol{\Pi}_{\boldsymbol{b(i)}}^*, \boldsymbol{x_i^*}\right)$ and $Profit_j^d\left(\boldsymbol{\Pi}_{\boldsymbol{b(j)}}^*, \boldsymbol{y_j^*}\right)$ are both non-negative. In other words, if $\pi_{b(i)0}^*$ and $\pi_{b(i)k}^*$ fail to cover generator i's costs, the production and reserve contributions of this generator will be 0, i.e., $g_{i0}^* = r_i^{g,up*} = r_i^{g,dn*} = 0$. Analogously, if $\pi_{b(j)0}^*$ and $\pi_{b(j)k}^*$ are not profitable for consumer j, the consumption and reserve contributions of this consumer will be 0, i.e., $d_{j0}^* = r_j^{d,up*} = r_j^{d,dn*} = 0$. Therefore:

$$Profit_i^g\left(\boldsymbol{\Pi}_{\boldsymbol{b(i)}}^*, \boldsymbol{x_i^*}\right) \geq 0, \ \forall i \in \mathcal{I} \tag{53}$$

$$Profit_j^d\left(\boldsymbol{\Pi}_{\boldsymbol{b(j)}}^*, \boldsymbol{y_j^*}\right) \geq 0, \ \forall j \in \mathcal{J} \tag{54}$$

Thus, the theorem holds. □

Theorem 4 (*System Revenue Adequacy*). *The total payment by consumers is equal to the sum of generation and transmission revenues.*

Proof. By the strong duality theorem, the optimal value of the primal objective function in (19) is equal to the optimal value of the LD function in (51). Thus, at the optimal solution $\boldsymbol{x}^*$, $\boldsymbol{y}^*$, and $\boldsymbol{\Pi}^*$, we have:

$$\begin{aligned}&\sum_{i\in\mathcal{I}} c_i g_{i0}^* + \sum_{i\in\mathcal{I}} q_i^{g,up} r_i^{g,up*} + \sum_{i\in\mathcal{I}} q_i^{g,dn} r_i^{g,dn*} - \sum_{j\in\mathcal{J}} w_j d_{j0}^* \\ &+ \sum_{j\in\mathcal{J}} q_j^{d,up} r_j^{d,up*} + \sum_{j\in\mathcal{J}} q_j^{d,dn} r_j^{d,dn*} = \\ &-\sum_{l\in\mathcal{L}}\left(|\pi_{l0}^{f*}| + \sum_{k\in\mathcal{K}}|\pi_{lk}^{f*}|\gamma_l\right)F_l^{ss} - \sum_{i\in\mathcal{I}}\Bigg[\left(\pi_{b(i)0}^* + \sum_{k\in\mathcal{K}}\pi_{b(i)k}^* - c_i\right)g_{i0}^* \\ &+ \left(\sum_{k\in\mathcal{K}}\pi_{b(i)k}^{+*} - q_i^{g,up}\right)r_i^{g,up*} + \left(\sum_{k\in\mathcal{K}}\pi_{b(i)k}^{-*} - q_i^{g,dn}\right)r_i^{g,dn*} \\ &- \sum_{k\in\mathcal{K}_i^{\mathrm{OFF}}}\left(\pi_{b(i)k}^* g_{i0}^* + \pi_{b(i)k}^{+*} r_i^{g,up*} + \pi_{b(i)k}^{-*} r_i^{g,dn*}\right)\Bigg] \\ &- \sum_{j\in\mathcal{J}}\Bigg[\left(w_j - \pi_{b(j)0}^* - \sum_{k\in\mathcal{K}}\pi_{b(j)k}^*\right)d_{j0}^* \\ &+ \left(\sum_{k\in\mathcal{K}}\pi_{b(j)k}^{+*} - q_j^{d,up}\right)r_j^{d,up*} + \left(\sum_{k\in\mathcal{K}}\pi_{b(j)k}^{-*} - q_j^{d,dn}\right)r_j^{d,dn*}\Bigg]\end{aligned} \tag{55}$$

表25 两母线示例 – 拟议的节点能量和储备价格（$/MWh）。

b	Nodal Energy Price, p_b^e	Nodal Up-Spinning Reserve Price, p_b^{up}	Nodal Down-Spinning Reserve Price, p_b^{dn}
1	200	180	0
2	100	85	5

表26
两母线示例 – $\boldsymbol{\pi}^{f*}$ 的数值 $_0$，$\boldsymbol{\pi}_{\boldsymbol{k}}^{f*}$，以及传输价格（美元/兆瓦时）。

l	π_0^{f*}	$\boldsymbol{\pi}_{\boldsymbol{k}}^{f*}$				Transmission Price, p_l^f
		Outage of Generator 1	Outage of Generator 2	Outage of Generator 3	Outage of Line 1–2	
1–2	0	95	0	0	–	95

表27
双母线示例 – 提议的发电结算（$）。

i	$R_i^{g,e}$	$R_i^{g,up}$	$R_i^{g,dn}$	$C_i^{g,s}$	$R_i^{g,t}$	$C_i^{g,e}$	$C_i^{g,up}$	$C_i^{g,dn}$	$C_i^{g,t}$	$Profit_i^g$
1	15,000	0	0	13,500	1,500	1,500	0	0	1,500	0
2	3,000	2,550	25	0	5,575	1,500	150	25	1,675	3,900
3	1,500	2,975	0	0	4,475	1,500	350	0	1,850	2,625
Total	**19,500**	**5,525**	**25**	**13,500**	**11,550**	**4,500**	**500**	**25**	**5,025**	**6,525**

表28
双母线示例 – 提议的消费者结算（$）。

j	CP_j^e	$R_j^{d,up}$	$R_j^{d,dn}$	CP_j	U_j^d	$C_j^{d,up}$	$C_j^{d,dn}$	$C_j^{d,t}$	$Profit_j^d$
1	16,000	1,800	0	14,200	16,000	1,500	0	1,500	300
2	4,000	0	0	4,000	6,000	0	0	0	2,000
Total	**20,000**	**1,800**	**0**	**18,200**	**22,000**	**1,500**	**0**	**1,500**	**2,300**

表29
双母线示例 – 提议的系统结算（美元）。

Generation Revenue	11,550
Transmission Revenue	6,650
Consumer Payment	18,200
Balance	0

在Arroyo和Galiana（2005）中，从而导致发电机的能源收入和消费者的能源支付相同。相比之下，使用不同的节点价格用于调频上旋和下旋储备（表25）与基准定价系统中为两种服务定义的单一节点价格存在显著差异，从而导致相关收入产生实质性差异。对于这个特定例子，比较表27和表19显示，发电机2和3的调频上旋储备收入增加了6.25%，而发电机2的调频下旋储备收入减少了16倍。此外，由于发电机1负责调频上旋储备的需求，将对该发电机征收安全费用。

至于消费者，对于这个特定的例子（表28和表20），结算保持不变。请注意，唯一对安全有贡献的负荷是在母线1的负荷1，以旋转备用的形式存在。对于该母线，拟议的旋转备用价格恰好与基准方法产生的安全价格相同，从而导致相同的收入。

此外，与基准定价系统不同，所提出的输电价格 p_l^f（表 26）承认了应急情况下的线路拥塞，以及相应的输电收入 $p_l^f F_l^{ss}$，对于本案例，该收入为 $95 \times 70 = \$6,650$。

正如预期，总发电利润、总消费者利润和输电收入加起来就是上述的最优社会福利，即 $15,475。值得注意的是，与基准不同定价方案中，所提出的方法使总消费者支付等于发电收入和输电收入之和（表29），这是理想的。这些结果为所提出的市场清算问题定价方案在成本回收和系统收入充足性方面提供了实证支持。接下来提供了定理1和定理2的相应扩展及其证明。

定理 3（*Cost Recovery*）。*The proposed pricing framework ensures that generators' and consumers' profits are non-negative.*

证明。我们将等式 (51) 中的 LD 函数重写如下：

$$\begin{aligned}\phi\left(\boldsymbol{\Pi}^*\right) = &-\sum_{l\in\mathcal{L}}\left(\left|\pi_{l0}^{f*}\right| + \sum_{k\in\mathcal{K}}\left|\pi_{lk}^{f*}\right|\gamma_l\right)F_l^{ss} \\ &- \sum_{i\in\mathcal{I}}\max_{\boldsymbol{x}_i\in\mathcal{X}_i}\left[Profit_i^g\left(\boldsymbol{\Pi}_{\boldsymbol{b(i)}}^*, \boldsymbol{x}_i\right)\right] \\ &- \sum_{j\in\mathcal{J}}\max_{\boldsymbol{y}_j\in\mathcal{Y}_j}\left[Profit_j^d\left(\boldsymbol{\Pi}_{\boldsymbol{b(j)}}^*, \boldsymbol{y}_j\right)\right]\end{aligned} \tag{52}$$

$Profit_i^g\left(\boldsymbol{\Pi}_{\boldsymbol{b(i)}}^*, \boldsymbol{x}_i\right)$ 和 $Profit_j^d\left(\boldsymbol{\Pi}_{\boldsymbol{b(j)}}^*, \boldsymbol{y}_j\right)$ 分别表示发电机 i 和消费者 j 的利润。需要注意的是，$Profit_i^g\left(\boldsymbol{\Pi}_{\boldsymbol{b(i)}}^*, \boldsymbol{x}_i\right)$ 和 $Profit_j^d\left(\boldsymbol{\Pi}_{\boldsymbol{b(j)}}^*, \boldsymbol{y}_j\right)$ 是分别在 $\boldsymbol{x}_i$ 和 $\boldsymbol{y}_j$ 以及 $\boldsymbol{x}_i \geq \boldsymbol{0}$ 和 $\boldsymbol{y}_j \geq \boldsymbol{0}$ 上最大化的线性目标函数。因此，在最优解下，$Profit_i^g\left(\boldsymbol{\Pi}_{\boldsymbol{b(i)}}^*, \boldsymbol{x}_i^*\right)$ 和 $Profit_j^d\left(\boldsymbol{\Pi}_{\boldsymbol{b(j)}}^*, \boldsymbol{y}_j^*\right)$ 都是非负的。换句话说，如果 $\pi_{b(i)0}^*$ 和 $\pi_{b(i)k}^*$ 无法覆盖发电机 i 的成本，该发电机的发电和备用贡献将为 0，即 $g_{i0}^* = r_i^{g,up*} = r_i^{g,dn*} = 0$。类似地，如果 $\pi_{b(j)0}^*$ 和 $\pi_{b(j)k}^*$ 对消费者 j 没有利润，该消费者的消耗和备用贡献将为 0，即 $d_{j0}^* = r_j^{d,up*} = r_j^{d,dn*} = 0$。因此：

$$Profit_i^g\left(\boldsymbol{\Pi}_{\boldsymbol{b(i)}}^*, \boldsymbol{x}_i^*\right) \geq 0,\ \forall i \in \mathcal{I} \tag{53}$$

$$Profit_j^d\left(\boldsymbol{\Pi}_{\boldsymbol{b(j)}}^*, \boldsymbol{y}_j^*\right) \geq 0,\ \forall j \in \mathcal{J} \tag{54}$$

因此，该定理成立。□

定理4（*System Revenue Adequacy*）。*The total payment by consumers is equal to the sum of generation and transmission revenues.*

证明。根据强对偶定理，(19)式中原问题目标函数的最优值等于(51)式中对偶拉格朗日函数的最优值。因此，在最优解 $\boldsymbol{x}^*$、$\boldsymbol{y}^*$ 和 $\boldsymbol{\Pi}^*$ 处，我们有：

$$\begin{aligned}&\sum_{i\in\mathcal{I}} c_i g_{i0}^* + \sum_{i\in\mathcal{I}} q_i^{g,up} r_i^{g,up*} + \sum_{i\in\mathcal{I}} q_i^{g,dn} r_i^{g,dn*} - \sum_{j\in\mathcal{J}} w_j d_{j0}^* \\ &+ \sum_{j\in\mathcal{J}} q_j^{d,up} r_j^{d,up*} + \sum_{j\in\mathcal{J}} q_j^{d,dn} r_j^{d,dn*} = \\ &- \sum_{l\in\mathcal{L}}\left(|\pi_{l0}^{f*}| + \sum_{k\in\mathcal{K}}|\pi_{lk}^{f*}|\gamma_l\right)F_l^{ss} - \sum_{i\in\mathcal{I}}\Bigg[\left(\pi_{b(i)0}^* + \sum_{k\in\mathcal{K}}\pi_{b(i)k}^* - c_i\right)g_{i0}^* \\ &+ \left(\sum_{k\in\mathcal{K}}\pi_{b(i)k}^{+*} - q_i^{g,up}\right)r_i^{g,up*} + \left(\sum_{k\in\mathcal{K}}\pi_{b(i)k}^{-*} - q_i^{g,dn}\right)r_i^{g,dn*} \\ &- \sum_{k\in\mathcal{K}_i^{\mathrm{OFF}}}\left(\pi_{b(i)k}^* g_{i0}^* + \pi_{b(i)k}^{+*} r_i^{g,up*} + \pi_{b(i)k}^{-*} r_i^{g,dn*}\right)\Bigg] \\ &- \sum_{j\in\mathcal{J}}\Bigg[\left(w_j - \pi_{b(j)0}^* - \sum_{k\in\mathcal{K}}\pi_{b(j)k}^*\right)d_{j0}^* \\ &+ \left(\sum_{k\in\mathcal{K}}\pi_{b(j)k}^{+*} - q_j^{d,up}\right)r_j^{d,up*} + \left(\sum_{k\in\mathcal{K}}\pi_{b(j)k}^{-*} - q_j^{d,dn}\right)r_j^{d,dn*}\Bigg]\end{aligned} \tag{55}$$

By canceling out identical terms in both sides of Eq. (55) and rearranging terms, we simplify the expression as follows:

$$\begin{aligned}
&\sum_{j\in\mathcal{J}}\left[\left(\pi^{*}_{b(j)0}+\sum_{k\in\mathcal{K}}\pi^{*}_{b(j)k}\right)d^{*}_{j0}-\sum_{k\in\mathcal{K}}\pi^{+*}_{b(j)k}r^{d,up*}_{j}-\sum_{k\in\mathcal{K}}\pi^{-*}_{b(j)k}r^{d,dn*}_{j}\right]=\\
&\sum_{l\in\mathcal{L}}\left(|\pi^{f*}_{l0}|+\sum_{k\in\mathcal{K}}|\pi^{f*}_{lk}|\gamma_{l}\right)F^{ss}_{l}+\sum_{i\in\mathcal{I}}\left[\left(\pi^{*}_{b(i)0}+\sum_{k\in\mathcal{K}}\pi^{*}_{b(i)k}\right)g^{*}_{i0}\right.\\
&\quad+\sum_{k\in\mathcal{K}}\pi^{+*}_{b(i)k}r^{g,up*}_{i}+\sum_{k\in\mathcal{K}}\pi^{-*}_{b(i)k}r^{g,dn*}_{i}\\
&\quad\left.-\sum_{k\in\mathcal{K}^{\mathrm{OFF}}_{i}}\left(\pi^{*}_{b(i)k}g^{*}_{i0}+\pi^{+*}_{b(i)k}r^{g,up*}_{i}+\pi^{-*}_{b(i)k}r^{g,dn*}_{i}\right)\right]
\end{aligned}\tag{56}$$

Therefore, at the optimal solution, the sum of all consumer payments equals the sum of all generator and transmission line revenues. □

4. 300-Bus case study

To assess the feasibility and robustness of the proposed pricing framework, we examine a larger case study based on a modified version of the standard IEEE 300-bus test system (Matpower, 2026). This system comprises 300 buses, 69 generators, and 822 transmission lines. Thus, 891 contingency states are considered, corresponding to the outage of every single line and every single generator. For the sake of reproducibility, system data are available in Ribeiro et al. (2025).

The optimal market-clearing solution attained by the simplex algorithm of CPLEX features a total value of the objective function equal to −$8,087,856.74. In the pre-contingency state, the market-clearing process balances total generation and price-responsive demand at 17,991.93 MW. This value represents the total power cleared in the system, reflecting both the aggregate supply and the corresponding consumption level.

For this case, both the proposed approach and the benchmark pricing system described in Arroyo and Galiana (2005) yield identical energy prices, which are shown in Fig. 2. To provide insight into the effect of contingencies on pricing, we have also implemented the contingency-unconstrained market-clearing counterpart associated with an $N-0$ security criterion, for which no reserves are needed. The optimal $N-0$ dispatch, which is 85% cheaper, as expected, yields no transmission congestion and, hence, no congestion rents, giving rise to a uniform nodal energy price of $500/MWh. The comparison of this result with the nodal energy prices shown in Fig. 2 reveals that neglecting security does not necessarily lead to lower energy prices, as also reported in Arroyo and Galiana (2005).

In contrast to energy, reserves are priced differently by both pricing frameworks. It should be noted that the up- and down-spinning reserve prices identified by the proposed scheme all are non-negative, whereas the benchmark method used for assessment purposes gives rise to negative security prices for some buses. Table 30 summarizes the results for the units with the largest discrepancies in down-spinning reserve remuneration. To facilitate a direct comparison, results from both methods are reported side by side, with their respective framework labels indicated in the first row of the table. More specifically, for each unit, this table lists (1) the allocated down-spinning reserve, (2) the corresponding nodal security price and the down-spinning reserve revenue both resulting from the benchmark scheme, and (3) the respective nodal down-spinning reserve price and down-spinning reserve revenue determined by the proposed approach. In the benchmark pricing method, negative security prices arise from their definition as the sum of Lagrange multipliers across all contingency states (Table 11). When the system is primarily constrained by the need to absorb excess generation, i.e., when downward reserve requirements dominate, these multipliers become negative, potentially leading to negative security prices. Table 30 shows that, under the benchmark scheme, units 11 and 31 receive significantly higher remuneration, while units 16, 28, and 56 collect negative revenues. These results underscore the critical importance of one of the salient aspects of the proposed approach, namely separately pricing upward and downward reserves according to their respective opportunity costs, thereby avoiding the distortions associated with a single aggregated security price.

Table 30
300-bus example – Generation down-spinning reserve results.

i	$r^{g,dn}_{i}$ (MW)	Benchmark Pricing System		Proposed Pricing System	
		$p^{s}_{b(i)}$ ($/MWh)	$R^{g,dn}_{i}$ ($)	$p^{dn}_{b(i)}$ ($/MWh)	$R^{g,dn}_{i}$ ($)
11	255.45	34.12	8,715.75	15.00	3,831.73
31	200.00	33.00	6,600.00	17.00	3,400.00
16	318.32	−6.00	−1,909.92	6.00	1,909.92
56	159.77	−7.69	−1,002.20	12.00	1,564.33
28	72.50	−6.00	−958.60	6.00	958.60

Fig. 3 shows the impact of both pricing systems on generation profits. Note that, for each generator, the profit difference depicted in Fig. 3 is computed as the relative difference between the profit attained under the proposed pricing scheme and the profit determined by the benchmark pricing system, expressed as a percentage of the benchmark profit. For expository purposes, generators are sorted along the horizontal axis of Fig. 3 in descending order of this percentage difference. Moreover, generators with negligible benchmark profits are excluded from this figure, as small absolute differences would otherwise translate into disproportionately large percentage variations. According to Fig. 3, substantial profit disparities arise, including percentage differences close to −100%, which correspond to generators with non-negative profits under the proposed pricing scheme and negative profits under the benchmark framework. Thus, this example is useful to illustrate another key advantage of the proposed pricing method, namely cost recovery. As expected, the proposed approach yields non-negative profits for all generators, unlike the benchmark pricing system, which leads to negative profits for 24 units totaling −$24,529.63.

Fig. 4 depicts the security charge along with the energy and reserve revenues identified by the proposed pricing mechanism for each generator. In order to better illustrate the effect of the cost-causation principle, pre-contingency generation dispatch levels from highest to lowest are provided in the horizontal axis. It should be noted that generators operating at higher pre-contingency dispatch levels tend to face higher security charges, since their outage imposes greater stress on the system, thereby requiring larger reserve procurement. Relevant exceptions to this trend are shown in Fig. 4, though, mainly stemming from the crucial impact of the network topology on the extent to which a generator's failure restricts system operation.

As for transmission, the proposed approach identified 106 lines with non-zero transmission revenues. It is worth highlighting that 90 of those lines, i.e., 84.9%, featured flow capacity constraints that are non-binding in the pre-contingency state but become binding under contingency. This result further illustrates that the contingency-constrained market-clearing model under consideration ensures that the transmission system has capacity reserved to ensure feasibility under contingency conditions. Fig. 5 presents the non-zero transmission revenues identified by the proposed framework for such pre-contingency uncongested lines. For illustration purposes, the horizontal axis displays the corresponding line indices, ordered by revenue magnitude. Notably, these lines collect revenues by operating in the pre-contingency state below their capacity, which is then fully utilized to accommodate feasible post-contingency flows, thereby ensuring reserve deliverability throughout the network under contingency conditions. Thus, in contrast to the benchmark pricing system, the proposed pricing scheme captures the system's opportunity costs associated with the scarce capacity of these lines and, consequently, extends the concept of transmission rent allocation to contingency states under the general marginal cost principle.

通过消去方程（55）两边的相同项并重新排列各项，我们将表达式简化如下：

$$\sum_{j\in\mathcal{J}}\left[\left(\pi^{*}_{b(j)0}+\sum_{k\in\mathcal{K}}\pi^{*}_{b(j)k}\right)d^{*}_{j0}-\sum_{k\in\mathcal{K}}\pi^{+*}_{b(j)k}r^{d,up*}_{j}-\sum_{k\in\mathcal{K}}\pi^{-*}_{b(j)k}r^{d,dn*}_{j}\right]=$$
$$\sum_{l\in\mathcal{L}}\left(|\pi^{f*}_{l0}|+\sum_{k\in\mathcal{K}}|\pi^{f*}_{lk}|\gamma_{l}\right)F^{ss}_{l}+\sum_{i\in\mathcal{I}}\left[\left(\pi^{*}_{b(i)0}+\sum_{k\in\mathcal{K}}\pi^{*}_{b(i)k}\right)g^{*}_{i0}\right.$$
$$+\sum_{k\in\mathcal{K}}\pi^{+*}_{b(i)k}r^{g,up*}_{i}+\sum_{k\in\mathcal{K}}\pi^{-*}_{b(i)k}r^{g,dn*}_{i}$$
$$\left.-\sum_{k\in\mathcal{K}^{\mathrm{OFF}}_{i}}\left(\pi^{*}_{b(i)k}g^{*}_{i0}+\pi^{+*}_{b(i)k}r^{g,up*}_{i}+\pi^{-*}_{b(i)k}r^{g,dn*}_{i}\right)\right] \tag{56}$$

因此，在最优解处，所有消费者支付的总和等于所有发电机和输电线路收入的总和。 □

4. 300母线案例研究

为了评估所提出的定价框架的可行性和稳健性，我们基于标准 IEEE 300 母线测试系统的修改版本（Matpower, 2026）进行更大规模的案例研究。该系统包含 300 个母线、69 个发电机和 822 条输电线路。因此，考虑了 891 种应急状态，对应于每条线路和每台发电机的单独停运。为了便于重复性，系统数据可参见 Ribeiro 等（2025）。

CPLEX 单纯形算法获得的最优市场清算解的目标函数总值为 −$8,087,856.74。在预应急状态下，市场清算过程将总发电量与价格响应需求平衡在 17,991.93 兆瓦。该值表示系统中的总清算电力，反映了总供应量及相应的消费水平。

对于这种情况，所提出的方法和 Arroyo 与 Galiana (2005) 中描述的基准定价系统产生的能源价格相同，如图 2 所示。为了深入了解突发事件对定价的影响，我们还实现了与 $N-0$ 安全标准相关的无突发事件约束的市场清算方法，其中无需储备。最优的 $N-0$ 调度，如预期的那样便宜 85%，不会产生输电拥堵，因此也不会有拥堵租金，从而产生统一的节点能源价格 $500/MWh。将此结果与图 2 所示的节点能源价格进行比较表明，忽略安全性并不一定会导致能源价格降低，Arroyo 与 Galiana (2005) 也有类似报道。

与能源相比，备用容量在两种定价框架下的定价方式不同。需要注意的是，所提方案确定的上旋和下旋备用容量价格均为非负值，而用于评估的基准方法会导致某些母线的备用容量价格为负值。表30总结了下旋备用容量报酬差异最大的机组结果。为了便于直接比较，表中并列报告了两种方法的结果，并在表格第一行标明各自的框架标签。更具体地说，对于每个机组，本表列出了(1) 分配的下旋备用容量，(2) 基准方案下相应的节点安全价格和下旋备用容量收入，以及(3) 所提方法确定的对应节点下旋备用容量价格和下旋备用容量收入。在基准定价方法中

表30 300母线示例 – 发电下旋备用容量结果。

i	$r^{g,dn}_{i}$ (MW)	Benchmark Pricing System		Proposed Pricing System	
		$p^{s}_{b(i)}$ ($/MWh)	$R^{g,dn}_{i}$ ($)	$p^{dn}_{b(i)}$ ($/MWh)	$R^{g,dn}_{i}$ ($)
11	255.45	34.12	8,715.75	15.00	3,831.73
31	200.00	33.00	6,600.00	17.00	3,400.00
16	318.32	−6.00	−1,909.92	6.00	1,909.92
56	159.77	−7.69	−1,002.20	12.00	1,564.33
28	72.50	−6.00	−958.60	6.00	958.60

并且56项收取负收益。这些结果强调了所提出方法的一个显著方面的关键重要性，即根据各自的机会成本分别定价上行和下行备用，从而避免与单一聚合证券价格相关的扭曲。

图3显示了两种定价系统对发电利润的影响。请注意，对于每个发电机，图3中显示的利润差异是根据在拟议定价方案下获得的利润与基准定价系统确定的利润之间的相对差异计算的，并以基准利润的百分比表示。出于说明目的，发电机在图3的水平轴上按该百分比差异的降序排列。此外，基准利润可忽略不计的发电机被排除在此图之外，因为较小的绝对差异会导致比例上过大的百分比变化。根据图3，利润存在显著差异，包括接近−100%的百分比差异，这对应于在拟议定价方案下利润为非负而在基准框架下利润为负的发电机。因此，这个例子有助于说明

图 4 展示了所提出的定价机制为每个发电机确定的安全费用以及能源和备用收入。为了更好地说明成本归因原理的影响，水平轴上给出了从高到低的事故前发电机出力水平。需要注意的是，事故前出力较高的发电机往往面临更高的安全费用，因为它们的停运对系统施加的压力更大，从而需要更多的备用采购。然而，图 4 也显示了这一趋势的相关例外，主要源于网络拓扑对发电机故障限制系统运行程度的关键影响。

关于输电，所提出的方法识别出了106条具有非零输电收入的线路。值得强调的是，这些线路中有90条，即84.9%，在事故前状态下其流量容量约束不具约束性，但在事故发生时变为约束性。这一结果进一步说明，所考虑的受事故约束的市场清算模型确保输电系统预留容量，以保证在事故情况下的可行性。图5展示了所提出框架为这些事故前未拥堵线路识别出的非零输电收入。为了便于说明，横轴显示按收入大小排序的对应线路索引。值得注意的是，这些线路通过在事故前状态下低于其容量运行来获取收入，然后其容量被充分利用以容纳事故后可行的潮流，从而确保了备用容量。

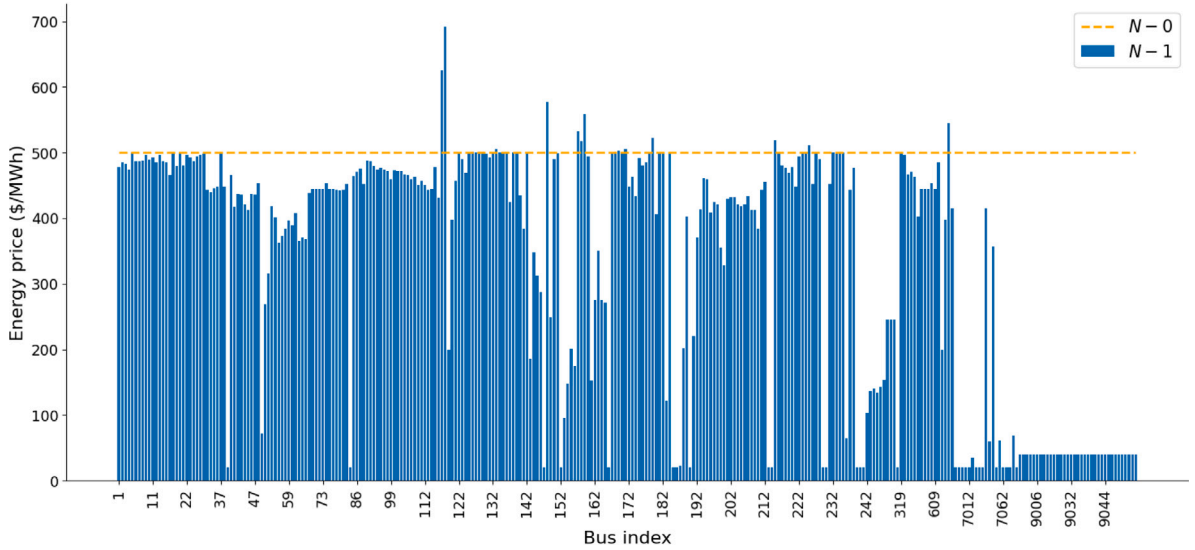


Fig. 2. 300-bus example – Nodal energy prices.

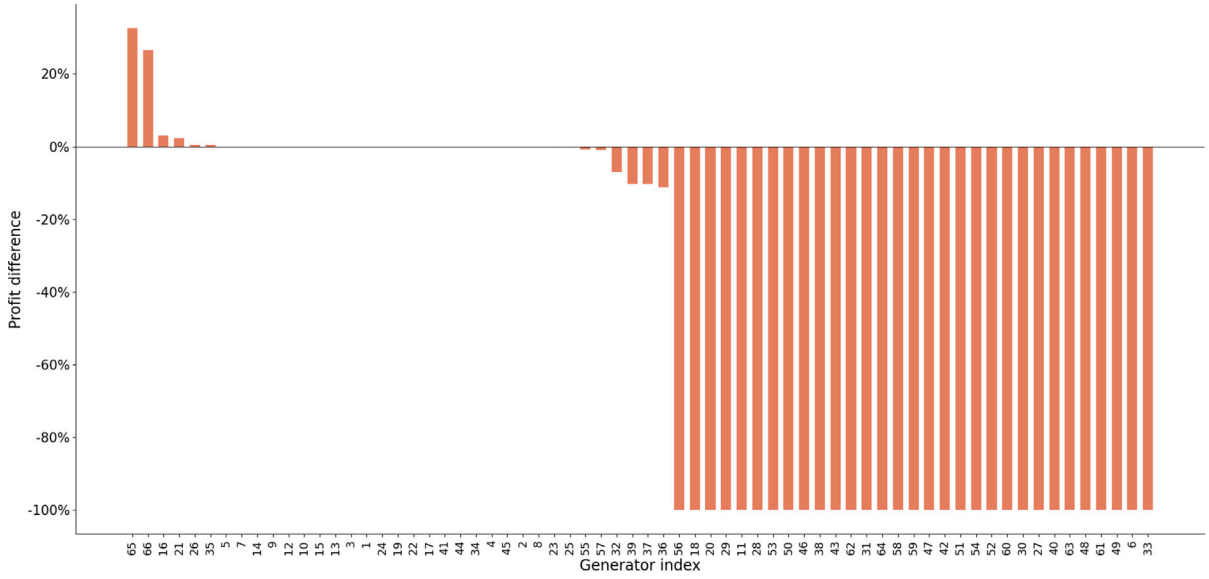


Fig. 3. 300-bus example – Percentage differences in generation profits.

Finally, Table 31 provides the system settlement for both pricing schemes. According to Table 31, consumers pay $9,957.69 less under the proposed pricing scheme than under the benchmark framework, while the total generation revenue is $208,229.00 lower, yet no unit incurs negative profit. These results are particularly relevant from the perspective of consumers. As previously discussed, the benchmark pricing scheme yields negative profits for a subset of generators, which would need to be compensated through payments in addition to those resulting from marginal pricing (e.g., make-whole payments), further increasing consumers' total payments beyond the market settlement given in Table 31. On the other hand, under the proposed pricing scheme, consumers pay less without requiring any payments beyond those associated with the marginal prices. Furthermore, the proposed pricing scheme also benefits the transmission system. As reported in Table 31, in the absence of transmission prices and the associated revenues, the benchmark approach produces a surplus that falls short by $198,271.31 of the total transmission revenue collected under the proposed scheme. In addition, as also shown in Table 31, the improvements introduced in the proposed pricing scheme allow the attainment of system revenue adequacy, unlike the benchmark method. Note that, as a desirable result, the proposed settlement yields a net balance of $0.00, i.e., the total consumer payment is fully distributed to generators and transmission assets, with neither surplus nor deficit.

5. Conclusion

This paper has presented a new causation-based pricing framework as an alternative to a previously reported scheme. Major salient features include explicitly defining prices for up-spinning reserves, down-spinning reserves, and transmission services, along with a novel security charge mechanism. These additions ensure a more comprehensive allocation of the costs associated with operating reserves and system security, resulting in efficient market operations. The proposed approach is rigorously grounded in the Lagrangian dual function, leveraging Lagrange multipliers to establish nodal prices and security charges. More importantly, the proposed pricing scheme yields a market settlement featuring two relevant and desirable properties, namely cost recovery

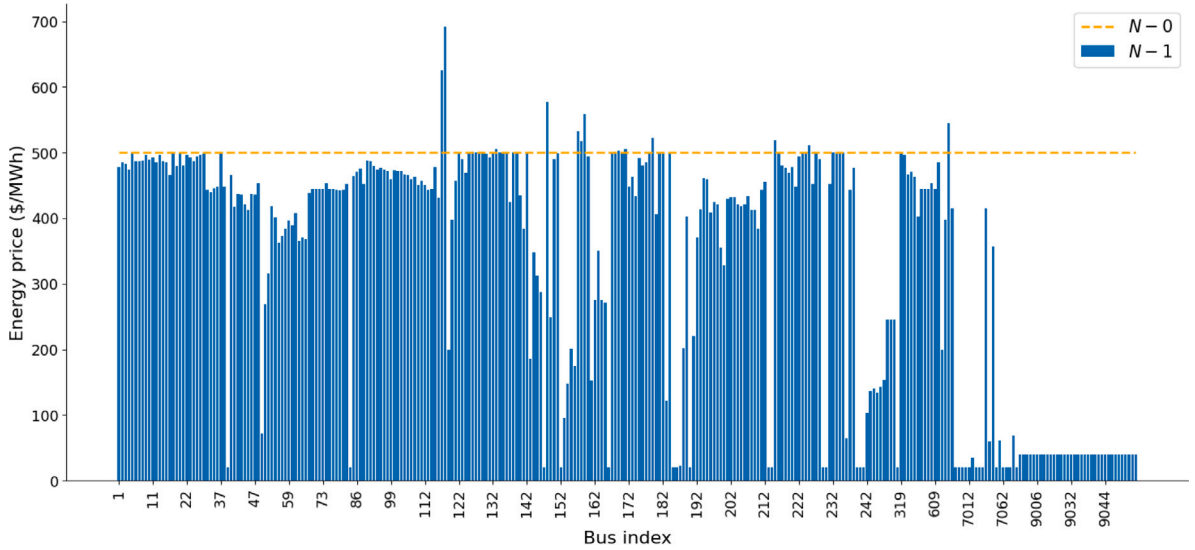


图 2. 300 总线示例 – 节点能量价格。

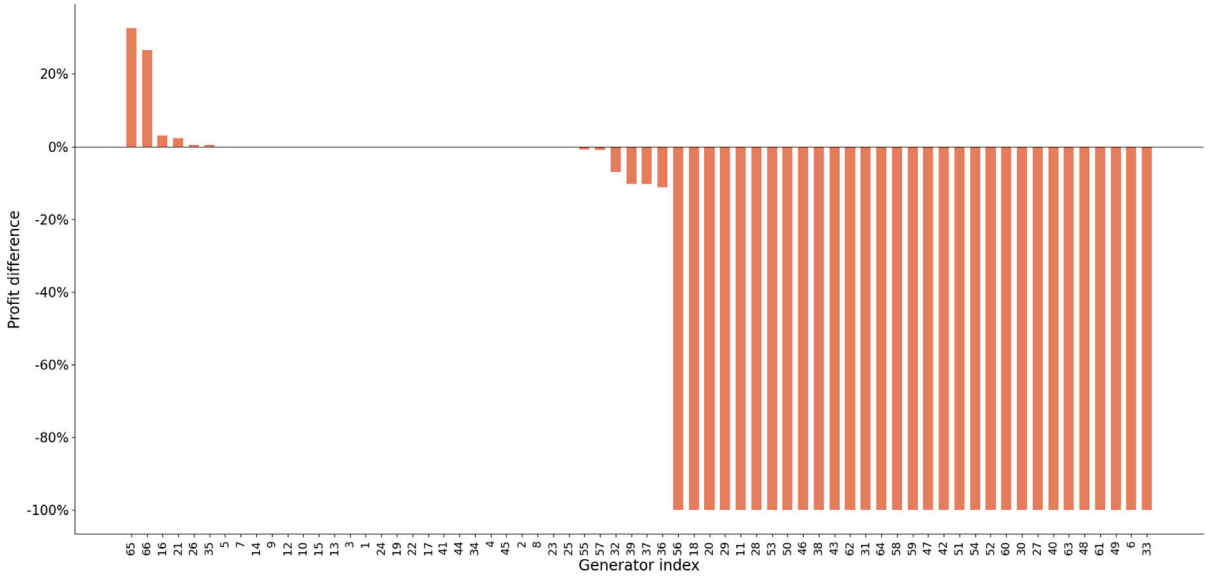


图 3. 300 母线示例 – 发电利润的百分比差异。

最后，表31提供了两种定价方案下的系统结算情况。根据表31，消费者在拟议的定价方案下支付的费用比基准框架下少9957.69美元，而总发电收入则低208,229.00美元，但没有任何机组产生负利润。这些结果从消费者的角度来看尤为重要。如前所述，基准定价方案会导致部分发电机组出现负利润，这些机组需要通过额外支付（例如全额补偿支付）来弥补，除了边际定价产生的支付外，这会进一步增加消费者的总支付额，超过表31中的市场结算金额。另一方面，在拟议的定价方案下，消费者支付更少，而不需要任何超出边际价格的额外支付。此外，拟议的定价方案也有利于传输系统。正如表31所示，在没有与基准方法不同，系统收入充足性方面的考虑。请注意，作为理想结果，所提出的结算产生了净余额为 $0.00，即总消费者支付完全分配给发电机和输电资产，没有盈余或赤字。

5. 结论

本文提出了一种基于因果关系的定价框架，作为先前报道方案的替代方案。主要显著特点包括明确界定上旋储备、下旋储备和输电服务的价格，以及一种新颖的安全收费机制。这些新增内容确保了与运行储备和系统安全相关成本的更全面分配，从而实现高效的市场运行。所提出的方法严格基于拉格朗日对偶函数，利用拉格朗日乘子建立节点价格和安全收费。更重要的是，所提出的定价方案产生了具有两个相关且可取特性的市场结算，即成本回收。

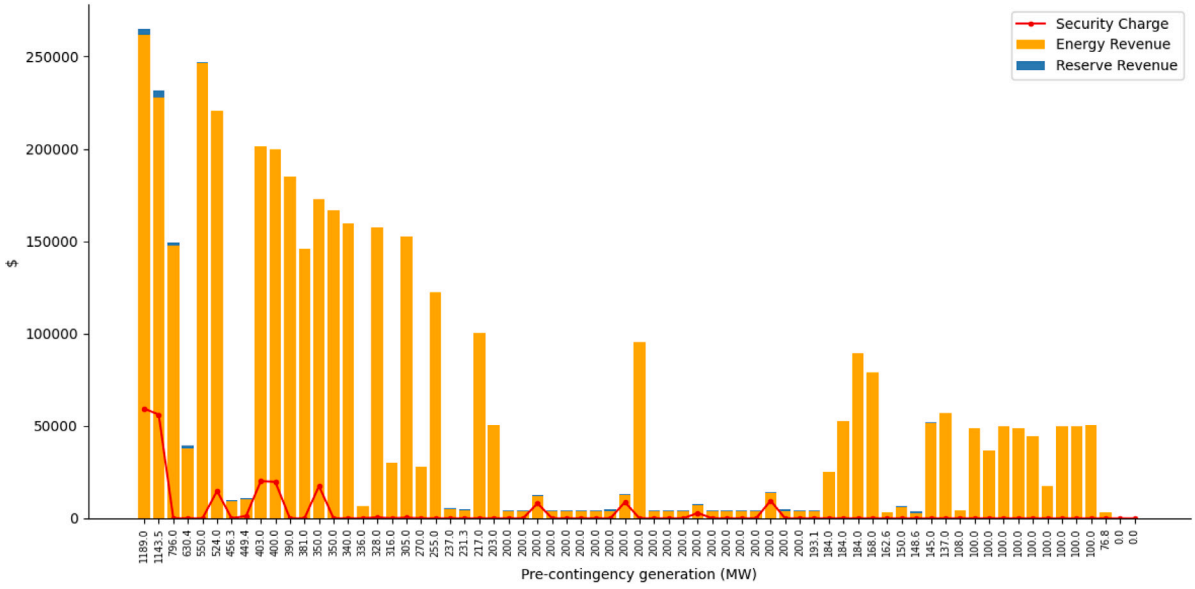


Fig. 4. 300-bus example – Generation revenues and security charges for the proposed framework.

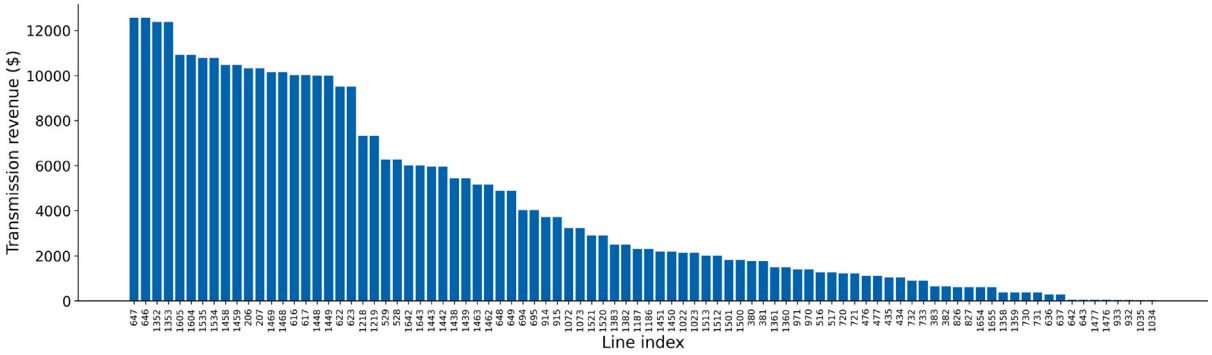


Fig. 5. 300-bus example – Non-zero transmission revenues identified by the proposed framework for the pre-contingency uncongested lines.

Table 31
300-bus example – System settlement.

Framework	Consumer Payment ($)	Generation Revenue ($)	Transmission Revenue ($)	Balance ($)
Benchmark Pricing System	7,464,201.68	4,048,997.62	0.00	3,415,204.06
Proposed Pricing System	7,454,243.99	3,840,768.62	3,613,475.37	0.00
Difference	9,957.69	208,229.00	−3,613,475.37	3,415,204.06

and system revenue adequacy, avoiding the need for any additional payments beyond those resulting from marginal pricing.

Numerical results corroborate the findings in the related literature, demonstrating that generators scheduled with higher pre-contingency output tend to bear higher security charges due to their significant contribution to the system-wide reserve requirements under contingency. Interestingly, based on the cost-causation principle, these charges remain consistent with the uniform pricing system largely used for energy and reserves. Additionally, the newly proposed pricing framework involves transmission assets, remunerating them based on energy and reserve utilization across all contingency states. This unification guarantees a consistent and transparent cost allocation framework. Overall, the proposed pricing system provides a sound extension of transmission rent allocation by capturing system's opportunity costs arising from scarce transmission capacity and consistently accounting for contingency states under the marginal cost principle.

The main limitation of the proposed approach is the adoption of a deterministic, single-period model relying on a convex characterization of system operation. Thus, this work paves the way for future research in this field. One potential direction is the extension to explicitly consider inter-temporal aspects and nonconvexities in market operations such as unit commitment constraints, which require binary variables, and an ac power flow model. The challenges associated with primal degeneracy, i.e., the existence of multiple vectors of Lagrange multipliers for the same optimal market-clearing solution, represent another interesting avenue of research. Moreover, the framework could be enhanced to model more complex, multi-stage security criteria, such as the $N-1-1$ standard. This modeling feature would involve incorporating different reserve products with varying ramp rates and activation times to represent the distinct requirements for immediate contingency response and subsequent system restoration. Furthermore, having established a consistent deterministic framework, a natural next step is to incorporate the uncertainty associated with component outages and nodal injections. This extension will be particularly relevant within the

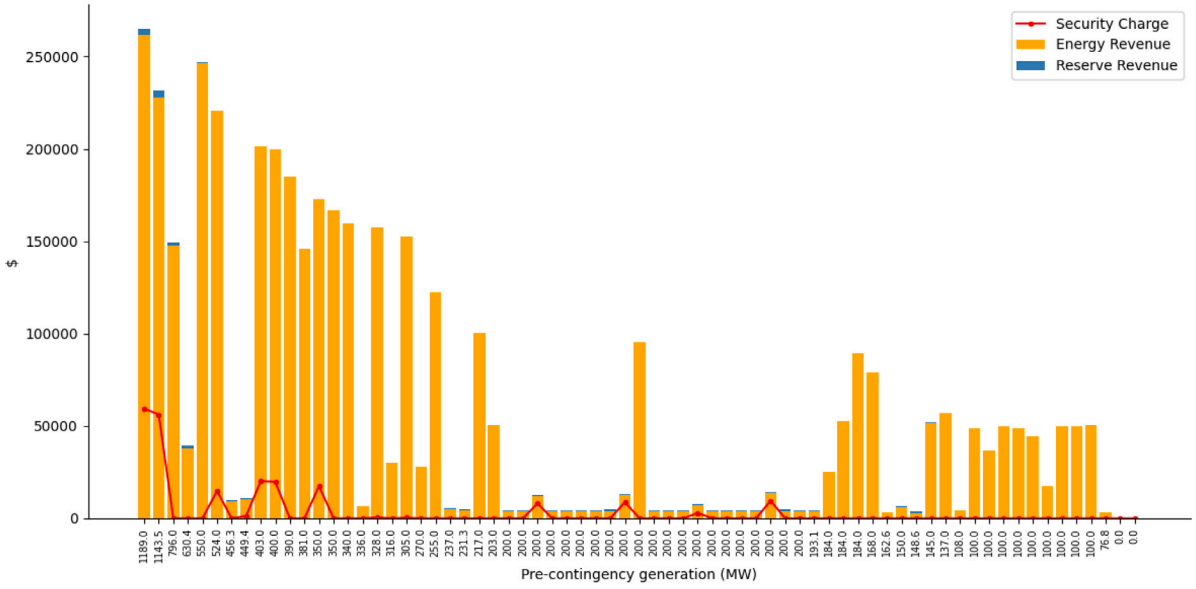


图4. 300母线示例——所提框架的发电收入与安全费用。

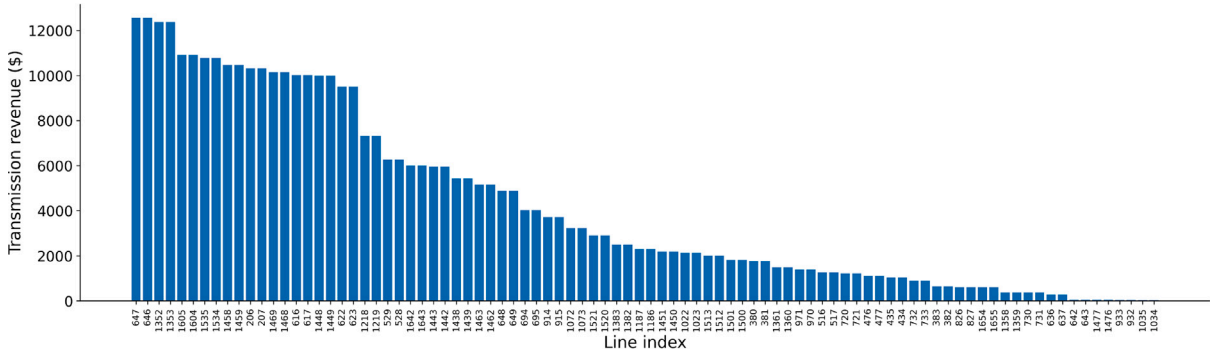


图5. 300母线示例——所提出的框架识别出的预事故无拥塞线路的非零输电收入。

表 31 300 节点示例——系统结算。

Framework	Consumer Payment ($)	Generation Revenue ($)	Transmission Revenue ($)	Balance ($)
Benchmark Pricing System	7,464,201.68	4,048,997.62	0.00	3,415,204.06
Proposed Pricing System	7,454,243.99	3,840,768.62	3,613,475.37	0.00
Difference	9,957.69	208,229.00	−3,613,475.37	3,415,204.06

以及系统收入的充足性，避免了除边际定价所产生的付款之外的任何额外支付的需要。

数值结果证实了相关文献中的研究发现，表明调度了较高预应急出力的发电机往往由于其在应急情况下对系统整体备用量的显著贡献而承担更高的安全费用。有趣的是，基于成本归因原则，这些费用在很大程度上与主要用于能源和备用的统一定价系统保持一致。此外，新提出的定价框架涉及输电资产，根据其在所有应急状态下的能源和备用利用情况进行报酬。这种统一确保了一个一致且透明的成本分配框架。总体而言，所提出的定价系统通过捕捉由于输电容量稀缺而产生的系统机会成本，并在边际成本原则下持续考虑应急状态，为输电租金分配提供了合理的扩展。

所提出方法的主要限制在于采用了基于系统运行凸性特征的确定性单周期模型。因此，这项工作为该领域的未来研究铺平了道路。一个潜在方向是扩展到明确考虑市场运营中的跨期问题和非凸性，例如需要二进制变量的机组组合约束，以及交流潮流模型。与初始退化相关的挑战，即针对同一最优市场清算解存在多个拉格朗日乘子向量，也是另一个有趣的研究方向。此外，该框架还可以增强以模拟更复杂的多阶段安全标准，如$N-1-1$标准。该建模特性将涉及结合具有不同爬坡速率和激活时间的不同备用产品，以表示对即时应急响应的不同需求。

future low-inertia renewables-dominated paradigm, complementing the existing security-based approach with a more granular assessment of operational risks. In this context, the modeling framework could be generalized to represent not only full generator outages but also partial derating of generating units as well as consumer contingencies giving rise to demand security charges. Additionally, further investigation into economic incentives could provide deeper insights into how market participants, including smaller generators and consumers, strategically respond to security charges and reserve pricing while also considering the potential exercise of market power.

Finally, it is worth emphasizing that explicitly charging generators according to a security criterion may be regarded as a nontrivial departure from current market designs, as most system operators do not directly internalize such charges at the generation level. In this regard, a detailed discussion of the associated regulatory and market design implications is beyond the scope of this paper. Interestingly, the proposed framework is agnostic with respect to the ultimate allocation of these costs, which may be charged to generators or alternatively passed through to consumers according to prevailing regulatory arrangements. Nevertheless, the proposed pricing framework should be interpreted as a methodological advancement in identifying transmission and reserve price signals and components, consistently with the dual prices associated with each contingency embedded in the security criterion. In this sense, the proposed pricing system can be of great value to system operators and regulators in identifying the critical components that contribute to reserve costs and to the value of transmission for system security. Given the regulatory nature of this subject, it can be viewed as a relevant, policy-based extension of this work—including the study of alternative allocation rules, such as those based on realized profits—and is therefore left for future research.

CRediT authorship contribution statement

Luíza Ribeiro: Writing – review & editing, Writing – original draft. **Alexandre Street:** Writing – review & editing, Project administration, Methodology, Funding acquisition. **José M. Arroyo:** Writing – review & editing, Supervision. **Rodrigo Moreno:** Validation, Supervision, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The work of Alexandre Street and Luíza Ribeiro was partly funded by the World Bank's Project BR-CCEE-TDR-14-21-PRECO-CS-QBS, within the scope of the META II initiative and with the support of the Brazilian Electricity Trading Chamber (CCEE) and PSR Energy. The work of Alexandre Street was partially supported by (1) CNPq (PQ fellowships grants 304604/2021-6 and 318623/2025-0, and International Cooperation PUC-Rio/Stanford–MCTI/CNPq 16/2025 grant 446718/2025-4), (2) by FAPERJ (grant E-26/200.035/2026), and (3) by the Center for Latin American Studies (CLAS) and the Program on Energy and Sustainable Development (PESD) at Stanford University. The work of J. M. Arroyo was supported in part by (1) the EU Horizon Europe Programme under GA ID: 101160614 (EU-DREAM Project, DOI: 10.3030/101160614), GA ID: 101230578 (INNO-TREC Project, DOI: 10.3030/101230578), and MSCA Doctoral Network GA ID: 101168796 (FITNESS Project, DOI: 10.3030/101168796); (2) Grants PID2021-122579OB-I00, PID2021-126566OB-I00, PID2024-159031OB-I00, and PID2025-171604OB-I00 funded by MICIU/AEI/10.13039/501100011033 and by ERDF/EU; and (3) Grant 2025-GRIN-38264, funded by the Universidad de Castilla-La Mancha and by ERDF.

Appendix. Nomenclature

This section lists the main notation used throughout the paper. Note that superscript "*" stands for optimal value.

Sets and indices:

b	Bus index.
$\mathcal{I}$	Set of indices i of generators.
$\mathcal{J}$	Set of indices j of loads.
$\mathcal{K}$	Set of indices k of contingencies.
$\mathcal{K}_i^{OFF}$	Set of indices k of the contingencies involving the loss of generator i.
$\mathcal{L}$	Set of indices l of lines.
$\mathcal{X}$	Generator feasibility set.
$\mathcal{X}_i$	Subset of $\mathcal{X}$ related to generator i.
$\mathcal{Y}$	Consumer feasibility set.
$\mathcal{Y}_j$	Subset of $\mathcal{Y}$ related to consumer j.

Constants:

$\boldsymbol{\gamma}$	Vector of contingency flow-relief factors, with each element denoted by γ_l.
$\boldsymbol{A}$	Incidence matrix in the pre-contingency state.
$\boldsymbol{A_k}$	Incidence matrix under contingency k.
$\boldsymbol{a_k^g}$	Vector of generator availability statuses under contingency k, with each element denoted by a_{ik}^g.
$\boldsymbol{c}$	Vector of cost rates offered by generators to provide energy in the pre-contingency state, with each element denoted by c_i.
d	Inelastic demand.
$\overline{\boldsymbol{D}}$	Vector of upper bounds on the power consumed, with each element denoted by $\overline{D}_j$.
$\boldsymbol{F^{ss}}$	Vector of steady-state line power flow capacities, with each element denoted by F_l^{ss}.
$\boldsymbol{F^{st}}$	Vector of short-term line power flow capacities, with each element denoted by F_l^{st}.
$\overline{\boldsymbol{G}}$	Vector of generator capacities, with each element denoted by $\overline{G}_i$.
$\boldsymbol{H}$	Matrix relating power flows to nodal phase angles in the pre-contingency state.
$\boldsymbol{H_k}$	Matrix relating power flows to nodal phase angles under contingency k.
$\boldsymbol{M^d}$	Demand-bus mapping matrix.
$\boldsymbol{M^g}$	Generator-bus mapping matrix.
$\boldsymbol{q^{d,dn}}$	Vector of cost rates offered by consumers to provide down-spinning reserve, with each element denoted by $q_j^{d,dn}$.
$\boldsymbol{q^{d,up}}$	Vector of cost rates offered by consumers to provide up-spinning reserve, with each element denoted by $q_j^{d,up}$.
$\boldsymbol{q^{g,dn}}$	Vector of cost rates offered by generators to provide down-spinning reserve, with each element denoted by $q_i^{g,dn}$.
$\boldsymbol{q^{g,up}}$	Vector of cost rates offered by generators to provide up-spinning reserve, with each element denoted by $q_i^{g,up}$.
$\overline{\boldsymbol{R}}^{\boldsymbol{d,dn}}$	Vector of upper bounds for consumer down-spinning reserves, with each element denoted by $\overline{R}_j^{d,dn}$.
$\overline{\boldsymbol{R}}^{\boldsymbol{d,up}}$	Vector of upper bounds for consumer up-spinning reserves, with each element denoted by $\overline{R}_j^{d,up}$.
$\overline{\boldsymbol{R}}^{\boldsymbol{g,dn}}$	Vector of upper bounds for generator down-spinning reserves, with each element denoted by $\overline{R}_i^{g,dn}$.

未来以低惯性可再生能源为主导的模式，通过对操作风险进行更细化的评估，补充现有的基于安全性的方式。在此背景下，该建模框架可以推广，不仅用于表示发电机的全面停机，还可以表示发电单元的部分降额以及导致需求安全费用的消费者突发情况。此外，对经济激励的进一步研究可以更深入地了解包括小型发电机和消费者在内的市场参与者在面对安全费用和备用电价时的战略响应，同时也考虑市场力量可能的行使。

最后，值得强调的是，根据安全标准明确向发电机收取费用可能被视为对现有市场设计的非平凡偏离，因为大多数系统运营商不会在发电层面直接内化此类费用。在这方面，关于相关监管和市场设计影响的详细讨论超出了本文的范围。有趣的是，所提出的框架对于这些成本的最终分配方式是中立的，这些成本可以向发电机收取，也可以根据现行监管安排转嫁给消费者。然而，所提出的定价框架应被理解为在识别传输和备用价格信号及其组成部分方面的方法论进展，与嵌入安全标准的每个应急事件相关的对偶价格一致。从这个意义上说，所提出的定价系统可以为

CRediT 作者贡献声明

Luíza Ribeiro：撰写 – 审阅与编辑，撰写 – 原始草稿。Alexandre Street：撰写 – 审阅与编辑，项目管理，方法学，资金获取。José M. Arroyo：撰写 – 审阅与编辑，监督。Rodrigo Moreno：验证，监督，方法学。

利益冲突声明

作者声明，他们没有已知的竞争性财务利益或个人关系，这些可能会影响本文所报告工作的结果。

致谢

亚历山大·斯特里特（Alexandre Street）和卢伊扎·里贝罗（Luíza Ribeiro）的工作部分由世界银行项目 BR-CCEE-TDR-14-21-PRECO-CS-QBS 资助，该项目属于 META II 计划，并且得到了巴西电力交易商会（CCEE）和 PSR Energy 的支持。亚历山大·斯特里特的工作部分得到了以下支持：(1) CNPq（PQ 奖学金资助 304604/2021-6 和 318623/2025-0，以及国际合作 PUC-Rio/Stanford–MCTI/CNPq 16/2025 资助 446718/2025-4），(2) FAPERJ（资助 E-26/200.035/2026），以及 (3) 斯坦福大学拉丁美洲研究中心（CLAS）和能源与可持续发展项目（PESD）。J. M. Arroyo 的工作部分得到了以下支持：(1) 欧盟 Horizon Europe 计划，资助协议 ID：101160614（EU-DREAM 项目，DOI: 10.3030/101160614）、资产协议 ID：101230578（INNO-TREC 项目，DOI: 10.3030/101230578）及 MSCA 博士网络资助协议 ID：101168796（FITNESS 项目，DOI:

附录. 名称法

本部分列出了整篇论文中使用的主要符号。注意，上标“*”表示最优值。

集合与索引：

b 总线索引。
$\mathcal{I}$ 生成元的索引集合 i。
$\mathcal{J}$ 负载的索引集合 j。
$\mathcal{K}$ 偶发事件的索引集合 k。
$\mathcal{K}_i^{OFF}$ 涉及发电机 i 失效的应急情况的索引集合 k。

$\mathcal{L}$ 行的索引集合 l。
$\mathcal{X}$ 发电机可行性集合。
$\mathcal{X}_i$ 是与发电机 i 相关的 $\mathcal{X}$ 子集。
$\mathcal{Y}$ 消费者可行性集合。
$\mathcal{Y}_j$ $\mathcal{Y}$ 的子集，与消费者 j 相关。
常数：$\boldsymbol{\gamma}$ 应急流量缓解因子的向量，每个元素表示为 γ_l。

$\boldsymbol{A}$ 预应急状态下的关联矩阵。
$\boldsymbol{A_k}$ 在列联表 k 下的发生矩阵。
$\boldsymbol{a_k^g}$ 在应急情况 k 下发电机可用性状态的向量，每个元素由 a_{ik}^g 表示。
$\boldsymbol{c}$ 发电机在预应急状态下提供能源所报价的成本率向量，每个元素用 c_i 表示。

d 非弹性需求。
$\boldsymbol{\overline{D}}$ 功率消耗上限向量，每个元素用 $\overline{D}_j$ 表示。

$\boldsymbol{F^{ss}}$ 稳态线路潮流能力向量，每个元素记为 F_l^{ss}。

$\boldsymbol{F^{st}}$ 短期线路功率流容量向量，每个元素表示为 F_l^{st}。

$\boldsymbol{\overline{G}}$ 发电机容量向量，每个元素由 $\overline{G}_i$ 表示。

$\boldsymbol{H}$ 矩阵，将功率流与预事故状态下的节点相角相关联。

$\boldsymbol{H_k}$ 在应急情况下，将功率流与节点相角相关联的矩阵 k。
$\boldsymbol{M^d}$ 需求-总线映射矩阵。
$\boldsymbol{M^g}$ 发电机-母线映射矩阵。
$\boldsymbol{q^{d,dn}}$ 由消费者提供向下旋转备用的费用率向量，每个元素表示为 $q_j^{d,dn}$。

$\boldsymbol{q^{d,up}}$ 消费者提供上旋备转的成本率向量，每个元素表示为 $q_j^{d,up}$。

$\boldsymbol{q^{g,dn}}$ 发电机提供下降旋转备用的费用率向量，每个元素表示为 $q_i^{g,dn}$。

$\boldsymbol{q^{g,up}}$ 发电机提供上旋储备的报价成本率向量，每个元素表示为 $q_i^{g,up}$。

$\boldsymbol{\overline{R}^{d,dn}}$ 消费者向下调节储备的上限向量，每个元素用 $\overline{R}_j^{d,dn}$ 表示。
$\boldsymbol{\overline{R}^{d,up}}$ 消费者上旋储备的上限向量，每个元素表示为 $\overline{R}_j^{d,up}$。
上限向量用于发电机降速备用容量，每个元素表示为 $\overline{R}_i^{g,dn}$。

$\overline{\boldsymbol{R}}^{g,up}$ Vector of upper bounds for generator up-spinning reserves, with each element denoted by $\overline{R}_i^{g,up}$.
$\boldsymbol{w}$ Vector of rates bid by consumers to buy energy, with each element denoted by w_j.
$\boldsymbol{X}$ Diagonal matrix of line reactances.

Variables:

$\boldsymbol{\theta_0}$ Vector of pre-contingency nodal phase angles.
$\boldsymbol{\theta_k}$ Vector of nodal phase angles under contingency k.
$\boldsymbol{d}_0$ Vector of pre-contingency nodal consumption levels, with each element denoted by d_{b0}.
$\boldsymbol{d_k}$ Vector of nodal consumption levels under contingency k, with each element denoted by d_{bk}.
$\boldsymbol{g}_0$ Vector of generator power outputs in the pre-contingency state, with each element denoted by g_{i0}.
$\boldsymbol{g_k}$ Vector of generator power outputs under contingency k, with each element denoted by g_{ik}.
$\boldsymbol{r^{d,dn}}$ Vector of down-spinning reserves provided by consumers, with each element denoted by $r_j^{d,dn}$.
$\boldsymbol{r^{d,up}}$ Vector of up-spinning reserves provided by consumers, with each element denoted by $r_j^{d,up}$.
$\boldsymbol{r^{g,dn}}$ Vector of down-spinning reserves provided by generators, with each element denoted by $r_i^{g,dn}$.
$\boldsymbol{r^{g,up}}$ Vector of up-spinning reserves provided by generators, with each element denoted by $r_i^{g,up}$.
$\boldsymbol{x}$ Vector of optimization variables related to generators.
$\boldsymbol{x_i}$ Vector of optimization variables related to generator i.
$\boldsymbol{y}$ Vector of optimization variables related to consumers.
$\boldsymbol{y_j}$ Vector of optimization variables related to consumer j.

Lagrange multipliers:

$\boldsymbol{\Pi}$ Vector of all Lagrange multipliers.
$\boldsymbol{\Pi_b}$ Subvector of $\boldsymbol{\Pi}$ related to bus b.
$\boldsymbol{\pi_0}$ Vector of Lagrange multipliers associated with the pre-contingency nodal power balance constraints, with each element denoted by π_{b0}.
π_0 Lagrange multiplier associated with the pre-contingency power balance constraint in the single-bus model.
$\boldsymbol{\pi_k}$ Vector of Lagrange multipliers associated with the nodal power balance constraints under contingency k, with each element denoted by π_{bk}.
π_k Lagrange multiplier associated with the nodal power balance constraint under contingency k in the single-bus model.
$\boldsymbol{\pi_0^{f+}}$ Vector of Lagrange multipliers associated with the lower bounds for the pre-contingency line flows, with each element denoted by π_{l0}^{f+}.
$\boldsymbol{\pi_0^{f-}}$ Vector of Lagrange multipliers associated with the upper bounds for the pre-contingency line flows, with each element denoted by π_{l0}^{f-}.
$\boldsymbol{\pi_k^{f+}}$ Vector of Lagrange multipliers associated with the lower bounds for the line flows under contingency k, with each element denoted by π_{lk}^{f+}.
$\boldsymbol{\pi_k^{f-}}$ Vector of Lagrange multipliers associated with the upper bounds for the line flows under contingency k, with each element denoted by π_{lk}^{f-}.

Functions:

$\phi(\cdot)$ Lagrangian dual function.
$\phi_\theta(\cdot)$ Term of the Lagrangian dual function associated with $\boldsymbol{\theta_0}$ and $\boldsymbol{\theta_k}$.
$\psi_{jk}^d(\cdot)$ Consumer j's payment fraction under contingency k.
$\psi_{ik}^g(\cdot)$ Generator i's revenue fraction under contingency k.

Others:

π_{bk}^+ Contribution to p_b^{up} due to contingency k.
π_{bk}^- Contribution to p_b^{dn} due to contingency k.
$\boldsymbol{\pi_0^f}$ Vector resulting from summing $\boldsymbol{\pi_0^{f+}}$ and $\boldsymbol{\pi_0^{f-}}$, with each element denoted by π_{l0}^f.
$\boldsymbol{\pi_k^f}$ Vector resulting from summing $\boldsymbol{\pi_k^{f+}}$ and $\boldsymbol{\pi_k^{f-}}$, with each element denoted by π_{lk}^f.
$b(i)$ Bus of generator i.
$b(j)$ Bus of consumer j.
$C_j^{d,dn}$ Down-spinning reserve offer cost of consumer j.
$C_j^{d,t}$ Total cost of consumer j.
$C_j^{d,up}$ Up-spinning reserve offer cost of consumer j.
$C_i^{g,dn}$ Down-spinning reserve offer cost of generator i.
$C_i^{g,e}$ Energy cost of generator i.
$C_i^{g,s}$ Security charge of generator i.
$C_i^{g,t}$ Total cost of generator i.
$C_i^{g,up}$ Up-spinning reserve offer cost of generator i.
CP Consumer payment.
CP_j Payment of consumer j.
CP_j^e Energy payment of consumer j.
p_b^{dn} Down-spinning reserve price at bus b.
p^e Energy price.
p_b^e Energy price at bus b.
p_l^f Transmission price for line l.
p^s Security price.
p_b^s Security price at bus b.
p^{up} Up-spinning reserve price.
p_b^{up} Up-spinning reserve price at bus b.
$Profit_j^d$ Profit of consumer j.
$Profit_i^g$ Profit of generator i.
$R_j^{d,dn}$ Down-spinning reserve revenue of consumer j.
$R_j^{d,up}$ Up-spinning reserve revenue of consumer j.
$R_i^{g,dn}$ Down-spinning reserve revenue of generator i.
$R_i^{g,e}$ Energy revenue of generator i.
$R_i^{g,t}$ Total revenue of generator i.
$R_i^{g,up}$ Up-spinning reserve revenue of generator i.
U_j^d Utility of consumer j.

References

Aganagic, M., Abdul-Rahman, K. H., & Waight, J. G. (1998). Spot pricing of capacities for generation and transmission of reserve in an extended Poolco model. *IEEE Transactions on Power Systems*, *13*(3), 1128–1135. http://dx.doi.org/10.1109/59.709110.

Ahunbay, M. Ş., Bichler, M., Dobos, T., & Knörr, J. (2024). Solving large-scale electricity market pricing problems in polynomial time. *European Journal of Operational Research*, *318*(2), 605–617. http://dx.doi.org/10.1016/j.ejor.2024.05.020.

Arroyo, J. M., & Galiana, F. D. (2005). Energy and reserve pricing in security and network-constrained electricity markets. *IEEE Transactions on Power Systems*, *20*(2), 634–643. http://dx.doi.org/10.1109/TPWRS.2005.846221.

Australian Energy Market Operator (AEMO) (2023). Wholesale electricity market design summary. Retrieved April 28, 2026, from https://aemo.com.au/-/media/files/initiatives/wem-reform-program/wem-reform-market-design-summary.pdf.

Badesa, L., Matamala, C., & Strbac, G. (2025). Who should pay for frequency-containment ancillary services? Making responsible units bear the cost to shape investment in generation and loads. *Energy Policy*, *196*, Article 114379. http://dx.doi.org/10.1016/j.enpol.2024.114379.

Badesa, L., Matamala, C., Zhou, Y., & Strbac, G. (2023). Assigning shadow prices to synthetic inertia and frequency response reserves from renewable energy sources. *IEEE Transactions on Sustainable Energy*, *14*(1), 12–26. http://dx.doi.org/10.1109/TSTE.2022.3198324.

Bertsimas, D., & Tsitsiklis, J. N. (1997). *Introduction to linear optimization*. Belmont, MA, USA: Athena Scientific.

Billimoria, F., Mancarella, P., & Poudineh, R. (2020). Market design for system security in low-carbon electricity grids: From the physics to the economics. *The Oxford Institute for Energy Studies*, EL 41. http://dx.doi.org/10.26889/9781784671600.

$\boldsymbol{R}^{\overline{g,up}}$ 发电机上调调节储备的上限向量，每个元素表示为 $R_i^{g,up}$。 $\boldsymbol{w}$ 消费者购买能源的出价率向量，每个元素表示为 w_j。 $\boldsymbol{X}$ 线路电抗的对角矩阵。 变量： $\boldsymbol{\theta_0}$ 事故前节点相角向量。 $\boldsymbol{\theta_k}$ 事故 k 发生时的节点相角向量。 $\boldsymbol{d}_0$ 事故前节点负荷水平向量，每个元素表示为 d_{b0}。 $\boldsymbol{d_k}$ 事故 k 发生时的节点负荷水平向量，每个元素表示为 d_{bk}。 $\boldsymbol{g}_0$ 事故前发电机功率输出向量，每个元素表示为 g_{i0}。 $\boldsymbol{g_k}$ 事故 k 发生时的发电机功率输出向量，每个元素表示为 g_{ik}。 $\boldsymbol{r^{d,dn}}$ 消费者提供的下调储备向量，每个元素表示为 $r_j^{d,dn}$。 $\boldsymbol{r^{d,up}}$ 消费者提供的上调储备向量，每个元素表示为 $r_j^{d,up}$。

$_{lk}$.

函数:

$\phi(\cdot)$	拉格朗日对偶函数
$\phi_\theta(\cdot)$	与 $\boldsymbol{\theta_0}$ 和 $\boldsymbol{\theta_k}$ 相关的拉格朗日对偶函数的项。
$\psi_{jk}^d(\cdot)$	消费者 j 在应急情况下 k 的支付比例。
$\psi_{ik}^g(\cdot)$	发电机 i 在应急情况 k 下的收入份额。

Others:

π_{bk}^+	由于应急情况 k 对 p_b^{up} 的贡献。
π_{bk}^-	由于应急情况 k 对 p_b^{dn} 的贡献。
$\boldsymbol{\pi_0^f}$	由将 $\boldsymbol{\pi_0^{f+}}$ 和 $\boldsymbol{\pi_0^{f-}}$ 求和得到的向量，每个元素表示为 π_{l0}^f。
$\boldsymbol{\pi_k^f}$	由将 $\boldsymbol{\pi_k^{f+}}$ 和 $\boldsymbol{\pi_k^{f-}}$ 求和得到的向量，每个元素表示为 π_{lk}^f。
$b(i)$	发电机 i 的母线。
$b(j)$	消费者 j 的巴士。
$C_j^{d,dn}$	向下旋转备用的消费者出价成本 j。
$C_j^{d,t}$	消费者 j 的总成本。
$C_j^{d,up}$	向上旋转备用的消费者出价成本 j。
$C_i^{g,dn}$	发电机 i 的下调备用出价成本。
$C_i^{g,e}$	发电机 i 的能耗。
$C_i^{g,s}$	发电机 i 的安全费用。
$C_i^{g,t}$	发电机 i 的总成本。
$C_i^{g,up}$	发电机 i 的调频备用出价成本。
CP	消费者支付。
CP_j	消费者 j 的支付。
CP_j^e	消费者 j 的能源支付。
p_b^{dn}	母线 b 的下旋备用价格。
p^e	能源价格
p_b^e	母线 b 的能量价格。
p_l^f	线路 l 的传输价格。
p^s	证券价格
p_b^s	公交车 b 上的安全价格。
p^{up}	上涨的备用价格。
p_b^{up}	在总线 b 提高旋转备用价格。
$Profit_j^d$	消费者的利润 j。
$Profit_i^g$	发电机 i 的利润。
$R_j^{d,dn}$	消费者 j 的下旋储备收入。
$R_j^{d,up}$	消费者 j 的上旋储备收入。
$R_i^{g,dn}$	发电机 i 的下调备用容量收入。
$R_i^{g,e}$	发电机 i 的能源收入。
$R_i^{g,t}$	发电机 i 的总收入。
$R_i^{g,up}$	发电机 i 的上旋备用收入。
U_j^d	消费者 j 的效用。

参考文献

Aganagic, M., Abdul-Rahman, K. H., & Waight, J. G. (1998)。在扩展的Poolco模型中，容量现货定价用于发电和备用传输。*IEEE Transactions on Power Systems*, *13*(3), 1128–1135。http://dx.doi.org/10.1109/59.709110。Ahunbay, M. ., Bichler, M., Dobos, T., & Knörr, J. (2024)。以多项式时间解决大规模电力市场定价问题。*European Journal of Operational Research*, *318*(2), 605–617。http://dx.doi.org/10.1016/j.ejor.2024.05.020。Arroyo, J. M., & Galiana, F. D. (2005)。在受安全性和网络约束的电力市场中，能源和备用定价。*IEEE Transactions on Power Systems*, *20*(2), 634–643。http://dx.doi.org/10.1109/TPWRS.2005.846221。澳大利亚能源市场运营商 (AEMO) (2023)。批发电力市场设计摘要。2026年4月28日检索自 https://aemo.com.au/-/media/files/initiatives/wem-reform-program/wem-reform-market-design-summary.pdf。Badesa, L., Mata

Bouffard, F., Galiana, F. D., & Conejo, A. J. (2005). Market-clearing with stochastic security—Part I: Formulation. *IEEE Transactions on Power Systems*, *20*(4), 1818–1826. http://dx.doi.org/10.1109/TPWRS.2005.857016.

Byers, C., & Hug, G. (2023). Long-run optimal pricing in electricity markets with non-convex costs. *European Journal of Operational Research*, *307*(1), 351–363. http://dx.doi.org/10.1016/j.ejor.2022.07.052.

California ISO (2019). 2019 annual report on market issues & performance. Retrieved April 28, 2026, from http://www.caiso.com/Documents/2019AnnualReportonMarketIssuesandPerformance.pdf.

California ISO (2025). Business practice manual for reliability requirements. Retrieved April 28, 2026, from https://www.caiso.com/documents/business-practice-manual-for-reliability-requirements-version-dame.pdf.

Chen, Y., Gribik, P., & Gardner, J. (2014). Incorporating post zonal reserve deployment transmission constraints into energy and ancillary service co-optimization. *IEEE Transactions on Power Systems*, *29*(2), 537–549. http://dx.doi.org/10.1109/TPWRS.2013.2284791.

Conejo, A. J., Arroyo, J. M., Redondo, N. J., & Prieto, F. J. (1999). Lagrangian relaxation applications to electric power operations and planning problems. In Y.-H. Song (Ed.), *Modern optimisation techniques in power systems* (pp. 173–203). Dordrecht, The Netherlands: Springer, http://dx.doi.org/10.1007/978-94-015-9189-8.

European Union Agency for the Cooperation of Energy Regulators (ACER) (2026). Capacity allocation and congestion management. Retrieved April 28, 2026, from https://www.acer.europa.eu/electricity/market-rules/capacity-allocation-and-congestion-management.

Federal Energy Regulatory Commission (FERC) (2011). Recent ISO software enhancements and future software and modeling plans. Retrieved April 9, 2026, from https://www.ferc.gov/sites/default/files/2020-05/rto-iso-soft-2011.pdf.

Frank, S., & Rebennack, S. (2016). An introduction to optimal power flow: Theory, formulation, and examples. *IIE Transactions*, *48*(12), 1172–1197. http://dx.doi.org/10.1080/0740817X.2016.1189626.

Galiana, F. D., Bouffard, F., Arroyo, J. M., & Restrepo, J. F. (2005). Scheduling and pricing of coupled energy and primary, secondary, and tertiary reserves. *Proceedings of the IEEE*, *93*(11), 1970–1983. http://dx.doi.org/10.1109/JPROC.2005.857492.

Gan, D., & Litvinov, E. (2003). Energy and reserve market designs with explicit consideration to lost opportunity costs. *IEEE Transactions on Power Systems*, *18*(1), 53–59. http://dx.doi.org/10.1109/TPWRS.2002.807052.

Karangelos, E., & Bouffard, F. (2012). Towards full integration of demand-side resources in joint forward energy/reserve electricity markets. *IEEE Transactions on Power Systems*, *27*(1), 280–289. http://dx.doi.org/10.1109/TPWRS.2011.2163949.

Kaya, A., Conejo, A. J., & Rebennack, S. (2026). Fifty years of power systems optimization. *European Journal of Operational Research*, *329*(1), 1–23. http://dx.doi.org/10.1016/j.ejor.2025.05.022.

Kirby, B., & Hirst, E. (2003). Allocating the costs of contingency reserves. *The Electricity Journal*, *16*(10), 39–47. http://dx.doi.org/10.1016/j.tej.2003.09.009.

Kirschen, D. S., & Strbac, G. (2019). *Fundamentals of power system economics* (2nd ed.). Chichester, UK: John Wiley & Sons, Ltd, http://dx.doi.org/10.1002/0470020598.

Li, X., & Hedman, K. W. (2019). Enhanced energy management system with corrective transmission switching strategy—Part II: Results and discussion. *IEEE Transactions on Power Systems*, *34*(6), 4503–4513. http://dx.doi.org/10.1109/TPWRS.2019.2922881.

Liu, S., Xiang, M., Yu, S., & Yang, Z. (2023). Cost allocation for frequency regulation mileage based on contributions to system power unbalance. In *2023 IEEE 6th International Electrical and Energy Conference* (pp. 1703–1707). Hefei, China: http://dx.doi.org/10.1109/CIEEC58067.2023.10166726.

Martin, N. C., & Fanzeres, B. (2023). Stochastic risk-averse energy and reserve scheduling and pricing schemes with non-convexities and revenue caps. *Electric Power Systems Research*, *225*, Article 109858. http://dx.doi.org/10.1016/j.epsr.2023.109858.

Matamala, C., Badesa, L., Moreno, R., & Strbac, G. (2024). Cost allocation for inertia and frequency response ancillary services. *IEEE Transactions on Energy Markets, Policy, and Regulation*, *2*(3), 328–338. http://dx.doi.org/10.1109/TEMPR.2024.3355291.

Matpower (2026). Case300: IEEE 300-bus power flow test case. Retrieved April 28, 2026, from https://matpower.org/docs/ref/matpower5.0/case300.html.

Mays, J., Morton, D. P., & O'Neill, R. P. (2021). Investment effects of pricing schemes for non-convex markets. *European Journal of Operational Research*, *289*(2), 712–726. http://dx.doi.org/10.1016/j.ejor.2020.07.026.

Morales, J. M., Conejo, A. J., Liu, K., & Zhong, J. (2012). Pricing electricity in pools with wind producers. *IEEE Transactions on Power Systems*, *27*(3), 1366–1376. http://dx.doi.org/10.1109/TPWRS.2011.2182622.

New York ISO (2026). Manual 04: Installed capacity manual. Retrieved April 28, 2026, from https://www.nyiso.com/documents/20142/2923301/icap_mnl.pdf.

North American Electric Reliability Corporation (NERC) (2026). Reliability standards for the bulk electric systems of North America. Retrieved April 28, 2026, from https://www.nerc.com/pa/Stand/Reliability%20Standards%20Complete%20Set/RSCompleteSet.pdf.

O'Neill, R. P., Castillo, A., Eldridge, B., & Hytowitz, R. B. (2017). Dual pricing algorithm in ISO markets. *IEEE Transactions on Power Systems*, *32*(4), 3308–3310. http://dx.doi.org/10.1109/TPWRS.2016.2614891.

O'Neill, R. P., Sotkiewicz, P. M., Hobbs, B. F., Rothkopf, M. H., & Stewart, W. R., Jr. (2005). Efficient market-clearing prices in markets with nonconvexities. *European Journal of Operational Research*, *164*(1), 269–285. http://dx.doi.org/10.1016/j.ejor.2003.12.011.

PJM Interconnection, L. L. C. (2025). PJM manual 11: Energy & ancillary services market operations. Retrieved April 28, 2026, from https://www.pjm.com/-/media/documents/manuals/m11.ashx.

Ribeiro, L., Arroyo, J. M., Street, A., & Moreno, R. (2025). Pricing of energy and reserves in electricity markets with endogenously defined reserve requirements: 300-bus case study data. Retrieved April 17, 2026, from http://dx.doi.org/10.21227/fcqy-6y53.

Ribeiro, L., Street, A., Valladão, D., Freire, A. C., & Barroso, L. (2023). Technical and economical aspects of wholesale electricity markets: An international comparison and main contributions for improvements in Brazil. *Electric Power Systems Research*, *220*, Article 109364. http://dx.doi.org/10.1016/j.epsr.2023.109364.

Roald, L. A., Pozo, D., Papavasiliou, A., Molzahn, D. K., Kazempour, J., & Conejo, A. (2023). Power systems optimization under uncertainty: A review of methods and applications. *Electric Power Systems Research*, *214*, Article 108725. http://dx.doi.org/10.1016/j.epsr.2022.108725.

Schweppe, F. C., Caramanis, M. C., Tabors, R. D., & Bohn, R. E. (1988). *Spot pricing of electricity*. Norwell, MA, USA: Kluwer Academic Publishers, http://dx.doi.org/10.1007/978-1-4613-1683-1.

Shi, J., Guo, Y., Tong, L., Wu, W., & Sun, H. (2023). A scenario-oriented approach to energy-reserve joint procurement and pricing. *IEEE Transactions on Power Systems*, *38*(1), 411–426. http://dx.doi.org/10.1109/TPWRS.2022.3165635.

Strbac, G., & Kirschen, D. S. (2000). Who should pay for reserve? *The Electricity Journal*, *13*(8), 32–37. http://dx.doi.org/10.1016/S1040-6190(00)00144-5.

Street, A., Moreira, A., Arroyo, J. M., Alguacil, N., & Barroso, L. (2025). Robustness: The missing ingredient in generation scheduling. *IEEE Power and Energy Magazine*, *23*(3), 31–43. http://dx.doi.org/10.1109/MPE.2025.3542381.

Wang, F., & Chen, Y. (2021). Market implications of short-term reserve deliverability enhancement. *IEEE Transactions on Power Systems*, *36*(2), 1504–1514. http://dx.doi.org/10.1109/TPWRS.2020.3017021.

Wang, J., Shahidehpour, M., & Li, Z. (2009). Contingency-constrained reserve requirements in joint energy and ancillary services auction. *IEEE Transactions on Power Systems*, *24*(3), 1457–1468. http://dx.doi.org/10.1109/TPWRS.2009.2022983.

Wong, S., & Fuller, J. D. (2007). Pricing energy and reserves using stochastic optimization in an alternative electricity market. *IEEE Transactions on Power Systems*, *22*(2), 631–638. http://dx.doi.org/10.1109/TPWRS.2007.894867.

Wood, A. J., Wollenberg, B. F., & Sheblé, G. B. (2014). *Power generation, operation, and control* (3rd ed.). Hoboken, NJ, USA: John Wiley & Sons, Inc.

Wu, T., Rothleder, M., Alaywan, Z., & Papalexopoulos, A. D. (2004). Pricing energy and ancillary services in integrated market systems by an optimal power flow. *IEEE Transactions on Power Systems*, *19*(1), 339–347. http://dx.doi.org/10.1109/TPWRS.2003.820701.

Xiang, M., Yang, Z., Yu, J., & Wang, G. (2023). Determination and cost allocation for regulation reserve with renewables: A data-driven assisted approach. *IEEE Transactions on Sustainable Energy*, *14*(2), 813–825. http://dx.doi.org/10.1109/TSTE.2022.3226255.

Bouffard, F., Galiana, F. D., & Conejo, A. J. (2005). 随机安全情况下的市场清算——第一部分：公式化。*IEEE Transactions on Power Systems*, *20*(4), 1818–1826. http://dx.doi.org/10.1109/TPWRS.2005.857016. Byers, C., & Hug, G. (2023). 电力市场中具有非凸成本的长期最优定价。*European Journal of Operational Research*, *307*(1), 351–363. http://dx.doi.org/10.1016/j.ejor.2022.07.052. 加利福尼亚独立系统运营商 (California ISO) (2019). 2019年市场问题与绩效年度报告。检索于2026年4月28日，从 http://www.caiso.com/Documents/2019AnnualReportonMarketIssuesandPerformance.pdf. 加利福尼亚独立系统运营商 (California ISO) (2025). 可靠性要求业务操作手册。检索于2026年4月28日，从 https://www.caiso.com/documents/business-practice-manual-for-reliability-requirements-version-dame.pdf. 陈毅, Gribik, P., & Gardner, J. (2014). 将区域后储备部署传输约束纳入能源与辅助服务联合优化。Matpower (2026)。Case300：IEEE 300 节点潮流测试案例。于 2026 年 4 月 28 日从 https://matpower.org/docs/ref/matpower5.0/case300.html 获取。Mays, J., Morton, D. P., & O’Neill, R. P. (2021)。非凸市场定价方案的投资影响。*European Journal of Operational Research*, *289*(2), 712–726。http://dx.doi.org/10.1016/j.ejor.2020.07.026。Morales, J. M., Conejo, A. J., Liu, K., & Zhong, J. (2012)。在包含风电生产商的电力池中定价电力。*IEEE Transactions on Power Systems*, *27*(3), 1366–1376。http://dx.doi.org/10.1109/TPWRS.2011.2182622。纽约 ISO (2026)。手册 04：装机容量手册。于 2026 年 4 月 28 日从 https://www.nyiso.com/documents/20142/2923301/icap_mnl.pdf 获取。北美电力可靠性公司 (NERC) (2026)。北美大电网系统的可靠性标准。于 2026 年 4 月 28 日从 https://www.nerc.com/pa/Stand/Reliability%20Standards%20Complete%20Set/RSCompleteSet.pdf 获取。O’N